AP Calculus Unit 2: Differentiation: Definition and Fundamental Properties

Lee

Name: _____ Date: _____

Contents

1 Derivative Rules **2**
1.1 Power rule, sums and constant multiples . . . 2
1.2 Derivatives of $\sin x$, $\cos x$, e^x, $\ln x$ and b^x . . . 5
1.3 The product rule . . . 7
1.4 The quotient rule . . . 10
1.5 Derivatives of $\tan x$, $\cot x$, $\sec x$ and $\csc x$ . . . 14
1.6 Rules with values from a table . . . 18
Exercise 1 . . . 21

2 What the Derivative Means **28**
2.1 The algebraic definition of the derivative . . . 28
2.2 Limits that are derivatives . . . 30
2.3 The graphical meaning of the derivative . . . 35
2.4 Derivatives and rates of change . . . 54
2.5 Rates of change in context . . . 63
2.6 Derivatives on the calculator . . . 67
Exercise 2 . . . 73

3 Differentiability and Continuity **82**
3.1 Differentiability from a graph . . . 82
3.2 Differentiability from a formula . . . 91
3.3 Differentiable implies continuous . . . 97
3.4 Making a piecewise function differentiable . . . 104
Exercise 3 . . . 107

4 Detailed solutions **117**
Practice . . . 117
Exercises . . . 142

1 Derivative Rules

1.1 Power rule, sums and constant multiples

Notes

- The derivative of f is a new function f'. The number $f'(a)$ is the slope of the graph of f at $x = a$ (the next section explains why). For $y = f(x)$ these all mean the derivative:

$$f'(x), \qquad y', \qquad \frac{dy}{dx}, \qquad \frac{d}{dx}f(x).$$

- The rules (c and n are constants):

$$\frac{d}{dx}(c) = 0, \qquad \frac{d}{dx}\big(x^n\big) = nx^{n-1},$$

$$\frac{d}{dx}\big(cf(x)\big) = cf'(x), \qquad \frac{d}{dx}\big(f(x) \pm g(x)\big) = f'(x) \pm g'(x).$$

Example 1 The power rule

Find the derivative of each function.

(a) $y = 6\sqrt{x} - \dfrac{4}{x^2} + \dfrac{5}{3x} + \pi^2$

(b) $y = \big(3x^2 - 1\big)^2$

(c) $f(x) = \dfrac{x^2 - 4x + 3}{\sqrt{x}}$. Find $f'(1)$.

Solution

(a)

$$y = 6x^{\frac{1}{2}} - 4x^{-2} + \frac{5}{3}x^{-1} + \pi^2 \qquad \text{rewrite every term as } cx^n$$

$$\frac{dy}{dx} = 6 \cdot \frac{1}{2}x^{-\frac{1}{2}} - 4(-2)x^{-3} + \frac{5}{3}(-1)x^{-2} + 0 \qquad \text{power rule; } \pi^2 \text{ is a constant}$$

$$= \frac{3}{\sqrt{x}} + \frac{8}{x^3} - \frac{5}{3x^2}.$$

(b) A bracket is not x, so the power rule does not apply to it. Expand first.

$$y = 9x^4 - 6x^2 + 1 \qquad \text{expand}$$

$$\frac{dy}{dx} = 36x^3 - 12x.$$

The trap answer $2\big(3x^2 - 1\big)$ treats the bracket as x.

(c) One term on the bottom: split. Then differentiate, then substitute.

$$f(x) = \frac{x^2}{x^{\frac{1}{2}}} - \frac{4x}{x^{\frac{1}{2}}} + \frac{3}{x^{\frac{1}{2}}} = x^{\frac{3}{2}} - 4x^{\frac{1}{2}} + 3x^{-\frac{1}{2}} \qquad \text{split; subtract the powers}$$

$$f'(x) = \frac{3}{2}x^{\frac{1}{2}} - 2x^{-\frac{1}{2}} - \frac{3}{2}x^{-\frac{3}{2}}$$

$$f'(1) = \frac{3}{2} - 2 - \frac{3}{2} = -2. \qquad \text{every power of 1 is 1}$$

Solution (continued)

The trap answer is $f(1) = \dfrac{1-4+3}{1} = 0$.

Rewrite first, then differentiate: every term must look like cx^n before the power rule is used.

Practice

P1 Find the derivative of each function.

(a) $y = 8\sqrt[3]{x} + \dfrac{2}{x^3} - \dfrac{7}{4x} + \sqrt{3}$

(b) $y = \left(2x^2 - 1\right)^2$

(c) $f(x) = \dfrac{x^3 - 2x + 5}{x^2}$. Find $f'(2)$.

P2 [BC 2013 Alt] If $y = \frac{1}{2}x^{\frac{4}{5}} - \frac{3}{x^5}$, then $\frac{dy}{dx} =$

(A) $\frac{2}{5x^{\frac{1}{5}}} + \frac{15}{x^6}$

(B) $\frac{2}{5x^{\frac{1}{5}}} + \frac{15}{x^4}$

(C) $\frac{2}{5x^{\frac{1}{5}}} - \frac{3}{5x^4}$

(D) $\frac{2x^{\frac{1}{5}}}{5} + \frac{15}{x^6}$

P3 [AB 2014 Intl] If $f(x) = 4x^{-2} + \frac{1}{4}x^2 + 4$, then $f'(2) =$

(A) -58 (B) -3 (C) 0 (D) 1

1.2 Derivatives of $\sin x$, $\cos x$, e^x, $\ln x$ and b^x

Notes

- The derivatives (angles in radians; $b > 0$ is a constant):

$$\frac{d}{dx}(\sin x) = \cos x, \qquad \frac{d}{dx}(\cos x) = -\sin x,$$
$$\frac{d}{dx}\big(e^x\big) = e^x, \qquad \frac{d}{dx}(\ln x) = \frac{1}{x}, \qquad \frac{d}{dx}\big(b^x\big) = b^x \ln b.$$

- ln of a constant times x: split it first, $\ln(ax) = \ln a + \ln x$. Then $\ln a$ is just a constant, so $\dfrac{d}{dx}\ln(4x) = \dfrac{d}{dx}(\ln 4 + \ln x) = \dfrac{1}{x}$.
- Numbers such as e^2, $\ln 7$ and $\sin\dfrac{\pi}{6}$ are constants: their derivative is 0.

Traps.

- The minus sign is on the derivative of $\cos x$, not of $\sin x$.
- b^x is not a power of x: $\dfrac{d}{dx}\big(5^x\big) = 5^x \ln 5$, not $x \cdot 5^{x-1}$.
- $f'(a)$, not $f(a)$: for $f(x) = \sin x$, $f'\Big(\dfrac{\pi}{3}\Big) = \cos\dfrac{\pi}{3} = \dfrac{1}{2}$, while $f\Big(\dfrac{\pi}{3}\Big) = \dfrac{\sqrt{3}}{2}$.

Example 2 The basic functions

Find the derivative of each function.

(a) $f(x) = 3\sin x - 2\cos x + 4e^x$. Find $f'(0)$.

(b) $y = 5\ln x - \dfrac{2}{x} + \ln 7$

(c) $y = 3^x + x^3 + e^3$

(d) $f(x) = \ln(4x) - \sqrt{x}$. Find $f'(1)$.

(e) $f(x) = 2\sin x + \cos x$. Find $f'\Big(\dfrac{\pi}{3}\Big)$.

Solution

(a) Differentiate first, then substitute.

$$f'(x) = 3\cos x - 2(-\sin x) + 4e^x = 3\cos x + 2\sin x + 4e^x$$
$$f'(0) = 3(1) + 2(0) + 4(1) = 7. \qquad \cos 0 = 1,\ \sin 0 = 0,\ e^0 = 1$$

The trap answer is $f(0) = 0 - 2 + 4 = 2$.

(b)

$$y = 5\ln x - 2x^{-1} + \ln 7 \qquad \ln 7 \text{ is a constant}$$
$$\frac{dy}{dx} = \frac{5}{x} + 2x^{-2} + 0 = \frac{5}{x} + \frac{2}{x^2}.$$

(c) 3^x is an exponential, x^3 is a power and e^3 is a constant.

$$\frac{dy}{dx} = 3^x \ln 3 + 3x^2 + 0 = 3^x \ln 3 + 3x^2.$$

Solution (continued)

(d) Split the ln first.

$$f(x) = \ln 4 + \ln x - x^{\frac{1}{2}} \qquad \ln(4x) = \ln 4 + \ln x$$

$$f'(x) = 0 + \frac{1}{x} - \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{x} - \frac{1}{2\sqrt{x}} \qquad \ln 4 \text{ is a constant}$$

$$f'(1) = 1 - \frac{1}{2} = \frac{1}{2}.$$

The trap $\dfrac{d}{dx}\ln(4x) = \dfrac{1}{4x}$ gives $\dfrac{1}{4} - \dfrac{1}{2} = -\dfrac{1}{4}$.

(e) Differentiate first, then substitute.

$$f'(x) = 2\cos x - \sin x$$

$$f'\left(\frac{\pi}{3}\right) = 2 \cdot \frac{1}{2} - \frac{\sqrt{3}}{2} = 1 - \frac{\sqrt{3}}{2}.$$

Practice

P4 Find the derivative of each function.

(a) $f(x) = 4\cos x - 5\sin x + 2e^x - e^2$. Find $f'(0)$.

(b) $y = 3\ln x + \dfrac{4}{x} - \ln 2$

(c) $y = 2^x - x^2 + e$

(d) $f(x) = \ln(5x) + 4\sqrt{x}$. Find $f'(4)$.

(e) $f(x) = 3\cos x - 2\sin x$. Find $f'\left(\dfrac{\pi}{6}\right)$.

1.3 The product rule

Notes

- Product rule: differentiate one factor and leave the other alone; then differentiate the other factor and leave the first alone; add.

$$\frac{d}{dx}\big(f(x)g(x)\big) = \frac{d}{dx}\big(f(x)\big)\cdot g(x) + \frac{d}{dx}\big(g(x)\big)\cdot f(x) = f'(x)g(x) + g'(x)f(x).$$

- Three factors (or more): the same idea. Differentiate one factor at a time and leave the others alone, once for each factor, then add. There is one term for each factor.

$$\frac{d}{dx}\big(f(x)g(x)h(x)\big) = \frac{d}{dx}\big(f(x)\big)\cdot g(x)h(x) + \frac{d}{dx}\big(g(x)\big)\cdot f(x)h(x) + \frac{d}{dx}\big(h(x)\big)\cdot f(x)g(x).$$

- Write the first line of the working in this form, so that you can see which factor is being differentiated. For $y = x^5 \ln x$:

$$\frac{dy}{dx} = \frac{d}{dx}\big(x^5\big)\cdot \ln x + \frac{d}{dx}(\ln x)\cdot x^5 = 5x^4 \ln x + \frac{1}{x}\cdot x^5 = 5x^4 \ln x + x^4.$$

- Tidy the answer by factoring out a common factor; answer choices are often factored, e.g. $2xe^x + x^2e^x = xe^x(x+2)$.
- Two polynomials: expanding first is just as good, and often quicker. With e^x, b^x, $\sin x$, $\cos x$ or $\ln x$, use the product rule.
- A constant times a function is not a product rule question: $\dfrac{d}{dx}(5\sin x) = 5\cos x$.

Trap.

- The derivative of a product is not the product of the derivatives: $\dfrac{d}{dx}\big(x^2e^x\big) \neq 2x\cdot e^x$.

Example 3 The product rule

Find the derivative of each function.

(a) $y = x^3 \cos x$

(b) $f(x) = e^x\big(x^2 - 3x\big)$. Find $f'(1)$.

(c) $y = \sqrt{x}\,\ln x$

(d) $y = \big(x^2 + 1\big)(3x - 5)$

(e) $f(x) = x^2 e^x \ln x$. Find $f'(1)$.

Solution

(a)

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\big(x^3\big)\cdot\cos x + \frac{d}{dx}(\cos x)\cdot x^3 && \text{product rule}\\ &= 3x^2\cos x + (-\sin x)\,x^3\\ &= 3x^2\cos x - x^3\sin x.\end{aligned}$$

(b)

$$\begin{aligned}f'(x) &= \frac{d}{dx}\big(e^x\big)\cdot\big(x^2 - 3x\big) + \frac{d}{dx}\big(x^2 - 3x\big)\cdot e^x && \text{product rule}\\ &= e^x\big(x^2 - 3x\big) + (2x - 3)e^x\\ &= e^x\big(x^2 - 3x + 2x - 3\big) = e^x\big(x^2 - x - 3\big) && \text{factor out } e^x\\ f'(1) &= e(1 - 1 - 3) = -3e.\end{aligned}$$

The trap answer $-e$ comes from the product of the derivatives, $e^x(2x - 3)$.

(c)

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\big(x^{\frac{1}{2}}\big)\cdot\ln x + \frac{d}{dx}(\ln x)\cdot\sqrt{x} && \text{product rule}\\ &= \frac{1}{2\sqrt{x}}\cdot\ln x + \frac{1}{x}\cdot\sqrt{x}\\ &= \frac{\ln x}{2\sqrt{x}} + \frac{1}{\sqrt{x}} && \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}}\\ &= \frac{\ln x + 2}{2\sqrt{x}}.\end{aligned}$$

(d) Two polynomials: either method works.

Product rule

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\big(x^2 + 1\big)\cdot(3x - 5)\\ &\qquad + \frac{d}{dx}(3x - 5)\cdot\big(x^2 + 1\big)\\ &= 2x(3x - 5) + 3\big(x^2 + 1\big)\\ &= 9x^2 - 10x + 3.\end{aligned}$$

Expand first

$$\begin{aligned}y &= 3x^3 - 5x^2 + 3x - 5\\ \frac{dy}{dx} &= 9x^2 - 10x + 3.\end{aligned}$$

Solution (continued)

(e) Three factors: three terms. Each time, differentiate one factor and leave the other two alone.

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(x^2\right) \cdot e^x \ln x + \frac{d}{dx}\left(e^x\right) \cdot x^2 \ln x + \frac{d}{dx}(\ln x) \cdot x^2 e^x && \text{product rule, three factors} \\ &= 2xe^x \ln x + e^x \cdot x^2 \ln x + \frac{1}{x} \cdot x^2 e^x \\ &= 2xe^x \ln x + x^2 e^x \ln x + xe^x \\ f'(1) &= 0 + 0 + e = e. && \ln 1 = 0 \end{aligned}$$

Differentiate one factor at a time and leave the others alone: one term for each factor. Factor the answer to match the choices.

Practice

P5 Find the derivative of each function.

(a) $y = x^4 \sin x$

(b) $f(x) = e^x\left(x^2 + 2x\right)$. Find $f'(1)$.

(c) $y = \sqrt[3]{x}\, \ln x$

(d) $y = (2x - 1)\left(x^2 + 3\right)$

(e) $f(x) = x^3\, 2^x \ln x$. Find $f'(1)$.

1.4 The quotient rule

Notes

- Quotient rule:

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}\big(f(x)\big)\cdot g(x) - \frac{d}{dx}\big(g(x)\big)\cdot f(x)}{\big(g(x)\big)^2} = \frac{f'(x)g(x) - g'(x)f(x)}{\big(g(x)\big)^2}.$$

  In words: differentiate the top and leave the bottom alone, minus differentiate the bottom and leave the top alone, all over the bottom squared. The top comes first.

- Simplify the numerator; leave the bottom as a square, not expanded.
- Quicker without the quotient rule:
  - one term on the bottom (x^2, $\sqrt{x}$, $3x$): split and use the power rule, e.g. $\dfrac{x^2+3\sqrt{x}}{x} = x+3x^{-\frac{1}{2}}$;
  - a constant on the bottom: $\dfrac{x^2+\sin x}{5} = \dfrac{1}{5}\big(x^2+\sin x\big)$.

Traps.

- The order in the numerator matters: $g'(x)f(x) - f'(x)g(x)$ gives the negative of the answer.
- Square the bottom. And $\dfrac{f'(x)}{g'(x)}$ is not the derivative.

Example 4 The quotient rule

Find the derivative of each function.

(a) $y = \dfrac{e^x}{x^2+1}$

(b) $y = \dfrac{\sin x}{1+\cos x}$

(c) $y = \dfrac{x^2+3\sqrt{x}}{x}$

(d) $f(x) = \dfrac{3^x}{x}$. Find $f'(1)$.

(e) $f(x) = \dfrac{\ln x}{x+1}$. Find $f'(1)$.

Solution

(a)

$$\frac{dy}{dx} = \frac{\frac{d}{dx}\big(e^x\big)\cdot\big(x^2+1\big) - \frac{d}{dx}\big(x^2+1\big)\cdot e^x}{\big(x^2+1\big)^2} \qquad \text{quotient rule}$$

$$= \frac{e^x\big(x^2+1\big) - 2x\,e^x}{\big(x^2+1\big)^2}$$

$$= \frac{e^x\big(x^2-2x+1\big)}{\big(x^2+1\big)^2} = \frac{e^x(x-1)^2}{\big(x^2+1\big)^2}. \qquad \text{factor out } e^x$$

Solution (continued)

(b)

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{d}{dx}(\sin x)\cdot(1+\cos x) - \frac{d}{dx}(1+\cos x)\cdot\sin x}{(1+\cos x)^2} && \text{quotient rule}\\ &= \frac{\cos x(1+\cos x) - (-\sin x)\sin x}{(1+\cos x)^2}\\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \frac{\cos x + 1}{(1+\cos x)^2} && \sin^2 x + \cos^2 x = 1\\ &= \frac{1}{1+\cos x}.\end{aligned}$$

(c) One term on the bottom: split instead.

$$\begin{aligned}y &= \frac{x^2}{x} + \frac{3x^{\frac{1}{2}}}{x} = x + 3x^{-\frac{1}{2}} && \text{split}\\ \frac{dy}{dx} &= 1 - \frac{3}{2}x^{-\frac{3}{2}}.\end{aligned}$$

(d) The bottom is one term, but 3^x is not a power of x, so splitting does not help. Use the quotient rule.

$$\begin{aligned}f'(x) &= \frac{\frac{d}{dx}\left(3^x\right)\cdot x - \frac{d}{dx}(x)\cdot 3^x}{x^2} && \text{quotient rule}\\ &= \frac{3^x\ln 3\cdot x - 3^x}{x^2} = \frac{3^x(x\ln 3 - 1)}{x^2}\\ f'(1) &= \frac{3(\ln 3 - 1)}{1} = 3\ln 3 - 3.\end{aligned}$$

(e)

$$\begin{aligned}f'(x) &= \frac{\frac{d}{dx}(\ln x)\cdot(x+1) - \frac{d}{dx}(x+1)\cdot\ln x}{(x+1)^2} && \text{quotient rule}\\ &= \frac{\frac{1}{x}(x+1) - \ln x}{(x+1)^2}\\ f'(1) &= \frac{1\cdot 2 - 0}{2^2} = \frac{1}{2}. && \ln 1 = 0\end{aligned}$$

Bottom-squared denominator, top-first numerator. Before using the rule, check whether splitting is quicker.

Practice

P6 Find the derivative of each function.

(a) $y = \dfrac{x}{e^x}$

(b) $y = \dfrac{\cos x}{1 - \sin x}$

(c) $y = \dfrac{4x^3 - \sqrt{x}}{x^2}$

(d) $f(x) = \dfrac{2^x}{x}$. Find $f'(2)$.

(e) $f(x) = \dfrac{\ln x}{2x - 1}$. Find $f'(1)$.

P7 [AB 2017 Alt] If $f(x) = \dfrac{5-x}{x^3+2}$, then $f'(x) =$

(A) $\dfrac{-4x^3+15x^2-2}{\left(x^3+2\right)^2}$

(B) $\dfrac{-2x^3+15x^2+2}{\left(x^3+2\right)^2}$

(C) $\dfrac{2x^3-15x^2-2}{\left(x^3+2\right)^2}$

(D) $\dfrac{4x^3-15x^2+2}{\left(x^3+2\right)^2}$

P8 [BC 2008 Form B] If $f(x) = \dfrac{3x-2}{2x+3}$, then $f'(x) =$

(A) $-\dfrac{13}{(2x+3)^2}$

(B) $\dfrac{5}{(2x+3)^2}$

(C) $\dfrac{13}{(2x+3)^2}$

(D) $\dfrac{12x+5}{(2x+3)^2}$

P9 [BC 2021] If $f(x) = x^2 + 4x - 2$, $g(x) = 4x + 2$ and $h(x) = \dfrac{f(x)}{g(x)}$, then $h'(1) =$

(A) $-\dfrac{2}{3}$ (B) $\dfrac{2}{3}$ (C) $\dfrac{4}{3}$ (D) $\dfrac{3}{2}$

1.5 Derivatives of $\tan x$, $\cot x$, $\sec x$ and $\csc x$

Notes

- The derivatives:
$$\frac{d}{dx}(\tan x) = \sec^2 x, \qquad \frac{d}{dx}(\sec x) = \sec x \tan x,$$
$$\frac{d}{dx}(\cot x) = -\csc^2 x, \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x.$$
The "co" functions (cot, csc) get a minus sign, like cos.
- Where they come from: write the function with sin and cos, then use the quotient rule (Example 5).
- Simplify first when you can: $\cos x \tan x = \sin x$, $\sin x \sec x = \tan x$, $\sin x \cot x = \cos x$.
- Values: $\sec x = \dfrac{1}{\cos x}$, $\csc x = \dfrac{1}{\sin x}$, $\cot x = \dfrac{1}{\tan x}$; e.g. $\sec \dfrac{\pi}{4} = \sqrt{2}$, $\csc \dfrac{\pi}{6} = 2$.

Example 5 The other trig functions

(a) Use the quotient rule to show that $\frac{d}{dx}(\tan x) = \sec^2 x$ and $\frac{d}{dx}(\sec x) = \sec x \tan x$.

(b) Find $\frac{dy}{dx}$ for $y = 3\tan x - 2\sec x + \csc x + 4\cos x - 6\sqrt{x} + 5\ln x$.

(c) Find $\frac{dy}{dx}$ for $y = e^x \tan x$.

(d) Let $f(x) = \sin x \sec x$. Find $f'\left(\frac{\pi}{3}\right)$.

(e) Find $\frac{dy}{dx}$ for $y = 2^x \cot x$.

(f) Let $f(x) = \sec x + \tan x$. Find $f'\left(\frac{\pi}{4}\right)$.

Solution

(a) Write each with sin and cos.

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\frac{d}{dx}(\sin x)\cdot\cos x - \frac{d}{dx}(\cos x)\cdot\sin x}{\cos^2 x} \qquad \text{quotient rule}$$

$$= \frac{\cos x\cos x - (-\sin x)\sin x}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} = \sec^2 x. \qquad \sin^2 x + \cos^2 x = 1$$

$$\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{\frac{d}{dx}(1)\cdot\cos x - \frac{d}{dx}(\cos x)\cdot 1}{\cos^2 x} \qquad \text{quotient rule}$$

$$= \frac{0\cdot\cos x - (-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} \qquad \text{the top 1 has derivative 0}$$

$$= \frac{1}{\cos x}\cdot\frac{\sin x}{\cos x} = \sec x\tan x.$$

(b) One rule for each term.

$$\frac{dy}{dx} = 3\sec^2 x - 2\sec x\tan x - \csc x\cot x - 4\sin x - \frac{3}{\sqrt{x}} + \frac{5}{x}.$$

(c)

$$\frac{dy}{dx} = \frac{d}{dx}\left(e^x\right)\cdot\tan x + \frac{d}{dx}(\tan x)\cdot e^x \qquad \text{product rule}$$

$$= e^x\tan x + e^x\sec^2 x = e^x\left(\tan x + \sec^2 x\right).$$

(d) Simplify first.

$$f(x) = \sin x\cdot\frac{1}{\cos x} = \tan x \qquad \sec x = \frac{1}{\cos x}$$

$$f'(x) = \sec^2 x$$

$$f'\left(\frac{\pi}{3}\right) = 2^2 = 4. \qquad \sec\frac{\pi}{3} = \frac{1}{\cos\frac{\pi}{3}} = 2$$

(The product rule gives $\cos x\cdot\sec x + \sec x\tan x\cdot\sin x = 1 + \tan^2 x$, which is the same, $\sec^2 x$, but takes longer.)

Solution (continued)

(e)

$$\frac{dy}{dx} = \frac{d}{dx}\left(2^x\right) \cdot \cot x + \frac{d}{dx}(\cot x) \cdot 2^x \qquad \text{product rule}$$
$$= 2^x \ln 2 \cot x - 2^x \csc^2 x = 2^x\left(\ln 2 \cot x - \csc^2 x\right).$$

(f)

$$f'(x) = \sec x \tan x + \sec^2 x$$
$$f'\left(\frac{\pi}{4}\right) = \sqrt{2} \cdot 1 + \left(\sqrt{2}\right)^2 = 2 + \sqrt{2}. \qquad \sec\frac{\pi}{4} = \sqrt{2},\ \tan\frac{\pi}{4} = 1$$

Practice

P10 (a) Use the quotient rule to show that $\frac{d}{dx}(\cot x) = -\csc^2 x$ and $\frac{d}{dx}(\csc x) = -\csc x \cot x$.

(b) Find $\frac{dy}{dx}$ for $y = 4\cot x + 3\csc x - \sec x + 2\sin x + 8\sqrt{x} - 3\ln x$.

(c) Find $\frac{dy}{dx}$ for $y = e^x \sec x$.

(d) Let $f(x) = \cot x \sin x$. Find $f'\left(\frac{\pi}{6}\right)$.

(e) Find $\frac{dy}{dx}$ for $y = 5^x \tan x$.

(f) Let $f(x) = \csc x - \cot x$. Find $f'\left(\frac{\pi}{6}\right)$.

P11 If $y = \tan x - \cot x$, then $\dfrac{dy}{dx} =$

(A) $\sec x \csc x$ (B) $\sec x + \csc x$ (C) $\sec^2 x - \csc^2 x$ (D) $\sec^2 x + \csc^2 x$

P12 Which of the following correctly shows the derivation of $\dfrac{d}{dx}(\cot x)$?

(A) $\dfrac{d}{dx}(\cot x) = \dfrac{d}{dx}\left(\dfrac{1}{\tan x}\right) = \dfrac{1}{\frac{d}{dx}(\tan x)} = \dfrac{-1}{\sec^2 x}$

(B) $\dfrac{d}{dx}(\cot x) = \dfrac{d}{dx}\left(\dfrac{1}{\tan x}\right) = \dfrac{1}{\frac{d}{dx}(\tan x)} = \dfrac{1}{\sec^2 x}$

(C) $\dfrac{d}{dx}(\cot x) = \dfrac{d}{dx}\left(\dfrac{1}{\tan x}\right) = \dfrac{\frac{d}{dx}(1)\cdot \tan x - 1 \cdot \frac{d}{dx}(\tan x)}{\tan^2 x} = \dfrac{0 \cdot \tan x - \sec^2 x}{\tan^2 x} = -\dfrac{\sec^2 x}{\tan^2 x}$

(D) $\dfrac{d}{dx}(\cot x) = \dfrac{d}{dx}\left(\dfrac{1}{\tan x}\right) = \dfrac{\frac{d}{dx}(1)\cdot \tan x + 1 \cdot \frac{d}{dx}(\tan x)}{\tan^2 x} = \dfrac{0 \cdot \tan x + \sec^2 x}{\tan^2 x} = \dfrac{\sec^2 x}{\tan^2 x}$

P13 $\dfrac{d}{dx}(\cos x \tan x) =$

(A) $\sec x + \sin x \tan x$ (B) $\cos x$ (C) $-\sin x \sec^2 x$ (D) $\sin x$

1.6 Rules with values from a table

Notes

- When f and g are known only through a table or a few values, write the rule with letters first, then substitute $x = a$:

$$(fg)'(a) = f'(a)g(a) + g'(a)f(a), \qquad \left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - g'(a)f(a)}{\big(g(a)\big)^2}.$$

- Terms with x are differentiated as usual, e.g.

$$\frac{d}{dx}\big(x^3 f(x)\big) = \frac{d}{dx}\big(x^3\big) \cdot f(x) + \frac{d}{dx}\big(f(x)\big) \cdot x^3 = 3x^2 f(x) + f'(x)\, x^3$$

  and $\dfrac{d}{dx}\big(x - g(x)\big) = 1 - g'(x)$.

- A product or quotient needs both the values ($f(a)$, $g(a)$) and the derivatives ($f'(a)$, $g'(a)$). A sum needs only the derivatives.
- Read which row to use: a table often has several values of x, and each part may use a different row.

Trap.

- Read the table header: the columns are not always in the order f, f', g, g'.

Example 6 Rules with values from a table

The table gives values of the differentiable functions f and g and their derivatives at selected values of x.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	3	2	-1	4
1	2	-3	4	5

(a) $h(x) = 3f(x) - e^x g(x)\cos x + 2^x$. Find $h'(0)$.

(b) $h(x) = \dfrac{f(x)g(x)}{\sqrt{x} + \ln x}$. Find $h'(1)$.

(c) $h(x) = \dfrac{g(x) + \sin x}{\tan x + f(x)}$. Find $h'(0)$.

Solution

(a) A sum; the middle term is a product of three factors. Use the row $x = 0$.

$$\begin{aligned} h'(x) &= 3f'(x) - \Big[\frac{d}{dx}\big(e^x\big)\cdot g(x)\cos x + \frac{d}{dx}\big(g(x)\big)\cdot e^x\cos x \\ &\qquad + \frac{d}{dx}(\cos x)\cdot e^x g(x)\Big] + 2^x\ln 2 \\ &= 3f'(x) - \Big[e^x g(x)\cos x + g'(x)e^x\cos x - \sin x\, e^x g(x)\Big] + 2^x\ln 2 \\ h'(0) &= 3(2) - \big[(1)(-1)(1) + (4)(1)(1) - 0\big] + \ln 2 \qquad e^0 = 1,\ \cos 0 = 1,\ \sin 0 = 0 \\ &= 6 - 3 + \ln 2 = 3 + \ln 2. \end{aligned}$$

(b) A quotient whose top is a product. Use the row $x = 1$.

- Top: $f(x)g(x)$. At $x = 1$: $(2)(4) = 8$. Its derivative is $f'(x)g(x) + g'(x)f(x)$; at $x = 1$: $(-3)(4) + (5)(2) = -2$.
- Bottom: $\sqrt{x} + \ln x$. At $x = 1$: $1 + 0 = 1$. Its derivative is $\dfrac{1}{2\sqrt{x}} + \dfrac{1}{x}$; at $x = 1$: $\dfrac{1}{2} + 1 = \dfrac{3}{2}$.

$$\begin{aligned} h'(1) &= \frac{\frac{d}{dx}(\text{top})\cdot\text{bottom} - \frac{d}{dx}(\text{bottom})\cdot\text{top}}{(\text{bottom})^2} \qquad \text{quotient rule} \\ &= \frac{(-2)(1) - \frac{3}{2}(8)}{1^2} = -2 - 12 = -14. \end{aligned}$$

(c) A quotient. Use the row $x = 0$.

- Top: $g(x) + \sin x$. At $x = 0$: $-1 + 0 = -1$. Its derivative is $g'(x) + \cos x$; at $x = 0$: $4 + 1 = 5$.
- Bottom: $\tan x + f(x)$. At $x = 0$: $0 + 3 = 3$. Its derivative is $\sec^2 x + f'(x)$; at $x = 0$: $1 + 2 = 3$.

$$h'(0) = \frac{(5)(3) - (3)(-1)}{3^2} = \frac{15 + 3}{9} = 2. \qquad \text{quotient rule}$$

Find the value and the derivative of each piece first (top, bottom, each factor); then put them into the rule.

Practice

P14 The table gives values of the differentiable functions f and g and their derivatives at selected values of x.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	-2	3	1	-4
1	-1	4	3	-2

(a) $h(x) = 4g(x) - e^x f(x)\cos x - 3^x$. Find $h'(0)$.

(b) $h(x) = \dfrac{f(x)g(x)}{2\sqrt{x} + \ln x}$. Find $h'(1)$.

(c) $h(x) = \dfrac{f(x) - \sin x}{\tan x + g(x)}$. Find $h'(0)$.

Exercise 1

1.1 [AB 2013 Intl] If $f(x) = x^3 - x^2 + x - 1$, then $f'(2) =$

(A) 5 (B) 7 (C) 9 (D) 10

1.2 If $f(x) = x^3 + 2x$, then $\dfrac{d}{dx}\big(f(2)\big) =$

(A) 0 (B) 12 (C) 14 (D) 16

1.3 If $f(x) = e^x \sin x$, then $f'(x) =$

(A) $e^x \cos x$

(B) $-e^x \cos x$

(C) $e^x(\sin x + \cos x)$

(D) $e^x(\sin x - \cos x)$

1.4 If $f(x) = \dfrac{\cos x}{\ln x}$, then $f'(x) =$

(A) $\dfrac{x \sin x \ln x - \cos x}{x(\ln x)^2}$

(B) $\dfrac{-x \sin x \ln x - \cos x}{x(\ln x)^2}$

(C) $-x \sin x$

(D) $\dfrac{-x \sin x \ln x + \cos x}{x(\ln x)^2}$

1.5 $\dfrac{d}{dx}(\cot x) =$

(A) $-\tan x$ (B) $-\csc^2 x$ (C) $\csc x$ (D) $\sec^2 x$

1.6

(a) Find $\dfrac{dy}{dx}$ for $y = 4\sqrt{x} - \dfrac{3}{x^2} + 2\sin x - \tan x + 5e^x - \ln x + 2^x + e^2 + \ln 3$.

(b) Find $\dfrac{dy}{dx}$ for $y = x^3 e^x \cos x$.

(c) Let $f(x) = \dfrac{e^x + \sin x}{x^2 + \cos x}$. Find $f'(0)$.

(d) Let $f(x) = x \ln x - 3^x$. Find $f'(1)$ and $\dfrac{d}{dx}\big(f(1)\big)$.

1.7 [BC 2016 Practice FRQ] The table gives values of a function f and its derivative f' for selected values of x. Let g be the function given by $g(x) = (2x - 1)f(x)$. Find $g'(3)$.

x	1	2	3	4	5
$f(x)$	4	8	7	3	-1
$f'(x)$	5	-2	-1	6	2

1.8 [BC 2024] The table gives values of the differentiable functions f, g and h and their derivatives at $x = 0$. If $M(x) = \dfrac{f(x) + g(x)}{h(x)}$, what is the value of $M'(0)$?

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$	$h(x)$	$h'(x)$
0	-1	-2	3	5	2	7

(A) -2 (B) $-\dfrac{8}{49}$ (C) $\dfrac{3}{7}$ (D) 5

1.9 The table gives values of the differentiable functions f and g and their derivatives at selected values of x.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	2	-1	3	1
1	-1	2	4	-3

(a) $h(x) = \dfrac{f(x)g(x)}{e^x + x} + g(1)$. Find $h'(0)$.

(b) $k(x) = 3^x f(x) - \sqrt{x}\, g(x)$. Find $k'(1)$.

(c) Find $\dfrac{d}{dx}\big(f(0)g(0)\big)$.

1.10 Let f be a differentiable function with $f(8) = 2$ and $f'(8) = 5$. If $g(x) = \dfrac{f(x)}{\sqrt[3]{x}}$, what is the value of $g'(8)$?

(A) $\dfrac{59}{24}$ (B) $\dfrac{5}{2}$ (C) $\dfrac{61}{24}$ (D) 60

1.11 Find $\dfrac{dy}{dx}$. Use the quickest method each time.

(a) $y = \big(x^2 + 1\big)\big(x^2 - 1\big)$

(b) $y = \dfrac{x^3 - 4\sqrt{x}}{x}$

(c) $y = \dfrac{x}{x^2 + 1}$

(d) $y = \sin^2 x$

(e) $y = \dfrac{5}{2x^3}$

2 What the Derivative Means

2.1 The algebraic definition of the derivative

Notes

The definition of the derivative (first principles):

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Example 7 The definition of the derivative

(a) Let $f(x) = x^3 - 4x$. Write $f'(x)$ as a limit. Then write $f'(2)$ as a limit in two ways, with $h \to 0$ and with $x \to 2$. Do not evaluate the limits.

(b) Using first principles, find $f'(4)$ for $f(x) = \sqrt{x}$.

(c) Use the definition to find $f'(x)$ for $f(x) = \dfrac{1}{x}$. Then find $f'(2)$.

Solution

(a) Replace every x by $x + h$. For $f'(2)$, note that $f(2) = 8 - 8 = 0$.

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^3 - 4(x+h) - \big(x^3 - 4x\big)}{h},$$

$$f'(2) = \lim_{h \to 0} \frac{(2+h)^3 - 4(2+h) - 0}{h} = \lim_{x \to 2} \frac{x^3 - 4x - 0}{x - 2}.$$

(Check with the rules: $f'(x) = 3x^2 - 4$, so all three limits give $f'(2) = 8$.)

(b) Use $f'(a) = \lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a}$ with $a = 4$.

$$f'(4) = \lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4} \qquad f(4) = \sqrt{4} = 2$$

$$= \lim_{x \to 4} \frac{\sqrt{x} - 2}{\big(\sqrt{x} - 2\big)\big(\sqrt{x} + 2\big)} \qquad x - 4 = \big(\sqrt{x}\big)^2 - 2^2$$

$$= \lim_{x \to 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{4}.$$

(Check: $f'(x) = \dfrac{1}{2\sqrt{x}}$, $f'(4) = \dfrac{1}{4}$.)

Solution (continued)

(c) Combine the fractions on the top.

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\dfrac{1}{x+h} - \dfrac{1}{x}}{h} = \lim_{h\to 0} \frac{\dfrac{x-(x+h)}{x(x+h)}}{h} && \text{common denominator} \\ &= \lim_{h\to 0} \frac{-h}{h\,x(x+h)} = \lim_{h\to 0} \frac{-1}{x(x+h)} && \text{cancel } h \\ &= -\frac{1}{x^2}, && h = 0 \\ f'(2) &= -\frac{1}{4}. \end{aligned}$$

(Check: $\dfrac{d}{dx}\left(x^{-1}\right) = -x^{-2}$.)

Practice

P15

(a) Let $f(x) = 2x^3 + x$. Write $f'(x)$ as a limit. Then write $f'(-1)$ as a limit in two ways, with $h \to 0$ and with $x \to -1$. Do not evaluate the limits.

(b) Using first principles, find $f'(9)$ for $f(x) = \sqrt{x}$.

(c) Use the definition to find $f'(x)$ for $f(x) = \dfrac{1}{x+2}$. Then find $f'(2)$.

P16 [BC 2017 Alt] If $f(x) = 3x^2 + 2x$, then $f'(x) =$

(A) $\lim\limits_{h\to 0} \dfrac{\big(3x^2 + 2x + h\big) - \big(3x^2 + 2x\big)}{h}$

(B) $\lim\limits_{x\to 0} \dfrac{\big(3x^2 + 2x + h\big) - \big(3x^2 + 2x\big)}{h}$

(C) $\lim\limits_{h\to 0} \dfrac{\big(3(x+h)^2 + 2(x+h)\big) - \big(3x^2 + 2x\big)}{h}$

(D) $\lim\limits_{x\to 0} \dfrac{\big(3(x+h)^2 + 2(x+h)\big) - \big(3x^2 + 2x\big)}{h}$

2.2 Limits that are derivatives

Notes

- A limit in the form of the definition is a derivative: find f and a, then evaluate the derivative with the rules.
- The step does not have to be h. Anything that goes to 0 works, as long as the top and the bottom use the same step:

$$f'(a) = \lim_{h\to 0} \frac{f(a+2h) - f(a)}{2h} = \lim_{h\to 0} \frac{f\big(a+h^2\big) - f(a)}{h^2} = \lim_{h\to 0} \frac{f(a-3h) - f(a)}{-3h}.$$

Example 8 Limits that are derivatives

(a) Evaluate $\lim\limits_{x\to\frac{\pi}{3}} \dfrac{\sin x - \sin\dfrac{\pi}{3}}{x-\dfrac{\pi}{3}}$

(b) Evaluate $\lim\limits_{x\to e} \dfrac{\left(x^{20}-3x\right)-\left(e^{20}-3e\right)}{x-e}$

(c) Evaluate $\lim\limits_{h\to 0} \dfrac{\tan\left(\dfrac{\pi}{4}+h\right)-1}{h}$

(d) Evaluate $\lim\limits_{h\to 0} \dfrac{e^{-1-h}-e^{-1}}{h}$

(e) Evaluate $\lim\limits_{h\to 0} \dfrac{\cos\left(\dfrac{\pi}{2}+h^2\right)-\cos\left(\dfrac{\pi}{2}\right)}{h}$

(f) The differentiable function f has the values in the table. Find $\lim\limits_{x\to 3}\dfrac{8-2f(x)}{3-x}$.

x	1	3
$f(x)$	6	4
$f'(x)$	5	-2

(g) For values of h very close to 0, which function best approximates $f(x)=\dfrac{\tan(x+h)-\tan x}{h}$?

When $x=\dfrac{\pi}{4}$, what is the limit of $\dfrac{\tan(x+h)-\tan x}{h}$ as $h\to 0$?

Solution

(a) $f(x)=\sin x$ and $a=\dfrac{\pi}{3}$:

$$\lim_{x\to\frac{\pi}{3}} \frac{\sin x-\sin\frac{\pi}{3}}{x-\frac{\pi}{3}} = f'\left(\frac{\pi}{3}\right) = \cos\frac{\pi}{3} = \frac{1}{2}.$$

The trap answer $\dfrac{\sqrt{3}}{2}$ is $f\left(\frac{\pi}{3}\right)$.

(b) $f(x)=x^{20}-3x$ and $a=e$; the subtracted bracket is $f(e)$.

$$\lim_{x\to e}\frac{f(x)-f(e)}{x-e} = f'(e) = 20e^{19}-3. \qquad f'(x)=20x^{19}-3$$

The trap answer $e^{20}-3e$ is $f(e)$.

(c) The 1 is a hidden $f(a)$: $1=\tan\dfrac{\pi}{4}$. So $f(x)=\tan x$ and $a=\dfrac{\pi}{4}$:

$$\lim_{h\to 0}\frac{\tan\left(\frac{\pi}{4}+h\right)-\tan\frac{\pi}{4}}{h} = f'\left(\frac{\pi}{4}\right) = \sec^2\frac{\pi}{4} = \left(\sqrt{2}\right)^2 = 2.$$

(d) The step is $-h$, so the bottom must be $-h$ too. Divide top and bottom by -1:

$$\lim_{h\to 0}\frac{e^{-1-h}-e^{-1}}{h} = -\lim_{h\to 0}\frac{e^{-1+(-h)}-e^{-1}}{-h} \qquad f(x)=e^x,\ a=-1,\ \text{step } -h$$

$$= -f'(-1) = -e^{-1} = -\frac{1}{e}.$$

Solution (continued)

The trap answer $\dfrac{1}{e}$ forgets the minus sign.

(e) The step on the top is h^2, but the bottom is only h. Multiply the top and the bottom by h to make the bottom h^2:

$$\lim_{h\to 0}\frac{\cos\left(\frac{\pi}{2}+h^2\right)-\cos\frac{\pi}{2}}{h}=\lim_{h\to 0}h\cdot\frac{\cos\left(\frac{\pi}{2}+h^2\right)-\cos\frac{\pi}{2}}{h^2}\qquad f(x)=\cos x,\ a=\frac{\pi}{2},\ \text{step } h^2$$

$$=0\cdot f'\left(\frac{\pi}{2}\right)=0\cdot\left(-\sin\frac{\pi}{2}\right)=0.$$

(f) $f(3)=4$, so $8-2f(x)=-2\big(f(x)-f(3)\big)$ and $3-x=-(x-3)$.

$$\lim_{x\to 3}\frac{8-2f(x)}{3-x}=\lim_{x\to 3}\frac{-2\big(f(x)-f(3)\big)}{-(x-3)}=2\lim_{x\to 3}\frac{f(x)-f(3)}{x-3}$$

$$=2f'(3)=2(-2)=-4.$$

(The values at $x=1$ are not needed.)

(g) As $h\to 0$ the quotient tends to $\dfrac{d}{dx}(\tan x)=\sec^2 x$, so for h very close to 0 it is approximately $\sec^2 x$. At $x=\dfrac{\pi}{4}$ the limit is

$$\sec^2\frac{\pi}{4}=\left(\sqrt{2}\right)^2=2.$$

Practice

P17

(a) Evaluate $\lim\limits_{x\to\frac{\pi}{5}} \dfrac{\cos x - \cos\dfrac{\pi}{5}}{x - \dfrac{\pi}{5}}$

(b) If $f(x) = \ln x$, find $\lim\limits_{x\to 3} \dfrac{f(x) - f(3)}{x - 3}$.

(c) Evaluate $\lim\limits_{h\to 0} \dfrac{\sqrt{4+h} - 2}{h}$

(d) Evaluate $\lim\limits_{h\to 0} \dfrac{\ln(3-h) - \ln 3}{h}$

(e) Evaluate $\lim\limits_{h\to 0} \dfrac{\sin\left(\pi + h^2\right) - \sin\pi}{h}$

(f) For values of h very close to 0, which function best approximates $\dfrac{\ln(x+h) - \ln x}{h}$? When $x = 2e$, what is the limit of $\dfrac{\ln(x+h) - \ln x}{h}$ as $h \to 0$?

P18 [BC 2024 Alt FRQ] The function f is differentiable. Values of f and f' at selected values of x are given in the table. Evaluate $\lim_{x\to 4}\left(\frac{6-2f(x)}{4-x}\right)$ or determine that the limit does not exist. Justify your answer.

x	0	1	2	4
$f(x)$	11	6	-1	3
$f'(x)$	-2	-7	-6	14

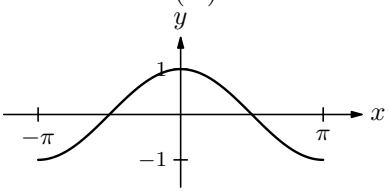
P19

(a) Evaluate $\lim_{h\to 0}\frac{(x+h)^2+4(x+h)-x^2-4x}{h}$. Which derivative is it?

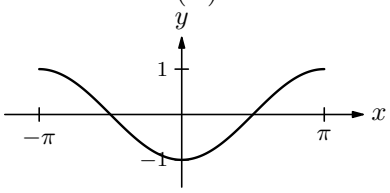
(b) Evaluate $\lim_{h\to 0}\frac{(2+h)^3+2(2+h)-(2)^3-2(2)}{h}$.

P20 [BC 2022 Intl] Which of the following could be the graph of $y = \dfrac{\cos(x + 0.001) - \cos(x)}{0.001}$?

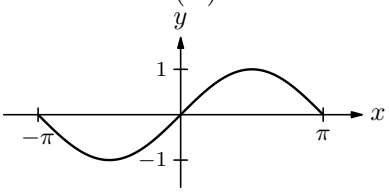
(A)



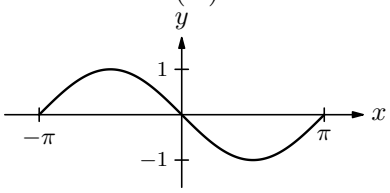
(B)



(C)

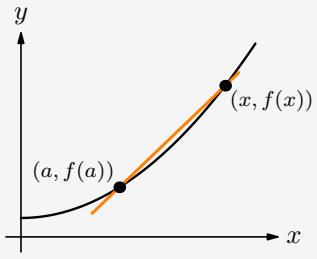


(D)



2.3 The graphical meaning of the derivative

Notes

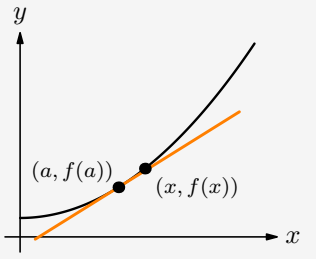


secant line

slope of the secant line

$$= \frac{f(x) - f(a)}{x - a}$$

$\xrightarrow{x \to a}$

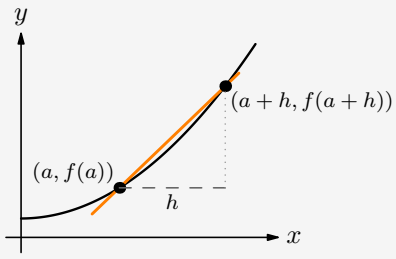


tangent line

slope of the tangent line

$$= \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

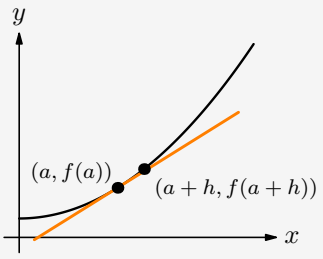
$$= f'(a)$$



secant line

slope of the secant line
$$= \frac{f(a+h)-f(a)}{h}$$

$\xrightarrow{h \to 0}$



tangent line

slope of the tangent line
$$= \lim_{h \to 0} \frac{f(a+h)-f(a)}{h}$$
$$= f'(a)$$

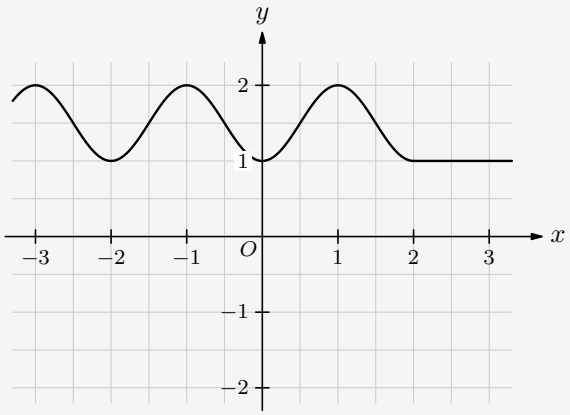
$f'(a)$ = slope of the tangent line

Example 9 Tangent lines

(a) Let $h(x) = x^3 - 2x + 1$. Write an equation of the line tangent to the graph of h at $x = 2$.

(b) $f(0) = 2$ and $f'(0) = -3$. The function g has derivative $g'(x) = e^{-2x}\big(3f(x) + 2f'(x)\big)$, and $g(0) = 4$. Write equations of the lines tangent to the graphs of f and of g at $x = 0$.

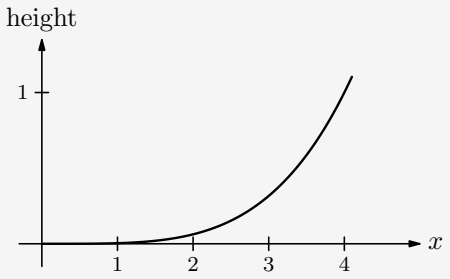
(c) $f(1) = 3$, and the graph of f', the derivative of f, is shown. Write an equation of the line tangent to the graph of f at $x = 1$.



Graph of f'

(d) [AB 2006 Form B FRQ] The graph below models the height of a skateboard ramp as a function of x. The function meets the following requirements.

1. At $x = 0$, the value of the function is 0, and the slope of the graph of the function is 0.
2. At $x = 4$, the value of the function is 1, and the slope of the graph of the function is 1.
3. Between $x = 0$ and $x = 4$, the function is increasing.



(i) Let $f(x) = ax^2$, where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement 2 above.

(ii) Let $g(x) = cx^3 - \dfrac{x^2}{16}$, where c is a nonzero constant. Find the value of c so that g meets requirement 2 above. Show the work that leads to your answer.

(iii) Let $h(x) = \dfrac{x^n}{k}$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement 2 above. Show that h also meets requirement 1 above.

Solution

(a) The point and the slope at $x = 2$:

$$h(2) = 8 - 4 + 1 = 5 \qquad \text{the point is } (2, 5)$$
$$h'(x) = 3x^2 - 2 \implies h'(2) = 12 - 2 = 10 \qquad \text{the slope}$$
$$y - 5 = 10(x - 2).$$

(b) The line tangent to f: point $(0, 2)$, slope -3, so $y - 2 = -3x$.
For g: substitute $x = 0$, $f(0) = 2$ and $f'(0) = -3$ into the formula for g'.

$$g'(0) = e^0\big(3f(0) + 2f'(0)\big) = 1 \cdot \big(3(2) + 2(-3)\big) = 0 \qquad e^0 = 1$$
$$y - 4 = 0(x - 0), \quad \text{i.e. } y = 4. \qquad \text{a horizontal tangent}$$

(c) Read the height of the graph of f' at $x = 1$: $f'(1) = 2$. The point is $\big(1, f(1)\big) = (1, 3)$.

$$y - 3 = 2(x - 1).$$

The graph of f' gives slopes, not values of f: the point $(1, 2)$ is not on the graph of f.

(d) Requirement 2 is two equations: value 1 and slope 1 at $x = 4$.

(i) $f'(x) = 2ax$.

$$f(4) = 16a = 1 \implies a = \frac{1}{16} \qquad \text{the value}$$
$$f'(4) = 8a = 1 \implies a = \frac{1}{8} \qquad \text{the slope}$$

The two equations need different values of a, so no value of a meets requirement 2.

(ii) $g'(x) = 3cx^2 - \dfrac{x}{8}$.

$$g(4) = 64c - 1 = 1 \implies c = \frac{1}{32} \qquad \text{the value}$$
$$g'(4) = 48c - \frac{1}{2} = 1 \implies c = \frac{1}{32} \qquad \text{the slope}$$

Both equations give $c = \dfrac{1}{32}$.

Solution (continued)

(iii) $h'(x) = \dfrac{nx^{n-1}}{k}$.

$$h(4) = \frac{4^n}{k} = 1 \implies k = 4^n \qquad \text{the value}$$

$$h'(4) = \frac{n \cdot 4^{n-1}}{k} = \frac{n \cdot 4^{n-1}}{4^n} = \frac{n}{4} = 1 \implies n = 4 \qquad \text{the slope}$$

So $n = 4$ and $k = 4^4 = 256$, i.e. $h(x) = \dfrac{x^4}{256}$.

Requirement 1: $h(0) = 0$ and $h'(0) = \dfrac{4 \cdot 0^3}{256} = 0$.

Every tangent line needs a point $\big(a, f(a)\big)$ and a slope $f'(a)$. "Tangent" means same point and same slope; "parallel" means same slope; "perpendicular" means slope $-\dfrac{1}{m}$.

Practice

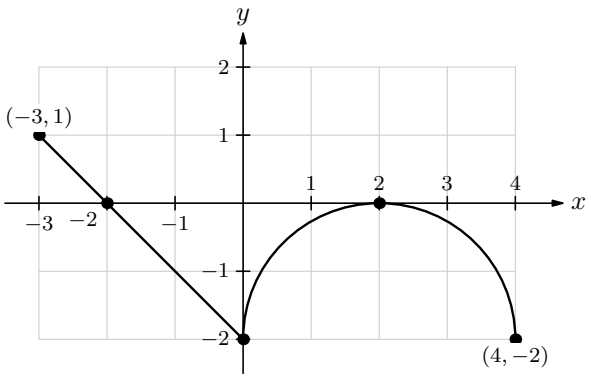
P21

(a) Let $g(x) = 2x^3 + x^2 - 5$. Write an equation of the line tangent to the graph of g at $x = -1$.

(b) $h(4) = -3$ and $h'(x) = \dfrac{x^2 - 2}{x}$ for $x \neq 0$. Write an equation of the line tangent to the graph of h at $x = 4$.

(c) $f(0) = -1$ and $f'(0) = 3$. The function g has derivative $g'(x) = \big(3f(x) + f'(x)\big)\cos x$, and $g(0) = 5$. Write equations of the lines tangent to the graphs of f and of g at $x = 0$.

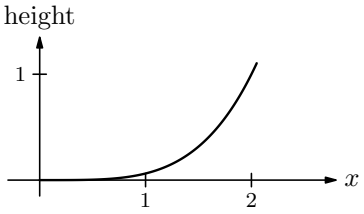
(d) The function f is defined on $-3 \leq x \leq 4$, and $f(0) = 3$. The graph of f', the derivative of f, consists of one line segment and a semicircle, as shown. Write an equation of the line tangent to the graph of f at the point $(0, 3)$.



Graph of f'

(e) The graph below models the height of a slide as a function of x. The function meets the following requirements.

1. At $x = 0$, the value of the function is 0, and the slope of the graph of the function is 0.
2. At $x = 2$, the value of the function is 1, and the slope of the graph of the function is 2.
3. Between $x = 0$ and $x = 2$, the function is increasing.



(i) Let $f(x) = ax^2$, where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement 2 above.

(ii) Let $g(x) = cx^3 - \dfrac{x^2}{4}$, where c is a nonzero constant. Find the value of c so that g meets requirement 2 above.

(iii) Let $h(x) = \dfrac{x^n}{k}$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement 2 above. Show that h also meets requirement 1 above.

(f) Let $y = f(x)$ be the solution of the differential equation $\dfrac{dy}{dx} = \dfrac{-xy^2}{2}$ with $f(-1) = 2$. Write an equation of the line tangent to the graph of f at $x = -1$.

(g) (i) The line tangent to the graph of f at $x = -2$ is $y = -3x + 1$. Find $f(-2)$ and $f'(-2)$. Must $f(0)$ be 1?

(ii) The line tangent to the graph of f at the point $(-1, 3)$ passes through the point $(3, 11)$. Find $f'(-1)$, and write an equation of this tangent line.

P22

(h) The line ℓ is tangent to the graph of $y = \dfrac{1}{x^2}$ at the point $P\left(w, \dfrac{1}{w^2}\right)$, where $w > 0$. The line ℓ crosses the x-axis at the point $R(k, 0)$. Find k in terms of w, and the value of k when $w = 3$.

(i) Let $f(x) = 4x^2 - x^3$. Show that the line $y = 18 - 3x$ is tangent to the graph of f at $x = 3$.

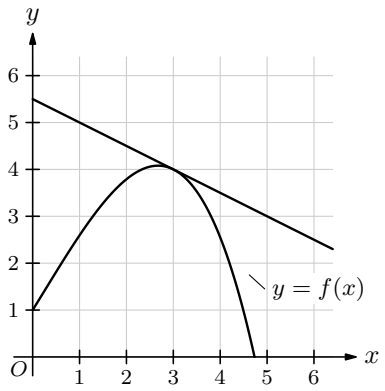
(j) (i) Find the points on the graph of $y = x^3 + 3x^2 - 5$ at which the tangent line is parallel to the line $y = 9x - 1$.
(ii) Let $f(x) = \sqrt{x}$ and $g(x) = 6 - x$. There is a point P on the graph of f at which the line tangent to the graph of f is perpendicular to the graph of g. Find the coordinates of P.

(k) (i) The line $y = 3x + k$ is tangent to the graph of $y = x^2 - 5x + 1$. Find k.
(ii) Let $g(x) = ax^2 + 3x$, where a is a constant. The line tangent to the graph of g at $x = 1$ is parallel to the line through $(0, 1)$ and $(2, -3)$. Find a.

P23 [AB 2003] If the line tangent to the graph of the function f at the point $(1, 7)$ passes through the point $(-2, -2)$, then $f'(1)$ is

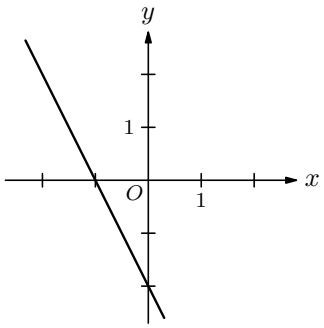
(A) 1 (B) 3 (C) 7 (D) undefined

P24 Shown is the graph of the differentiable function f along with the line tangent to the graph of f at $x = 3$. What is the value of $f'(3)$?



(A) $-\frac{1}{2}$ (B) -2 (C) 4 (D) $\frac{11}{2}$

P25 [BC 2019 Alt] Let f be a differentiable function. The figure shows the graph of the line tangent to the graph of f at $x = 0$. Of the following, which must be true?



(A) $f'(0) = -f(0)$ (B) $f'(0) < f(0)$ (C) $f'(0) = f(0)$ (D) $f'(0) > f(0)$

P26 [BC 2024 Alt] The function f satisfies the properties shown. What is the slope of the line tangent to the graph of f at $x = -1$?

$$f(0) = 0 \qquad \lim_{x \to -1} f(x) = \frac{2}{5} \qquad \lim_{x \to -1} \frac{f(x) - f(-1)}{x + 1} = -\frac{3}{5}$$

(A) $-\frac{3}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{2}{5}$ (D) 1

P27 [BC 2008] In the xy-plane, the line $x + y = k$, where k is a constant, is tangent to the graph of $y = x^2 + 3x + 1$. What is the value of k?

(A) -3 (B) -2 (C) -1 (D) 1

P28 [AB 2018 Practice] The function g is defined by $g(x) = x^2 + bx$, where b is a constant. If the line tangent to the graph of g at $x = -1$ is parallel to the line that contains the points $(0, -2)$ and $(3, 4)$, what is the value of b?

(A) -1 (B) 2 (C) $\frac{5}{2}$ (D) 4

Example 10 Derivatives from tables and graphs

(a) The temperature of water in a tub is modeled by a differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t in minutes. Selected values are given in the table.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

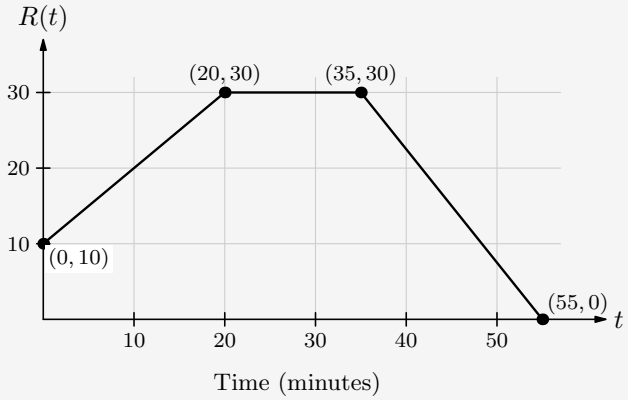
(i) Use the data in the table to estimate $W'(12)$. Indicate units of measure.

(ii) Estimate $W'(4)$.

(b) The radius of the circular cross section of a juice bottle at height h above its base is given by a differentiable function r, where h and $r(h)$ are measured in centimeters. Selected values are given in the table. Approximate $r'(3.1)$ using the average rate of change of r over $0 \le h \le 6.2$. Indicate units of measure.

h (cm)	0	6.2	13	20
$r(h)$ (cm)	2.5	2.7	4.3	2.6

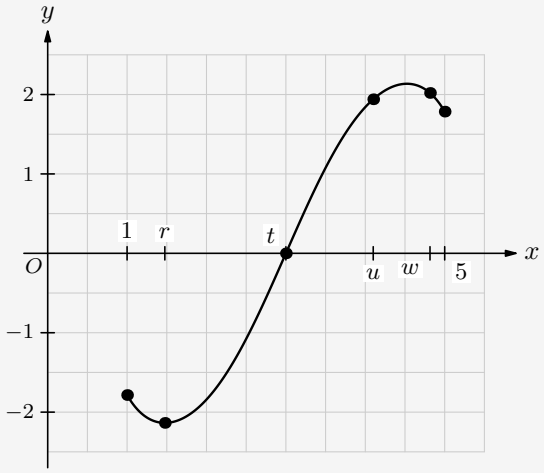
(c) The piecewise-linear graph shows the rate $R(t)$, in liters per minute, at which water is pumped into a tank.



(i) Find $R'(45)$ and $R'(10)$. Indicate units of measure.

(ii) Let $k(t) = t^2 R(t)$. Find $k'(10)$.

(d) The graph of the differentiable function f is shown for $1 \leq x \leq 5$.



Graph of f

(i) Put $f'(r)$, $f'(t)$ and $f'(w)$ in order, from least to greatest.

(ii) Which is greatest: $f(t)$, $f'(t)$, $f'(w)$ or $\dfrac{f(5) - f(1)}{5 - 1}$?

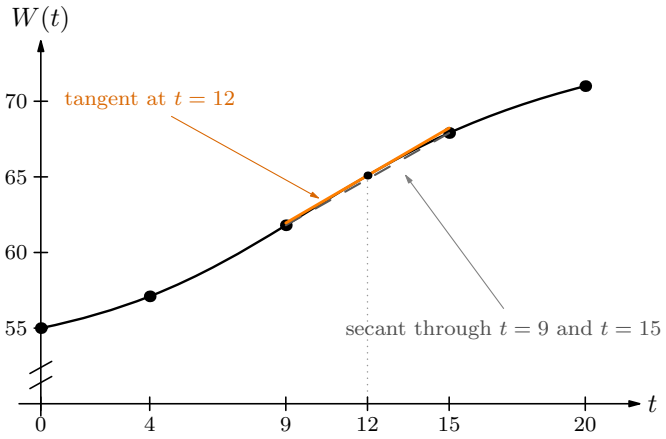
(iii) Which of r, t, u, w could be the x-coordinate of a point at which the line tangent to the graph of f is parallel to the secant line through $\big(1, f(1)\big)$ and $\big(5, f(5)\big)$?

(iv) How many such points are there on $1 < x < 5$?

Solution

(a) (i) $W'(12)$ is the slope of the tangent line at $t = 12$. The closest table values on each side of 12 are $t = 9$ and $t = 15$; the secant line through these two points has almost the same slope as the tangent line.

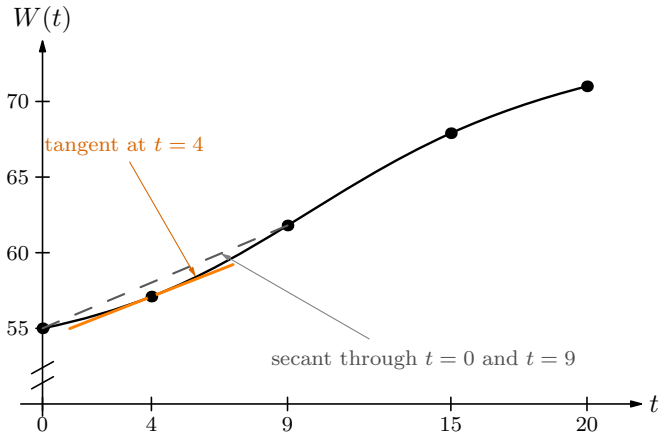
Solution (continued)



The table gives only the dots; the curve is one possible graph of W.

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} = \frac{6.1}{6} \approx 1.017 \text{ degrees Fahrenheit per minute.}$$

(ii) $t = 4$ is in the table; use its neighbours 0 and 9. The secant line through them has almost the same slope as the tangent line at $t = 4$.

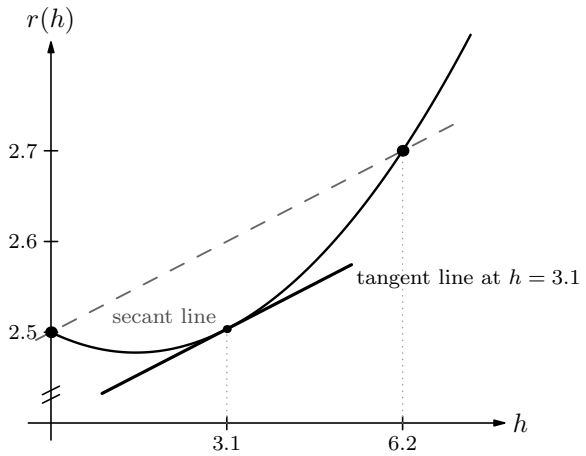


$$W'(4) \approx \frac{W(9) - W(0)}{9 - 0} = \frac{61.8 - 55.0}{9} = \frac{6.8}{9} \approx 0.756 \text{ degrees Fahrenheit per minute.}$$

(b) 3.1 is the midpoint of $[0, 6.2]$.

$$r'(3.1) \approx \frac{r(6.2) - r(0)}{6.2 - 0} = \frac{2.7 - 2.5}{6.2} = \frac{0.2}{6.2} = \frac{1}{31} \approx 0.032 \text{ centimeter per centimeter.}$$

The two table points closest to $h = 3.1$ on either side are $h = 0$ and $h = 6.2$. The secant line through them has almost the same slope as the tangent line at $h = 3.1$:



Solution (continued)

The table gives only the dots; the curve is one possible graph of r.
The two lines have almost the same slope, so the secant slope estimates $r'(3.1)$.

(c) (i) $t = 45$ is on the segment from $(35, 30)$ to $(55, 0)$; $t = 10$ is on the segment from $(0, 10)$ to $(20, 30)$.

$$R'(45) = \frac{0 - 30}{55 - 35} = -\frac{30}{20} = -1.5 \text{ liters per minute per minute,}$$
$$R'(10) = \frac{30 - 10}{20 - 0} = 1 \text{ liter per minute per minute.}$$

Use the segment that contains the point: the slope of the middle segment, 0, is not $R'(45)$.

(ii) Product rule, with values read from the graph of R: $R(10) = 20$ (halfway up the first segment) and $R'(10) = 1$ from (i).

$$k'(t) = \frac{d}{dt}\big(t^2\big) \cdot R(t) + \frac{d}{dt}\big(R(t)\big) \cdot t^2 = 2tR(t) + R'(t)\, t^2 \qquad \text{product rule}$$
$$k'(10) = 2(10)(20) + (1)(100) = 500.$$

(d) (i) At r the graph has a valley, so $f'(r) = 0$. At t the graph rises steeply, so $f'(t) > 0$. At w (after the peak) the graph falls, so $f'(w) < 0$.

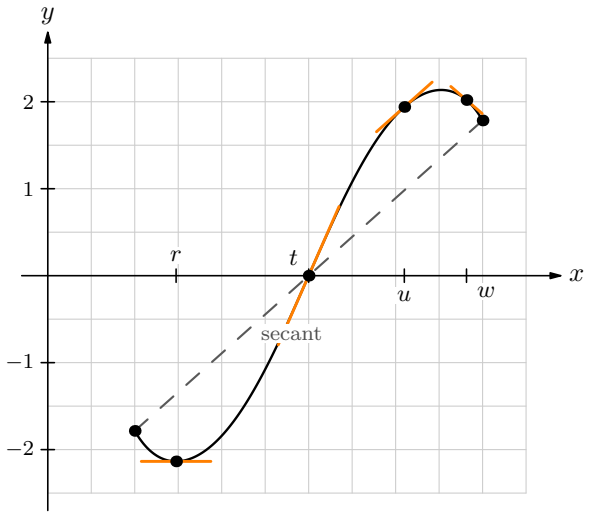
$$f'(w) < f'(r) < f'(t).$$

(ii) Read each number.

- $f(t) = 0$: the graph crosses the x-axis at t.
- $f'(w) < 0$: the graph falls at w.
- $\dfrac{f(5) - f(1)}{5 - 1} \approx \dfrac{1.8 - (-1.8)}{4} = 0.9$: the slope of the secant.
- $f'(t) \approx 2$: near t the graph climbs about 2 units from $x = 2.5$ to $x = 3.5$, much steeper than the secant.

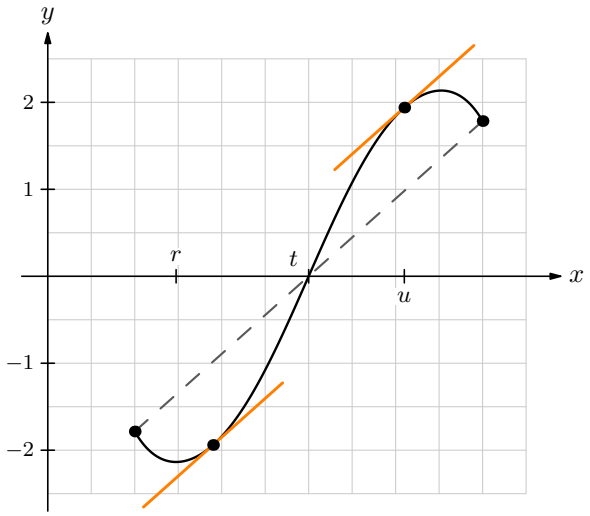
The greatest is $f'(t)$.

(iii) Draw the secant line (dashed), then the tangent line at each labelled point. Only the tangent at u runs parallel to the secant: the tangent at r is flat, the one at t is much steeper, and the one at w falls. The answer is u.



(iv) Slide a line parallel to the secant along the curve and mark where it just touches the curve as a tangent. That happens twice: at u, and at one point between r and t. There are 2 such points.

Solution (continued)



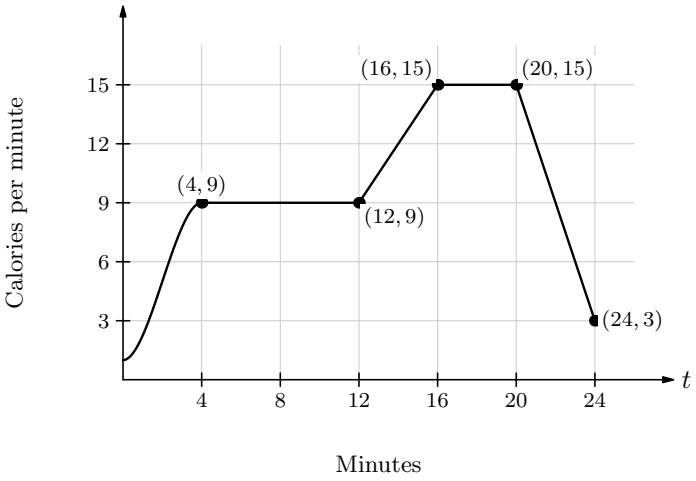
The two tangent lines (orange) are parallel to the secant (dashed).

From a table: the nearest values on each side, divided by their real gap. From a graph: the slope of the segment or tangent at the point, never the height.

Practice

P29

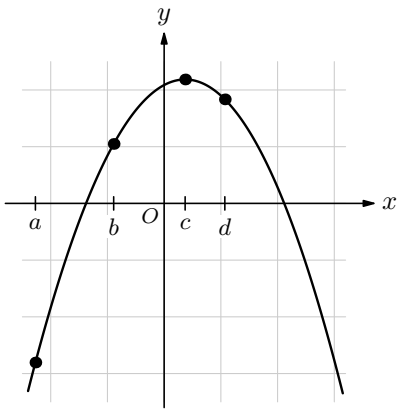
(a) The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function f. In the figure, $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$ for $0 \le t \le 4$, and f is piecewise linear for $4 \le t \le 24$.



(i) Find $f'(22)$, $f'(14)$ and $f'(2)$. Indicate units of measure.

(ii) Let $k(t) = t\,f(t)$. Find $k'(14)$.

(b) The figure shows the graph of a function f.



Graph of f

(i) Put $f'(b)$, $f'(c)$ and $f'(d)$ in order, from least to greatest.

(ii) Which of the following has the greatest value: $f(a)$, $f'(a)$, $f'(c)$, $f(d) - f(c)$ or $\dfrac{f(b) - f(a)}{b - a}$?

(iii) How many points on $a < x < d$ have a tangent line parallel to the secant line through $\big(a, f(a)\big)$ and $\big(d, f(d)\big)$? Between which two of the labelled points is each one?

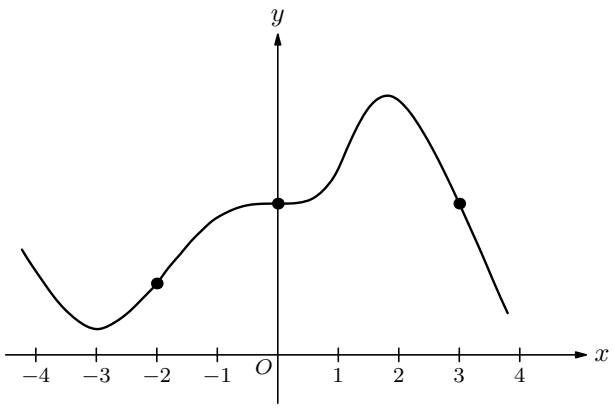
P30 [AB 2008 FRQ] Concert tickets went on sale at noon ($t = 0$). The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Estimate the rate at which the number of people waiting in line was changing at 5:30 P.M. ($t = 5.5$). Show the computations that lead to your answer. Indicate units of measure.

t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

P31 [AB 2011 FRQ] As a pot of tea cools, its temperature $H(t)$, in degrees Celsius, is modeled by a differentiable function H for $0 \leq t \leq 10$, where t is measured in minutes. Approximate the rate at which the temperature of the tea is changing at time $t = 3.5$. Show the computations that lead to your answer. Indicate units of measure.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43

P32 [AB 2017 Alt] The graph of a differentiable function f is shown. Which of the following is true?



Graph of f

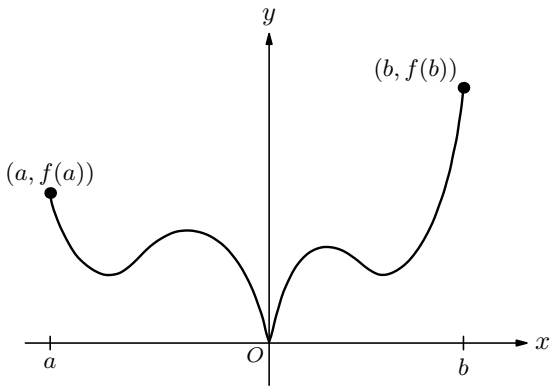
(A) $f'(-2) < f'(0) < f'(3)$

(B) $f'(-2) < f'(3) < f'(0)$

(C) $f'(3) < f'(-2) < f'(0)$

(D) $f'(3) < f'(0) < f'(-2)$

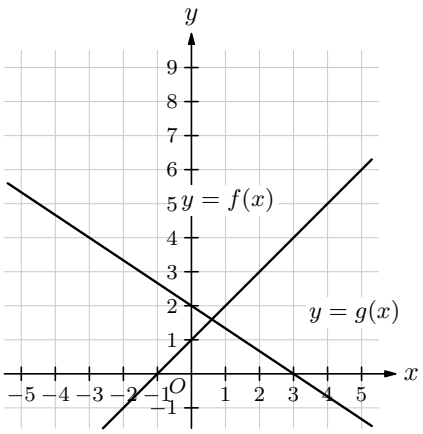
P33 The graph of a function f with $f(b) > f(a)$ is shown for $a \leq x \leq b$. The derivative of f exists for all x in the interval $a < x < b$ except $x = 0$. For how many values of c, for $a < c < b$, does $\lim_{x \to c} \dfrac{f(x) - f(c)}{x - c} = \dfrac{f(b) - f(a)}{b - a}$?



Graph of f

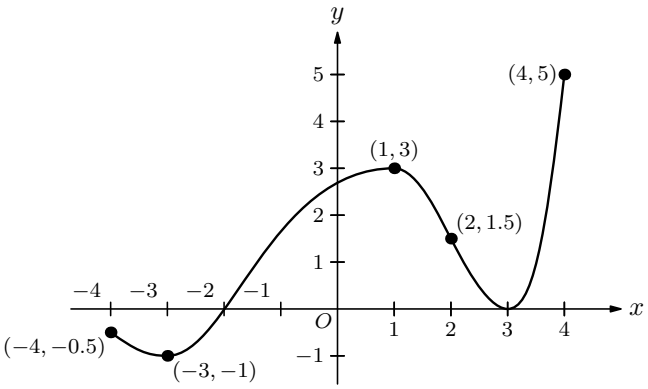
(A) Zero (B) Two (C) Three (D) Four

P34 (a) The graphs of the linear functions f and g are shown. If $h(x) = f(x) + g(x)$, then $h'(x) =$



(A) $\frac{5}{3}$ (B) 0 (C) $\frac{1}{3}$ (D) $\frac{1}{3}x + 3$

(b) [AB 2026 FRQ] Let f be a twice-differentiable function on the closed interval $[-4, 4]$ with $f(2) = 3$. The graph of f', the derivative of f, is shown. For $x > 0$, the function g is defined by $g(x) = f(x) - \ln x$. Find $g'(2)$. Show the work that leads to your answer.



Graph of f'

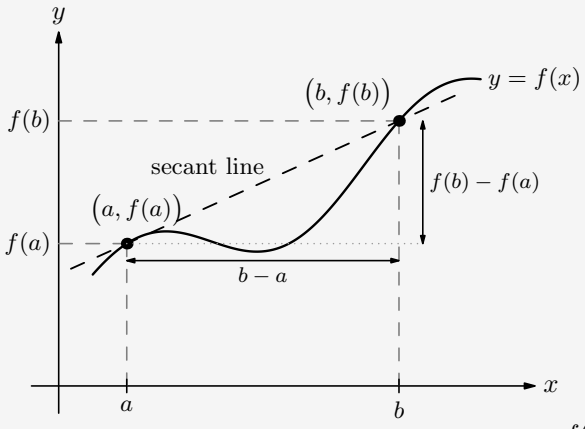
2.4 Derivatives and rates of change

Notes

- The average rate of change of f on $[a, b]$ is the change in f divided by the change in x:

$$\text{average rate of change} = \frac{f(b) - f(a)}{b - a}.$$

It is the slope of the secant line through $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$. Its units are the units of f per unit of x.



The average rate of change is the slope of the secant line: $\dfrac{f(b) - f(a)}{b - a}$.

Let the interval shrink to the point a:

average rate of change		instantaneous rate of change
$\dfrac{f(x) - f(a)}{x - a}$	$\xrightarrow{x \to a}$	$\lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a} = f'(a)$
$\dfrac{f(a + h) - f(a)}{h}$	$\xrightarrow{h \to 0}$	$\lim\limits_{h \to 0} \dfrac{f(a + h) - f(a)}{h} = f'(a)$

$f'(a)$ is the slope of the tangent line and the instantaneous rate of change, and you get it from secant slopes (average rates) by letting the second point move onto the first.

- The instantaneous rate of change equals the average rate of change on $[a, b]$: solve

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

and keep only the solutions with $a \le c \le b$.

Example 11 Average rate of change

(a) Find the average rate of change of $f(x) = e^x \cos x$ on the interval $0 \le x \le \pi$.

(b) The traffic flow at an intersection, in cars per minute, is modeled by $F(t) = 82 + 4\sin\left(\frac{t}{2}\right)$ for $0 \le t \le 30$, where t is measured in minutes. Find the average rate of change of the traffic flow over $10 \le t \le 15$. Indicate units of measure.

(c) Let $f(x) = \begin{cases} \sqrt{9-x^2} & \text{for } -3 \le x \le 0 \\ -x + 3\cos\left(\frac{\pi x}{2}\right) & \text{for } 0 < x \le 4. \end{cases}$ Find the average rate of change of f on the interval $-3 \le x \le 4$.

(d) Let $f(x) = x^2 + x$. The average rate of change of f over $[1, b]$ is 7, where $b > 1$. Find b.

Solution

(a) Only the values at the endpoints are needed (no product rule).

$$\frac{f(\pi) - f(0)}{\pi - 0} = \frac{e^\pi \cos\pi - e^0 \cos 0}{\pi}$$
$$= \frac{-e^\pi - 1}{\pi}. \qquad \cos\pi = -1,\ \cos 0 = 1,\ e^0 = 1$$

(b) The function is itself a rate (cars per minute), so the answer is in cars per minute per minute.

$$\frac{F(15) - F(10)}{15 - 10} = \frac{85.752 - 78.164}{5} \qquad \text{calculator, radians}$$
$$= 1.518 \text{ cars per minute per minute (or 1.517).}$$

(c) Put each endpoint into its own piece: $x = -3$ is in the first piece, $x = 4$ in the second.

$$f(-3) = \sqrt{9-9} = 0, \qquad f(4) = -4 + 3\cos(2\pi) = -4 + 3 = -1,$$
$$\frac{f(4) - f(-3)}{4 - (-3)} = \frac{-1 - 0}{7} = -\frac{1}{7}.$$

(The graph has a corner at $x = 0$; that does not matter.)

(d)

$$\frac{f(b) - f(1)}{b - 1} = 7 \qquad \text{average rate of change over } [1, b]$$
$$\frac{b^2 + b - 2}{b - 1} = 7 \qquad f(1) = 2$$
$$\frac{(b+2)(b-1)}{b-1} = 7$$
$$b + 2 = 7 \qquad b \ne 1$$
$$b = 5.$$

Practice

P35

(a) Find the average rate of change of $f(x) = e^x \sin x$ on the interval $0 \leq x \leq \dfrac{\pi}{2}$.

(b) Water flows into a tank at the rate $R(t) = 30 + 5\cos\left(\dfrac{t}{3}\right)$ gallons per hour, where t is measured in hours. Find the average rate of change of R over $2 \leq t \leq 8$. Indicate units of measure.

(c) Let $f(x) = x^2 - 3x$. The average rate of change of f over $[2, b]$ is 6, where $b > 2$. Find b.

P36 [AB 2015 Intl] What is the average rate of change of $y = \cos(2x)$ on the interval $\left[0, \frac{\pi}{2}\right]$?

(A) $-\frac{4}{\pi}$ (B) -1 (C) 0 (D) $\frac{4}{\pi}$

P37 [BC 2024 Alt FRQ] Let f be a differentiable function. Selected values of f and its derivative f' are shown in the table.

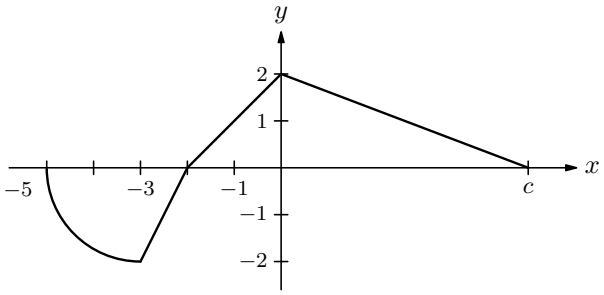
x	-2	3	5	12	15
$f(x)$	-10	-2	-1	15	36
$f'(x)$	12	5	0	9	2

(a) Find the average rate of change of f over the interval $3 \leq x \leq 12$.

(b) Find the average rate of change of f' over the interval $3 \leq x \leq 12$.

(c) On which of the intervals $[-2, 3]$, $[3, 5]$, $[5, 12]$, $[12, 15]$ is the average rate of change of f greatest?

P38 [AB 2016 FRQ] The function f is defined on $-5 \leq x \leq c$, where $c > 0$ and $f(c) = 0$. Its graph, shown below, consists of three line segments and a quarter of a circle with center $(-3, 0)$ and radius 2. Find the average rate of change of f over the interval $[-5, 0]$.



Graph of f

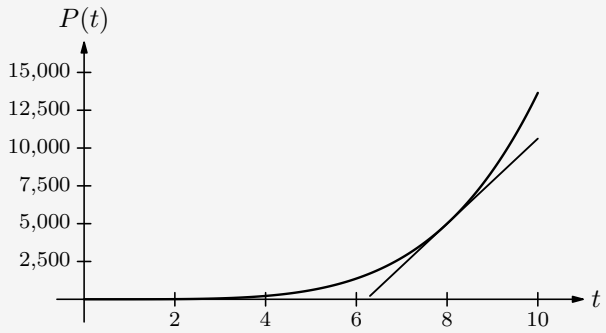
P39 [BC 2021] Let g be the function given by $g(x) = \begin{cases} x^2 + 1 & \text{for } x < 3 \\ 13 - x & \text{for } x \geq 3. \end{cases}$

What is the average rate of change of g on the closed interval $[1, 5]$?

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 5 (D) Undefined

Example 12 Instantaneous rate of change

(a) The size of a population of bacteria is modeled by the function P, where $P(t)$ gives the number of bacteria and t gives the number of hours after midnight for $0 \leq t \leq 10$. The graph of P and the line tangent to P at $t = 8$ are shown. Which of the following gives the best estimate for the instantaneous rate of change of P at $t = 8$?



(A) $P(8)$

(B) The slope of the line joining $\big(0, P(0)\big)$ and $\big(8, P(8)\big)$

(C) The slope of the line joining $\big(0, P(0)\big)$ and $\big(10, P(10)\big)$

(D) The slope of the line joining $\big(7.9, P(7.9)\big)$ and $\big(8.1, P(8.1)\big)$

(b) Let $f(x) = x^3 + 5x$. For what value of x in $[1, 3]$ does the instantaneous rate of change of f equal the average rate of change of f on $[1, 3]$?

Solution

(a) The instantaneous rate at $t = 8$ is $P'(8)$. The best estimate is the average rate of change over the shortest interval containing 8, here $[7.9, 8.1]$. The answer is (D).
(A) is a number of bacteria, not a rate; (B) and (C) are average rates over long intervals.

Solution (continued)

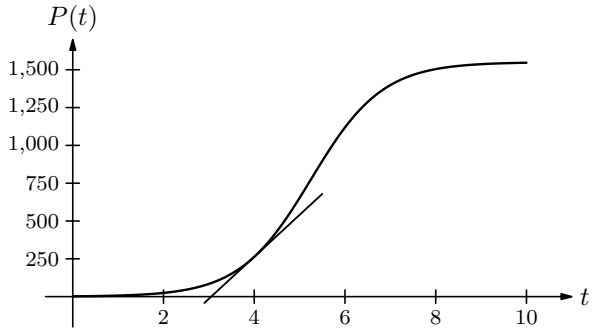
(b) Average rate of change: $\dfrac{f(3)-f(1)}{3-1} = \dfrac{42-6}{2} = 18$. Instantaneous rate: $f'(x) = 3x^2 + 5$.

$$3x^2 + 5 = 18 \implies x^2 = \frac{13}{3} \implies x = \pm\sqrt{\frac{13}{3}}.$$

Only $x = \sqrt{\dfrac{13}{3}} \approx 2.08$ is in $[1, 3]$; reject $-\sqrt{\dfrac{13}{3}}$.

Practice

P40 (a) The size of a population of fish in a pond is modeled by the function P, where $P(t)$ gives the number of fish and t gives the number of years after the fish were introduced to the pond, for $0 \le t \le 10$. The graph of P and the line tangent to P at $t = 4$ are shown. Which of the following gives the best estimate for the instantaneous rate of change of P at $t = 4$?



(A) $P(4)$

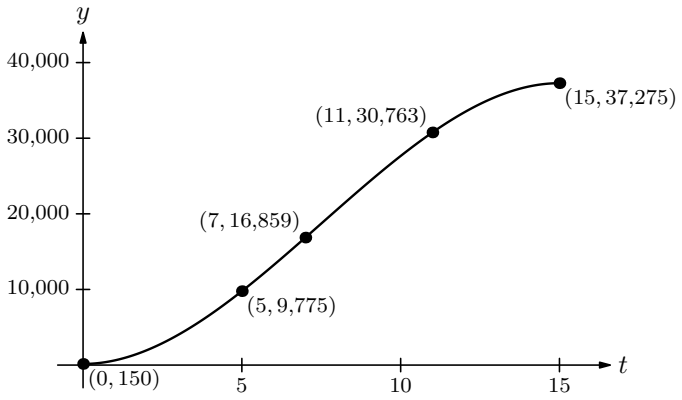
(B) The slope of the line joining $\big(0, P(0)\big)$ and $\big(4, P(4)\big)$

(C) The slope of the line joining $\big(0, P(0)\big)$ and $\big(8, P(8)\big)$

(D) The slope of the line joining $\big(3.9, P(3.9)\big)$ and $\big(4.1, P(4.1)\big)$

(b) Let $f(x) = x^3 - 2x^2 + 5x - 16$. For what value of x in $[0, 5]$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?

P41 [BC 2023] An airplane's height above sea level is modeled by the differentiable function $y = h(t)$, where t is the number of minutes after takeoff and $h(t)$ is measured in feet. The graph of h is shown for $0 \le t \le 15$. Of the following, which is the best estimate of the rate at which the plane is rising above sea level, in feet per minute, at time $t = 9$?



(A) $\dfrac{37{,}275 - 150}{15 - 0}$

(B) $\dfrac{37{,}275 - 9{,}775}{15 - 5}$

(C) $\dfrac{37{,}275 - 16{,}859}{15 - 7}$

(D) $\dfrac{30{,}763 - 16{,}859}{11 - 7}$

P42 [BC 2003] Let f be the function defined by $f(x) = x + \ln x$. What is the value of c for which the instantaneous rate of change of f at $x = c$ is the same as the average rate of change of f over $[1, 4]$?

(A) 0.456 (B) 1.244 (C) 2.164 (D) 2.342

2.5 Rates of change in context

Notes

The instantaneous rate of change of a quantity Q with respect to t at $t = a$ is the derivative $Q'(a)$. Its units are the units of Q per unit of t:

$$\text{rate of change of } Q \text{ at } t = a = Q'(a), \qquad \text{units of } Q'(a) = \frac{\text{units of } Q}{\text{units of } t}.$$

Example 13 Rates of change in context

(a) [AB 2011 Form B FRQ] A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S, where $S(t)$ is measured in millimeters and t is measured in days for $0 \le t \le 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$. Assuming no evaporation occurs, at what rate is the volume of water in the can changing at time $t = 7$? Indicate units of measure.

(b) [AB 2016 Alt FRQ] A company produces and sells chili powder. The company's weekly profit on the sale of x kilograms of chili powder is modeled by the function P given by $P(x) = 48x + 1.4x^2 - 0.05x^{2.8} - 270$, where $P(x)$ is in dollars and $0 \le x \le 80$. Find the rate, in dollars per kilogram, at which the company's weekly profit is changing when it sells 32 kilograms of chili powder.

Solution

(a) **Step 1: The volume in terms of the height.**

"At what rate is the volume changing" asks for $V'(7)$. The radius is fixed at 10 millimeters, so

$$V = \pi(10)^2 S = 100\pi S \implies V'(t) = 100\pi S'(t). \qquad \text{constant-multiple rule}$$

Step 2: Substitute $t = 7$.

$$V'(7) = 100\pi\big(2\sin(0.21) + 1.5\big) = 602.218. \qquad \text{radian mode}$$

The volume is changing at 602.218 cubic millimeters per day.

(b) The rate is $P'(32)$.

$$\begin{aligned} P'(x) &= 48 + 2.8x - 0.14x^{1.8} && \text{power rule: } 0.05 \cdot 2.8 = 0.14 \\ P'(32) &= 48 + 2.8(32) - 0.14(32)^{1.8} \\ &= 48 + 89.6 - 0.14(512) = 65.92. && 32^{1.8} = \big(2^5\big)^{1.8} = 2^9 = 512 \end{aligned}$$

The weekly profit is changing at 65.92 dollars per kilogram.

Practice

P43 Indicate units of measure in each answer.

(a) A cylindrical tank of radius 3 meters is being filled with water. The height of the water in the tank is modeled by the function h, where $h(t)$ is measured in meters and t is measured in hours. The rate at which the height of the water is rising is given by $h'(t) = 0.5 + 0.2\cos(0.1t)$. At what rate is the volume of water in the tank changing at time $t = 5$?

(b) A bakery's weekly revenue from selling x kilograms of bread is modeled by $R(x) = 60x + 0.8x^2 - 0.02x^{2.5} - 150$, where $R(x)$ is in dollars and $0 \leq x \leq 60$. Find the rate, in dollars per kilogram, at which the weekly revenue is changing when the bakery sells 25 kilograms of bread.

P44 [AB 2017 Alt] The number of gallons of water in a storage tank at time t, in minutes, is modeled by $w(t) = 25 - t^2$ for $0 \leq t \leq 5$. At what rate, in gallons per minute, is the amount of water in the tank changing at time $t = 3$ minutes?

(A) 66 (B) 16 (C) -3 (D) -6

P45 [AB 2015 Intl] The velocity v, in meters per second, of a certain type of wave is given by $v(h) = 3\sqrt{h}$, where h is the depth, in meters, of the water through which the wave moves. What is the rate of change, in meters per second per meter, of the velocity of the wave with respect to the depth of the water, when the depth is 2 meters?

(A) $-\dfrac{3}{4\sqrt{2}}$ (B) $\dfrac{3}{2\sqrt{2}}$ (C) $\dfrac{3}{\sqrt{2}}$ (D) $4\sqrt{2}$

2.6 Derivatives on the calculator

Notes

- Calculator in radian mode. Give answers correct to three decimal places; store long values (STO→) instead of rounding them halfway.
- $f'(a)$ on the TI-84: MATH 8 (nDeriv). The calculator differentiates any formula, composites included.

  nDeriv(f(X),X,a) or, on a MathPrint screen, $\left.\frac{d}{d\mathtt{X}}\big(f(\mathtt{X})\big)\right|_{\mathtt{X}=a}$.

  Quicker: type f into Y1 once and use nDeriv(Y1,X,a) (Y1 is under VARS, Y-VARS, Function).
- On the test, write the derivative, the value and the units, e.g. $r'(4) = -39.932$ gallons per hour per hour. For a piecewise function, use the piece that contains a.
- Solving $f'(x) = k$: graph

  $$\mathtt{Y2} = \mathtt{nDeriv(Y1,X,X)}, \qquad \mathtt{Y3} = k$$

  with the window set to the interval, then 2nd TRACE (CALC) 5:intersect. Or graph $f'(x) - k$ and use 2:zero. If $f'(x)$ is given, type it in directly. To count the solutions, count the crossings on the interval (zoom in where the graphs nearly touch).
- Which equation to solve:
  - tangent parallel to a line of slope m: $f'(x) = m$; horizontal tangent: $f'(x) = 0$;
  - tangent to f at $x = 2$ parallel to the tangent to g at $x = a$: $g'(a) = f'(2)$;
  - tangent at $\big(k, f(k)\big)$ through the origin, or with y-intercept b. The tangent is $y = f(k) + f'(k)(x - k)$; put $x = 0$:

    $$y\text{-intercept} = f(k) - kf'(k), \qquad \text{through the origin: } f(k) - kf'(k) = 0;$$

  - slope where the graph crosses the x-axis: solve $f(x) = 0$ first (2:zero); the root is stored in X, so nDeriv(Y1,X,X) on the home screen gives the slope there.

Example 14 Derivatives on the calculator

Give each answer correct to three decimal places.

(a) Let $f(x) = x^3 - 2x^2 + \sin(x^2)$. Find $f'(1.5)$. Of $x = -1$, 0, 1 and 2, at which does the line tangent to the graph of f have the greatest slope?

(b) Water flows into a tank at the rate

$$r(t) = \begin{cases} 40 + 30e^{-0.4t} & \text{for } 0 \le t \le 3 \\ 20 + 15\ln(t^2 - 5) & \text{for } 3 < t \le 8, \end{cases}$$

where $r(t)$ is measured in liters per hour and t in hours. Find $r'(4)$. Indicate units of measure.

(c) The graph of $y = e^{\cos x} - 2$ crosses the x-axis at one point in the interval $[0, 2]$. What is the slope of the graph at this point?

(d) The derivative of the function g is $g'(x) = 2\cos(x^2) + 1$.

(i) For how many values of x in $0 \le x \le 3$ is the line tangent to the graph of g parallel to the line $y = 2x + 5$?

(ii) Find all values of x in $0 < x < 2$ at which the graph of g has a horizontal tangent line.

(e) The line tangent to the graph of $y = \sin x$ at the point $(k, \sin k)$, where $\pi < k < 2\pi$, passes through the origin. Find k.

(f) The line tangent to the graph of $p(x) = \sqrt{x^2 + 5}$ at $x = 2$ is parallel to the line tangent to the graph of $q(x) = x^3 - x$ at $x = a$, where $a > 0$. Find a.

Solution

(a) *Scratch:* `Y1=X^3-2X^2+sin(X^2)`; then `nDeriv(Y1,X,1.5)`, and the same with -1, 0, 1, 2.

$$f'(1.5) = -1.135.$$

The slopes are $f'(-1) = 5.919$, $f'(0) = 0$, $f'(1) = 0.081$, $f'(2) = 1.385$. The greatest slope is at $x = -1$.

(b) $t = 4$ is in the second piece, $3 < t \le 8$. *Scratch:* `nDeriv(20+15ln(X^2-5),X,4)`.

$$r'(4) = 10.909 \text{ liters per hour per hour.}$$

The first piece at $t = 4$ gives -2.423, a trap answer.

(c) First find the point, then the slope there.

Step 1: Find where the graph crosses the x-axis.

Scratch: `Y1=e^(cos(X))-2`, window $0 \le x \le 2$, `2:zero`.

$$e^{\cos x} - 2 = 0 \implies x = 0.80495\ldots$$

Step 2: The slope there.

Scratch: home screen: `nDeriv(Y1,X,X)` (`X` holds the root).

$$\left.\frac{dy}{dx}\right|_{x=0.80495\ldots} = -1.442.$$

The trap answer 0.805 is the x-value, not the slope.

Solution (continued)

(d) (i) The line $y = 2x + 5$ has slope 2, so solve $g'(x) = 2$ on $[0, 3]$. *Scratch:* `Y2=2cos(X^2)+1`, `Y3= 2`, window $0 \le x \le 3$; count the crossings.

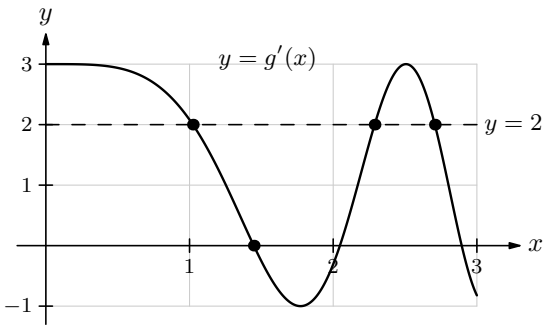
$$2\cos\left(x^2\right) + 1 = 2 \implies x = 1.023,\ 2.288,\ 2.707.$$

There are three values.

(ii) Horizontal tangent: $g'(x) = 0$.

$$2\cos\left(x^2\right) + 1 = 0 \implies x = 1.447 \quad \text{in } 0 < x < 2.$$

(The next zero, $x = 2.047$, is outside the interval.)



(e) Write the y-intercept of the tangent in terms of k, then solve.

Step 1: The y-intercept of the tangent.

With $f(x) = \sin x$, the tangent at $(k, \sin k)$ is $y = \sin k + \cos k\,(x - k)$. At $x = 0$:

$$y\text{-intercept} = \sin k - k\cos k.$$

Step 2: Through the origin.

$$\sin k - k\cos k = 0 \implies k = 4.493 \quad \text{in } \pi < k < 2\pi.$$

Scratch: `Y1=sin(X)-Xcos(X)`, window $\pi \le x \le 2\pi$, `2:zero`.

(f) Parallel lines have equal slopes.

Step 1: The slope of the tangent to p at $x = 2$.

$$p'(2) = 0.6667 \qquad \left(\text{exactly } \frac{2}{3}\right).$$

Step 2: Parallel lines have equal slopes.

$$\begin{aligned} q'(a) &= p'(2) \\ 3a^2 - 1 &= \frac{2}{3} && q'(x) = 3x^2 - 1 \\ a &= 0.745. && a > 0 \end{aligned}$$

Practice

P46 Let f be the function given by $f(x) = x^4 + \frac{1}{2}x^3 - 5x^2 + \tan\left(\frac{x}{2}\right)$. Of the following values of x, at which does the line tangent to the graph of f have the greatest slope?

(A) $x = -2$ (B) $x = -1$ (C) $x = 0$ (D) $x = 1$

P47 [AB 2012 Intl] The graph of $y = e^{\tan x} - 2$ crosses the x-axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.606 (B) 2.242 (C) 2.961 (D) 3.747

P48 [BC 2024 Alt] A 1000-gallon tank is used to collect rainwater. The tank is empty at time $t = 0$. Rainwater enters the tank at a rate modeled by the piecewise-defined function r defined by

$$r(t) = \begin{cases} 200e^{-0.21t} & \text{for } 0 \le t \le 2 \\ 115 + 25\sin\left(0.2t^2\right) & \text{for } 2 < t \le 5, \end{cases}$$

where $r(t)$ is measured in gallons per hour and t is measured in hours. Find $r'(4)$. Indicate units of measure.

P49 [BC 2021] The derivative of the function f is given by $f'(x) = 1 - 5\cos\left(3x^2 - 2\right)$. For how many values of x in the interval $0 \le x \le 2$ is the line tangent to the graph of $y = f(x)$ at the point $\big(x, f(x)\big)$ parallel to the line $y = 1 + 4x$?

(A) One (B) Two (C) Three (D) Four

P50 [BC 2014 Alt] Line ℓ is tangent to the graph of $y = \cos x$ at the point $(k, \cos k)$, where $0 < k < \pi$. For what value of k does line ℓ pass through the origin?

(A) 0.860

(B) 1.571

(C) 2.798

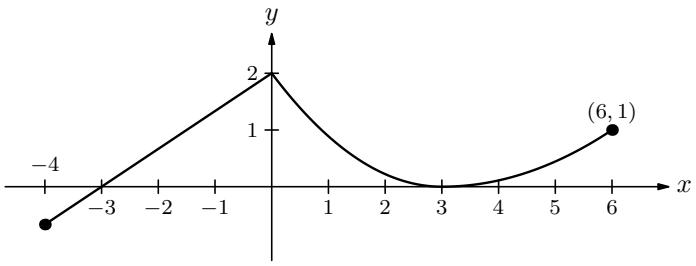
(D) There is no such value of k.

P51 [AB 2017 Alt] Let f be the function defined by $f(x) = \ln\left(x^2 + 1\right)$, and let g be the function defined by $g(x) = x^5 + x^3$. The line tangent to the graph of f at $x = 2$ is parallel to the line tangent to the graph of g at $x = a$, where a is a positive constant. What is the value of a?

(A) 0.246 (B) 0.430 (C) 0.447 (D) 0.790

Exercise 2

2.1 [AB 2009 Form B FRQ] A continuous function f is defined on the closed interval $-4 \leq x \leq 6$. The graph of f consists of a line segment and a curve that is tangent to the x-axis at $x = 3$, as shown. For how many values of a, $-4 \leq a < 6$, is the average rate of change of f on the interval $[a, 6]$ equal to 0? Give a reason for your answer.



Graph of f

2.2 Let $f(x) = x^4 - 5x$.

(a) Find the average rate of change of f on the closed interval $[0, 3]$.

(b) Find $\lim\limits_{h\to 0} \dfrac{f(1+h) - f(1)}{h}$.

(c) Find $\lim\limits_{x\to 2} \dfrac{f(x) - 6}{x - 2}$.

(d) Find $\dfrac{d}{dx}\big(f(2)\big)$.

(e) If $g(x) = \tan x$, find $\lim\limits_{x\to \frac{\pi}{4}} \dfrac{g(x) - g\left(\frac{\pi}{4}\right)}{x - \frac{\pi}{4}}$.

2.3★ Let f be a differentiable function with $f(2) = 3$ and $f'(2) = 5$. Which of the following limits are equal to 5?

I. $\lim\limits_{h\to 0} \dfrac{f(2+h)-3}{h}$

II. $\lim\limits_{x\to 2} \dfrac{3-f(x)}{2-x}$

III. $\lim\limits_{h\to 0} \dfrac{f(2+h)-f(2-h)}{h}$

(A) I only (B) I and II only (C) I and III only (D) I, II and III

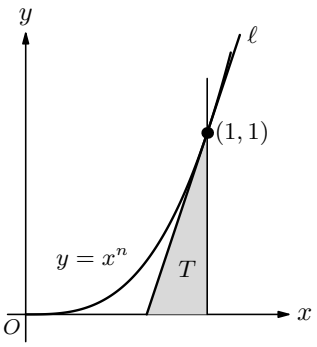
2.4 [Based on AB 1998 FRQ] Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $\dfrac{3x^2 + 1}{2y}$.

(a) Find the slope of the graph of f at the point where $x = 1$.

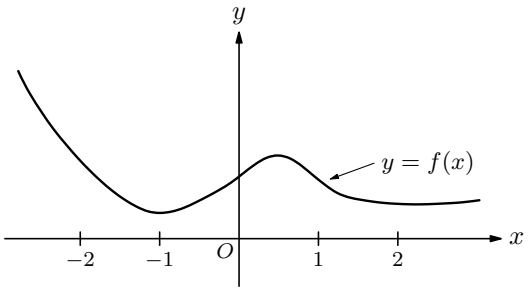
(b) Let $g(x) = x^2 f(x)$. Write an equation of the line tangent to the graph of g at $x = 1$.

(c) Find $\dfrac{d}{dx}\big(g(1)\big)$.

2.5★ [AB 2004 Form B FRQ] Let ℓ be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown. Let T be the triangular region bounded by ℓ, the x-axis, and the line $x = 1$. Show that the area of T is $\dfrac{1}{2n}$.



2.6 [AB 2013 Intl] The graph of a twice-differentiable function f is shown. Which of the following is true?



(A) $f'(-1) < f'(1) < f'(0)$

(B) $f'(-1) < f'(0) < f'(1)$

(C) $f'(1) < f'(-1) < f'(0)$

(D) $f'(1) < f'(0) < f'(-1)$

2.7 [Based on AB 2014 Intl] A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$.

(a) At what rate are shoppers entering the store 3 hours after the start of the sale?

(b) Find $\frac{d}{dt}\big(S(3)\big)$.

(c) Find all times t, for $0 < t < 12$, at which the rate at which shoppers enter the store equals the average rate of change of S over $0 \leq t \leq 12$.

2.8 [AB 2007 Form B] The wind chill is the temperature, in degrees Fahrenheit (°F), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v, in miles per hour (mph). If the air temperature is 32°F, then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \leq v \leq 60$.

(a) Find $W'(20)$. Indicate units of measure.

(b) Find the average rate of change of W over the interval $5 \leq v \leq 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \leq v \leq 60$.

2.9

(a) [AB 2010 Form B FRQ] The function g is defined for $x > 0$ with $g(1) = 2$ and $g'(x) = \sin\left(x + \frac{1}{x}\right)$. Find all values of x in the interval $0.12 \leq x \leq 1$ at which the graph of g has a horizontal tangent line.

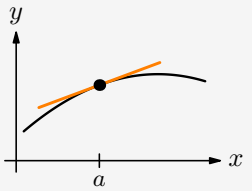
(b) [AB 2025 FRQ] Let $f(x) = x^2 - 2x$ and $h(x) = x + \sin(\pi x)$. It can be shown that $h'(x) = 1 + \pi\cos(\pi x)$. Find the value of x, for $0 < x < 1$, at which the line tangent to the graph of f is parallel to the line tangent to the graph of h.

3 Differentiability and Continuity

3.1 Differentiability from a graph

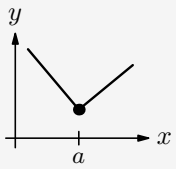
Notes

Graphically, $f'(a)$ exists when the graph is **smooth** at a and does not have a vertical tangent at a.

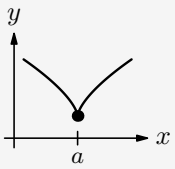


smooth at a: one tangent line

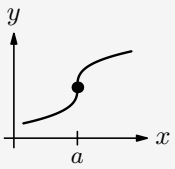
$f'(a)$ does not exist at a corner, a cusp or a vertical tangent, or where f is not continuous at a.



a corner



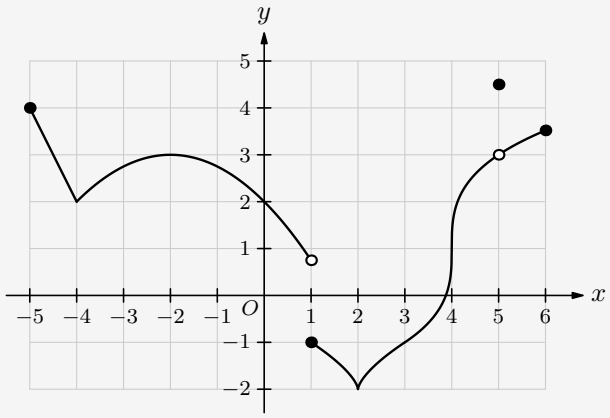
a cusp



a vertical tangent

Example 15 Differentiability from a graph

(a) The graph of the function f, defined for $-5 \leq x \leq 6$, is shown. The graph has a horizontal tangent at $x = -2$ and a vertical tangent at $x = 4$.

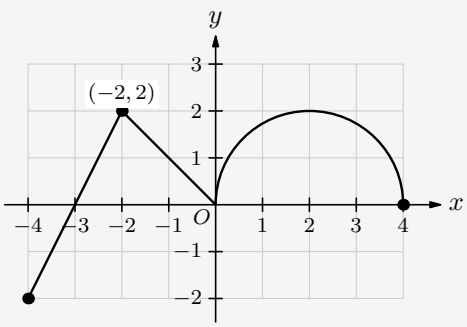


Graph of f

(i) At which values of x, $-5 < x < 6$, is f not differentiable?

(ii) At which of the values in (i) is f continuous?

(b) The continuous function f is defined on $-4 \leq x \leq 4$. Its graph consists of two line segments and a semicircle, as shown.



Graph of f

(i) Is f differentiable at $x = -2$? Is f differentiable at $x = 0$? Answer yes or no.

(ii) Write a piecewise-defined function for $f'(x)$ for $-4 < x < 0$, $x \neq -2$.

(iii) For each of $\lim_{x \to 0^-} f'(x)$ and $\lim_{x \to 0^+} f'(x)$, find the value or state that it does not exist.

Solution

(a) (i) f is not differentiable at

- $x = -4$: a corner (slope -2 on the left, slope $+1$ on the right);
- $x = 1$: a jump ($f(1) = -1$, but the graph comes in to the open dot at height 0.75);
- $x = 2$: a cusp;
- $x = 4$: a vertical tangent;
- $x = 5$: a hole ($f(5) = 4.5$, but the graph comes in to height 3 from both sides).

At $x = -2$ the tangent is horizontal, so $f'(-2) = 0$ exists. At $x = 3$ the two curves join smoothly.

(ii) Continuous at $x = -4$, 2 and 4 (corner, cusp, vertical tangent). The jump at $x = 1$ and the hole at $x = 5$ are discontinuities.

(b) (i) At $x = -2$: no, the graph has a corner (slope 2 on the left, -1 on the right).
At $x = 0$: no, on the right the semicircle has a vertical tangent at $(0, 0)$.

(ii) The slopes of the segments: $\dfrac{2-(-2)}{-2-(-4)} = 2$ and $\dfrac{0-2}{0-(-2)} = -1$.

$$f'(x) = \begin{cases} 2 & \text{for } -4 < x < -2 \\ -1 & \text{for } -2 < x < 0 \end{cases}$$

(iii) $\lim_{x \to 0^-} f'(x) = -1$, the slope of the segment.
$\lim_{x \to 0^+} f'(x) = \text{DNE}/\infty$: at $(0, 0)$ the semicircle has a vertical tangent, and its slope grows without bound as $x \to 0^+$.

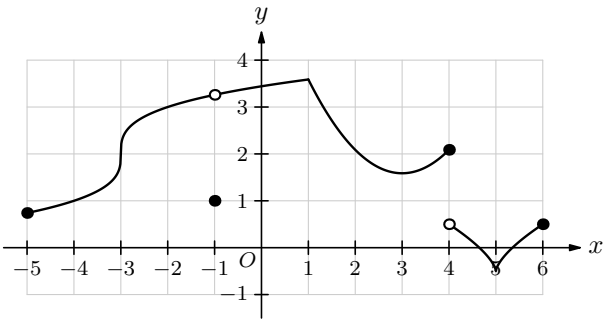
Practice

P52

(a) The graph of the function f, defined for $-5 \leq x \leq 6$, is shown below. The graph has a vertical tangent at $x = -3$ and a horizontal tangent at $x = 3$.

(i) At which values of x, $-5 < x < 6$, is f not differentiable?

(ii) At which of the values in (i) is f continuous?



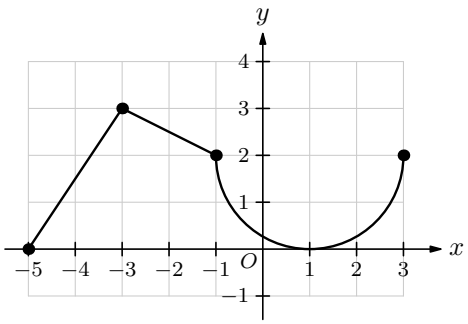
Graph of f

(b) The continuous function f is defined on $-5 \leq x \leq 3$. Its graph consists of two line segments and a semicircle, as shown below.

(i) Is f differentiable at $x = -3$? Is f differentiable at $x = -1$? Answer yes or no.

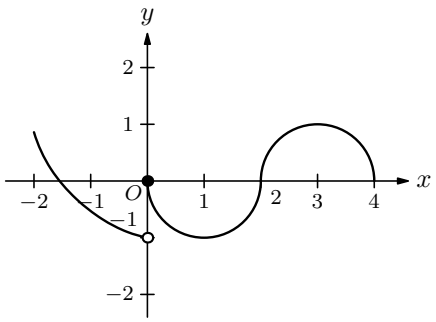
(ii) Write a piecewise-defined function for $f'(x)$ for $-5 < x < -1$, $x \neq -3$.

(iii) For each of $\lim_{x \to -1^-} f'(x)$ and $\lim_{x \to -1^+} f'(x)$, find the value or state that it does not exist.



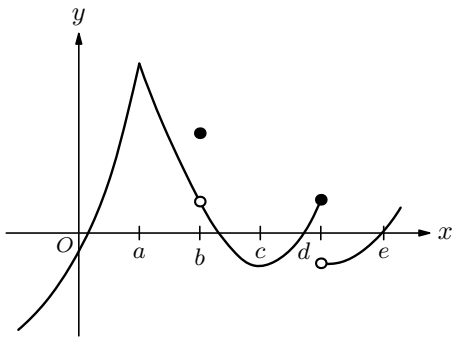
Graph of f

P53 [AB 1998] The graph of the function f shown in the figure has a vertical tangent at the point $(2, 0)$ and horizontal tangents at the points $(1, -1)$ and $(3, 1)$. For what values of x, $-2 < x < 4$, is f not differentiable?



(A) 0 only (B) 0 and 2 only (C) 1 and 3 only (D) 0, 1, 2, and 3

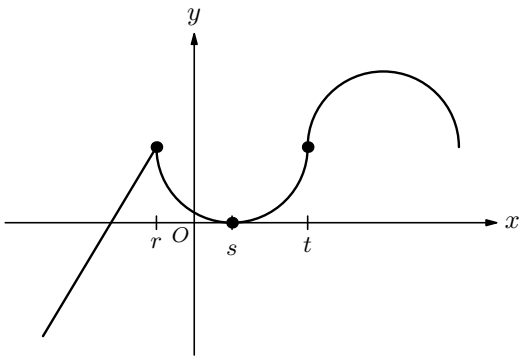
P54 [AB 2003] The graph of a function f is shown. At which value of x is f continuous, but not differentiable?



Graph of f

(A) a (B) b (C) c (D) d

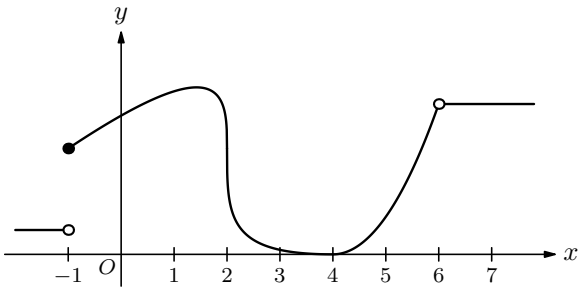
P55 The figure shows the graph of a continuous piecewise-defined function f. The graph consists of a line segment and two semicircles. For which of the following values of x is $f'(x)$ undefined?



Graph of f

(A) r only (B) t only (C) r and s (D) r and t

P56 The figure shows the graph of a function f, which has a vertical tangent at $x = 2$ and a horizontal tangent at $x = 4$. Which of the following statements is false?



Graph of f

(A) f is not differentiable at $x = -1$ because the graph of f has a jump discontinuity at $x = -1$.

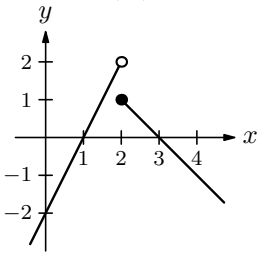
(B) f is not differentiable at $x = 2$ because the graph of f has a vertical tangent at $x = 2$.

(C) f is not differentiable at $x = 4$ because the graph of f has a horizontal tangent at $x = 4$.

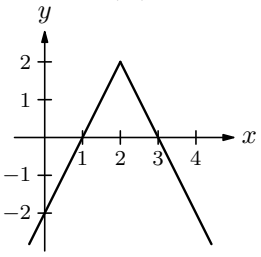
(D) f is not differentiable at $x = 6$ because the graph of f has a removable discontinuity at $x = 6$.

P57 [BC 2021] Let g be a function such that $\lim_{x\to 2} g(x) = g(2)$ and $\lim_{x\to 2^-} g'(x) \neq \lim_{x\to 2^+} g'(x)$. Which of the following could be the graph of g?

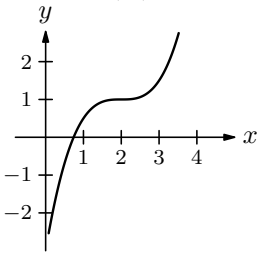
(A)



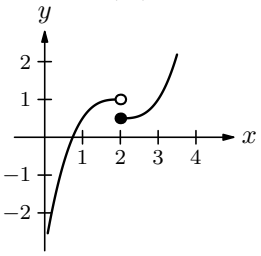
(B)



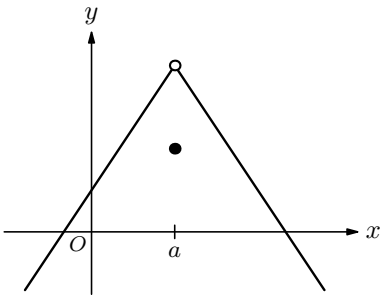
(C)



(D)



P58 [BC 2003] The graph of the function f is shown. Which of the following statements must be false?



Graph of f

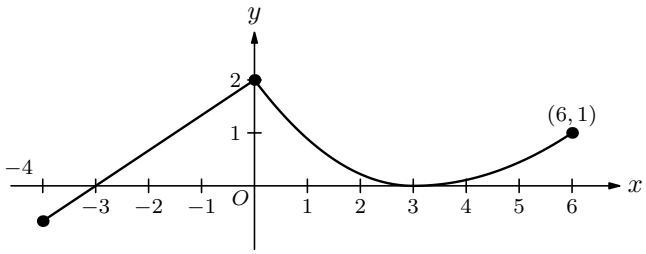
(A) $f(a)$ exists.

(B) f is not continuous at $x = a$.

(C) $\lim_{x \to a} f(x)$ exists.

(D) $\lim_{x \to a} f'(x)$ exists.

P59 [AB 2009 Form B FRQ] A continuous function f is defined on the closed interval $-4 \le x \le 6$. The graph of f consists of a line segment and a curve that is tangent to the x-axis at $x = 3$, as shown in the figure. Is f differentiable at $x = 0$? Use the definition of the derivative with one-sided limits to justify your answer.



Graph of f

3.2 Differentiability from a formula

Notes

f is differentiable at a when

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \quad \text{exists,}$$

or, the same thing,

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} \quad \text{exists.}$$

For a piecewise function, $f(a)$ comes from the piece that contains a; use the left formula for the left-hand limit and the right formula for the right-hand limit. If the pieces do not meet, one of the two limits blows up, even when the pieces have the same slope.

Example 16 Differentiability from a formula

(a) Let $f(x) = \left|x^2 - 4\right|$. Is f differentiable at $x = 2$? Justify your answer.

(b) Let $g(x) = 1 + \sqrt{|x-1|}$. Is g continuous at $x = 1$? Is g differentiable at $x = 1$? Justify your answers.

(c) Let $f(x) = \begin{cases} x^2 + 1 & \text{for } x \le 1 \\ 2x & \text{for } x > 1. \end{cases}$

Is f differentiable at $x = 1$? Justify your answer.

(d) [BC 2014] The function f is defined by

$$f(x) = \begin{cases} 2x + 5 & \text{for } x < -1 \\ -x^2 + 6 & \text{for } x \ge -1. \end{cases}$$

Is f differentiable at $x = -1$? Justify your answer.

Solution

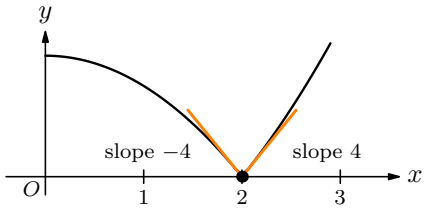
(a) We need to check whether $\lim\limits_{h\to 0} \dfrac{f(2+h) - f(2)}{h}$ exists, where $f(2) = 0$. Split it into the two one-sided limits.

$$\begin{aligned} &\lim_{h\to 0^-} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h\to 0^-} \frac{\left|(2+h)^2 - 4\right| - 0}{h} \\ &= \lim_{h\to 0^-} \frac{\left|4h + h^2\right|}{h} \\ &= \lim_{h\to 0^-} \frac{\left|h(h+4)\right|}{h} \\ &= \lim_{h\to 0^-} \frac{-h(h+4)}{h} \\ &= \lim_{h\to 0^-} (-4 - h) = -4 \end{aligned}$$

$$\begin{aligned} &\lim_{h\to 0^+} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h\to 0^+} \frac{\left|(2+h)^2 - 4\right| - 0}{h} \\ &= \lim_{h\to 0^+} \frac{\left|4h + h^2\right|}{h} \\ &= \lim_{h\to 0^+} \frac{\left|h(h+4)\right|}{h} \\ &= \lim_{h\to 0^+} \frac{h(h+4)}{h} \\ &= \lim_{h\to 0^+} (4 + h) = 4 \end{aligned}$$

(For h close to 0, $h + 4 > 0$, so $h(h+4)$ has the sign of h: negative on the left, so the absolute value changes its sign; positive on the right.)

$-4 \neq 4$, so $f'(2)$ does not exist: f is not differentiable at $x = 2$. The graph has a corner at $(2, 0)$:



(b) Continuity: $g(1) = 1 + 0 = 1$ and $\lim\limits_{x\to 1} g(x) = 1 + \sqrt{0} = 1 = g(1)$, so g is continuous at $x = 1$.

Differentiability: we need to check whether $\lim\limits_{h\to 0} \dfrac{g(1+h) - g(1)}{h}$ exists. Split it into the two one-sided limits.

Solution (continued)

$$\lim_{h\to 0^-}\frac{g(1+h)-g(1)}{h} = \lim_{h\to 0^-}\frac{1+\sqrt{|h|}-1}{h} = \lim_{h\to 0^-}\frac{\sqrt{-h}}{h} = \lim_{h\to 0^-}\frac{\sqrt{-h}}{-\left(\sqrt{-h}\right)^2} = \lim_{h\to 0^-}\frac{-1}{\sqrt{-h}} = \text{DNE}/-\infty$$

$$\lim_{h\to 0^+}\frac{g(1+h)-g(1)}{h} = \lim_{h\to 0^+}\frac{1+\sqrt{|h|}-1}{h} = \lim_{h\to 0^+}\frac{\sqrt{h}}{h} = \lim_{h\to 0^+}\frac{\sqrt{h}}{\left(\sqrt{h}\right)^2} = \lim_{h\to 0^+}\frac{1}{\sqrt{h}} = \text{DNE}/\infty$$

The one-sided limits are not finite, so $g'(1)$ does not exist: g is continuous but not differentiable at $x = 1$ (a cusp).

(c) We need to check whether $\lim_{h\to 0}\dfrac{f(1+h)-f(1)}{h}$ exists, where $f(1) = 1^2+1 = 2$ (the piece $x \le 1$). Split it into the two one-sided limits. For $h < 0$, $1+h < 1$: use x^2+1. For $h > 0$, $1+h > 1$: use $2x$.

$$\lim_{h\to 0^-}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0^-}\frac{(1+h)^2+1-2}{h} = \lim_{h\to 0^-}\frac{2h+h^2}{h} = \lim_{h\to 0^-}(2+h) = 2$$

$$\lim_{h\to 0^+}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0^+}\frac{2(1+h)-2}{h} = \lim_{h\to 0^+}\frac{2h}{h} = \lim_{h\to 0^+}2 = 2$$

Both one-sided limits are 2, so f is differentiable at $x = 1$, and $f'(1) = 2$.

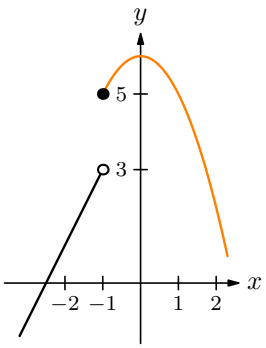
(d) $f(-1) = -(-1)^2+6 = 5$, from $-x^2+6$.

$$\lim_{x\to -1^-}\frac{f(x)-f(-1)}{x-(-1)} = \lim_{x\to -1^-}\frac{(2x+5)-5}{x+1} \qquad \text{left formula}$$

$$= \lim_{x\to -1^-}\frac{2x}{x+1} = \text{DNE}/\infty, \qquad \text{top} \to -2,\ \text{bottom} \to 0^-$$

$$\lim_{x\to -1^+}\frac{f(x)-f(-1)}{x-(-1)} = \lim_{x\to -1^+}\frac{\left(-x^2+6\right)-5}{x+1} = \lim_{x\to -1^+}(1-x) = 2.$$

The left limit does not exist, so f is not differentiable at $x = -1$. The pieces do not meet ($2(-1)+5 = 3 \neq 5$): the graph jumps. (The trap answer 2 compares only the slopes of the pieces, which are both 2.)



Practice

P60

(a) Let $f(x) = \left|9 - x^2\right|$. Is f differentiable at $x = -3$? Justify your answer.

(b) Let $g(x) = \sqrt{|x+2|} - 3$. Is g continuous at $x = -2$? Is g differentiable at $x = -2$? Justify your answers.

(c) Let $f(x) = \begin{cases} x^2 - x & \text{for } x \leq 3 \\ 5x - 9 & \text{for } x > 3. \end{cases}$

Is f differentiable at $x = 3$? Justify your answer.

(d) The function f is defined by

$$f(x) = \begin{cases} 3x + 1 & \text{for } x < 2 \\ x^2 - x + 1 & \text{for } x \geq 2. \end{cases}$$

Is f differentiable at $x = 2$? Justify your answer.

P61 If $f(x) = 2 + |x - 3|$ for all x, then the value of the derivative $f'(x)$ at $x = 3$ is

(A) -1 (B) 0 (C) 1 (D) nonexistent

P62 [AB 2003]

$$f(x) = \begin{cases} x + 2 & \text{if } x \leq 3 \\ 4x - 7 & \text{if } x > 3 \end{cases}$$

Let f be the function given above. Which of the following statements are true about f?

I. $\lim_{x \to 3} f(x)$ exists.

II. f is continuous at $x = 3$.

III. f is differentiable at $x = 3$.

(A) None (B) I only (C) I and II only (D) I, II, and III

P63 [BC 2008]

$$f(x) = \begin{cases} \dfrac{x^2 - 4}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Let f be the function defined above. Which of the following statements about f are true?

I. f has a limit at $x = 2$.

II. f is continuous at $x = 2$.

III. f is differentiable at $x = 2$.

(A) I only (B) II only (C) I and II only (D) I, II, and III

P64

$$f(x) = \begin{cases} \dfrac{2}{x} & \text{for } x < -1 \\ x^2 - 3 & \text{for } -1 \leq x \leq 2 \\ 4x - 3 & \text{for } x > 2 \end{cases}$$

Let f be the function defined above. At what values of x, if any, is f not differentiable?

(A) $x = -1$ only (B) $x = 2$ only

(C) $x = -1$ and $x = 2$ (D) f is differentiable for all values of x.

3.3 Differentiable implies continuous

Notes

$$f \text{ is differentiable at } a \implies f \text{ is continuous at } a, \text{ so } \lim_{x\to a} f(x) = f(a).$$

- The converse is false: $|x|$ is continuous at 0 but not differentiable there.
- Not continuous at $a \implies$ not differentiable at a.
- $f'(a)$ existing does not give $\lim_{x\to a} f'(x) = f'(a)$.

$$f \text{ is twice differentiable} \implies f' \text{ is differentiable} \implies f' \text{ is continuous,}$$

so the Intermediate Value Theorem applies to f'.

Example 17 Differentiable implies continuous

(a) The function f is differentiable at $x = 2$, with $f(2) = 5$ and $f'(2) = -3$. Which of the following statements must be true?

I. $\lim_{x\to 2} f(x) = 5$

II. $\lim_{h\to 0} \dfrac{f(2+h) - 5}{h} = -3$

III. $\lim_{x\to 2} f'(x) = -3$

(b) The function g is continuous at $x = 1$. Which of the following statements must be true?

I. $\lim_{x\to 1^-} g(x) = \lim_{x\to 1^+} g(x) = g(1)$

II. $g'(1)$ exists.

III. If g is not differentiable at $x = 1$, then $\lim_{x\to 1} \dfrac{g(x) - g(1)}{x - 1}$ does not exist.

(c) The function f is differentiable. The table gives values of f and f' at selected values of x. Find $\lim_{x\to 3} \dfrac{\big(f(x)\big)^2 - 16}{x - 3}$, or state that the limit does not exist.

x	1	3	5
$f(x)$	-2	4	7
$f'(x)$	6	-2	1

(d) The twice-differentiable function W models the volume of water in a tank at time t, where $W(t)$ is measured in liters and t is measured in minutes. The table gives values of $W'(t)$ at selected times.

t (minutes)	0	4	9	15
$W'(t)$ (liters per minute)	1.2	2.5	3.1	1.8

(i) Explain why there must be at least one time t, other than $t = 4$, such that $W'(t) = 2.5$ liters per minute.

(ii) Is there a time t, $0 \le t \le 4$, for which $W'(t) = 2$? Justify your answer.

Solution

(a) I and II only.

- I: f is differentiable at 2 $\implies$ f is continuous at 2 $\implies$ $\lim_{x\to 2} f(x) = f(2) = 5$. True.
- II: $5 = f(2)$, so this is the definition of $f'(2)$. True.
- III: this says f' is continuous at 2, which is not given. Could be false.

(b) I and III only.

- I: this is what "continuous at 1" means. True.
- II: continuous does not give differentiable: $g(x) = |x-1|$ is continuous at 1, but $g'(1)$ does not exist. Could be false.
- III: "not differentiable at 1" means exactly that this limit, the definition of $g'(1)$, does not exist. True.

(c) $f(3) = 4$, so $\big(f(x)\big)^2 - 16$ is a difference of squares with $16 = \big(f(3)\big)^2$. Factor it:

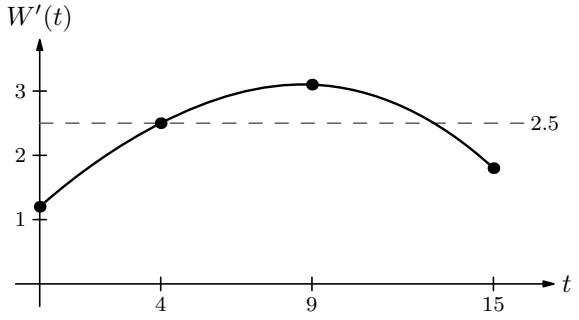
$$\begin{aligned}
\lim_{x\to 3} \frac{\big(f(x)\big)^2 - 16}{x-3} &= \lim_{x\to 3} \frac{\big(f(x)-4\big)\big(f(x)+4\big)}{x-3} \\
&= \lim_{x\to 3} \frac{f(x)-f(3)}{x-3} \cdot \lim_{x\to 3} \big(f(x)+4\big) \\
&= f'(3) \cdot \big(f(3)+4\big) && \text{definition of } f'(3);\ f \text{ is continuous at 3} \\
&= (-2)(4+4) = -16.
\end{aligned}$$

The second limit uses continuity: f is differentiable at 3, so $\lim_{x\to 3} f(x) = f(3) = 4$.

(d) (i) W is twice differentiable $\implies$ W' is differentiable $\implies$ W' is continuous on $[9, 15]$.

$$W'(15) = 1.8 < 2.5 < 3.1 = W'(9).$$

By the Intermediate Value Theorem there is a time t in $(9, 15)$ with $W'(t) = 2.5$; this t is not 4. (The interval $[0, 4]$ gives only $t = 4$ itself.)



One possible graph of W': it is continuous, so it crosses the line 2.5 between $t = 9$ and $t = 15$.

(ii) Yes. W' is continuous on $[0, 4]$ (as in (i)), and

$$W'(0) = 1.2 < 2 < 2.5 = W'(4).$$

By the Intermediate Value Theorem there is a time t in $(0, 4)$ with $W'(t) = 2$.

Practice

P65

(a) The function f is differentiable at $x = -1$, with $f(-1) = 2$ and $f'(-1) = 4$. Which of the following statements must be true?

I. $\lim\limits_{x \to -1^+} f(x) = 2$

II. $\lim\limits_{x \to -1} \dfrac{f(x) - 2}{x + 1} = 4$

III. $\lim\limits_{x \to -1} f'(x) = 4$

(b) The function g is continuous at $x = -2$. Which of the following statements must be true?

I. $\lim\limits_{x \to -2} g(x)$ exists.

II. $g'(-2)$ exists.

III. If $g'(-2)$ does not exist, then $\lim\limits_{h \to 0} \dfrac{g(-2 + h) - g(-2)}{h}$ does not exist.

(c) The function f is differentiable. The table gives values of f and f' at selected values of x. Find $\lim\limits_{x \to 2} \dfrac{\big(f(x)\big)^2 - 1}{x - 2}$, or state that the limit does not exist.

x	0	2	4
$f(x)$	3	-1	5
$f'(x)$	-4	2	3

(d) The twice-differentiable function P models the population of a town, where $P(t)$ is measured in people and t is measured in years. The table gives values of $P'(t)$ at selected times.

t (years)	0	2	5	8
$P'(t)$ (people per year)	40	65	90	55

(i) Explain why there must be at least one time t, other than $t = 2$, such that $P'(t) = 65$ people per year.

(ii) Is there a time t, $0 \le t \le 2$, for which $P'(t) = 50$? Justify your answer.

P66 [AB 2017 Alt] For any function f, which of the following statements must be true?

I. If f is defined at $x = a$, then $\lim_{x \to a} f(x) = f(a)$.

II. If f is continuous at $x = a$, then $\lim_{x \to a} f(x) = f(a)$.

III. If f is differentiable at $x = a$, then $\lim_{x \to a} f(x) = f(a)$.

(A) III only (B) I and II only (C) II and III only (D) I, II, and III

P67 [BC 2008 Form B] If f is a differentiable function such that $f(3) = 8$ and $f'(3) = 5$, which of the following statements could be false?

(A) $\lim_{x \to 3^+} f(x) = \lim_{x \to 3^-} f(x)$

(B) $\lim_{x \to 3} \dfrac{f(x) - 8}{x - 3} = 5$

(C) $\lim_{h \to 0} \dfrac{f(3 + h) - 8}{h} = 5$

(D) $\lim_{x \to 3} f'(x) = 5$

P68 [BC 2019 Alt] The function f is not differentiable at $x = 5$. Which of the following statements must be true?

(A) f is not continuous at $x = 5$.

(B) $\lim_{x \to 5} f(x)$ does not exist.

(C) $\lim_{x \to 5} \dfrac{f(x) - f(5)}{x - 5}$ does not exist.

(D) The graph of f has a corner at $x = 5$.

P69 [BC 2012] If the function f is continuous at $x = 3$, which of the following must be true?

(A) $\lim_{x \to 3^-} f(x) \neq \lim_{x \to 3^+} f(x)$

(B) $f(3) = \lim_{x \to 3^-} f(x) = \lim_{x \to 3^+} f(x)$

(C) The derivative of f at $x = 3$ exists.

(D) The derivative of f is positive for $x < 3$ and negative for $x > 3$.

P70 Which of the following statements about the function f, if true, cannot be used to conclude that f is defined at $x = 1$?

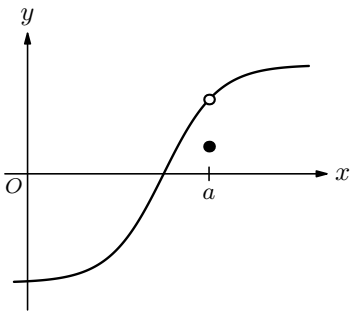
(A) $\lim_{x \to 1} f(x)$ exists.

(B) f is continuous at $x = 1$.

(C) f is differentiable at $x = 1$.

(D) The line tangent to the graph of f at $x = 1$ exists.

P71 [BC 2017 Alt] The graph of $y = f(x)$ is shown. Which of the following is true?



Graph of f

(A) $\lim_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$ exists.

(B) $\lim_{x \to a^+} f(x) \neq \lim_{x \to a^-} f(x)$

(C) $\lim_{x \to a} f(x) \neq f(a)$

(D) $\lim_{x \to a} f(x)$ does not exist.

P72 [AB 2026 FRQ] The function f is twice differentiable. The table gives values of f and its derivative f' at selected values of x. Find $\lim_{x \to 2} \dfrac{f(x)}{x}$, or state that the limit does not exist.

x	0	2	3	6
$f(x)$	-1	3	8	5
$f'(x)$	-5	4	9	-2

P73 Let f be the function defined below. Which of the following statements is true?

$$f(x) = \begin{cases} x^2 - 20 & \text{for } x < 5 \\ -x^2 + 20 & \text{for } x \geq 5 \end{cases}$$

(A) f is not differentiable at $x = 5$ because f is not continuous at $x = 5$.

(B) f is not differentiable at $x = 5$ because the graph of f has a sharp corner at $x = 5$.

(C) f is not differentiable at $x = 5$ because the graph of f has a vertical tangent at $x = 5$.

(D) f is not differentiable at $x = 5$ because f is not defined at $x = 5$.

3.4 Making a piecewise function differentiable

Notes

For $f(x) = \begin{cases} g(x), & x < a \\ k, & x = a \\ h(x), & x > a, \end{cases}$ there are two ways to check that f is differentiable at $x = a$ (and to find unknown constants).

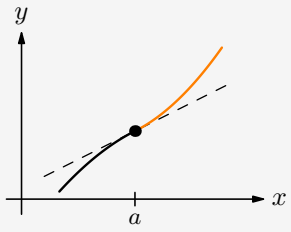
Method 1: match the value and the slope. Both are needed:

$$g(a) = k = h(a) \qquad \text{the value matches: } f \text{ is continuous at } a$$
$$g'(a) = h'(a) \qquad \text{the slopes match}$$

Then $f'(a) = g'(a) = h'(a)$. The slope equation often has only one unknown: solve it first.

Method 2: the definition. The two one-sided limits must be finite and equal:

$$\lim_{x \to a^-} \frac{g(x) - k}{x - a} = \lim_{x \to a^+} \frac{h(x) - k}{x - a}.$$



same point, same tangent line

Example 18 Matching the value and the slope

(a) Let $f(x) = \begin{cases} ax^2 + 3 & \text{if } x < 1 \\ k & \text{if } x = 1 \\ bx + 1 & \text{if } x > 1. \end{cases}$ Find the values of a, b and k for which f is differentiable at $x = 1$.

(b) Let $f(x) = \begin{cases} ae^x + b & \text{if } x \le 0 \\ 2\cos x + 3x & \text{if } x > 0. \end{cases}$ Find the values of a and b for which f is differentiable at $x = 0$.

(c) Let $f(x) = \begin{cases} x^2 - 2x, & x \le 4 \\ g(x), & x > 4. \end{cases}$ If f is differentiable, which of the following could define $g(x)$: $6x$, $x^{3/2}$, $x^2 - 8$, $\frac{1}{8}x^3$?

Solution

(a) **Method 1.** Match the value and the slope at $x = 1$.

$$a(1)^2 + 3 = k = b(1) + 1 \qquad \text{the value: } a + 3 = k \text{ and } b + 1 = k$$
$$2a(1) = b \qquad \text{the slopes: } g'(x) = 2ax,\ h'(x) = b$$

So $a + 3 = b + 1 = 2a + 1$, which gives $a = 2$, then $b = 4$ and $k = 5$.

Method 2. Use the definition with $f(1) = k$.

Solution (continued)

$$\lim_{x\to 1^-} \frac{ax^2+3-k}{x-1}$$

The bottom tends to 0, so the top must too: $a+3-k=0$, $k=a+3$.

$$= \lim_{x\to 1^-} \frac{ax^2-a}{x-1}$$
$$= \lim_{x\to 1^-} \frac{a(x-1)(x+1)}{x-1} = 2a$$

$$\lim_{x\to 1^+} \frac{bx+1-k}{x-1}$$

The top must tend to 0: $b+1-k=0$, $k=b+1$.

$$= \lim_{x\to 1^+} \frac{bx-b}{x-1}$$
$$= \lim_{x\to 1^+} \frac{b(x-1)}{x-1} = b$$

Equal: $2a=b$. These are the same three equations, so again $a=2$, $b=4$, $k=5$.

(b) Slopes first: they give a at once.

$$\lim_{x\to 0^-} f'(x) = \lim_{x\to 0^-} ae^x = a, \qquad \lim_{x\to 0^+} f'(x) = \lim_{x\to 0^+} (-2\sin x + 3) = 3,$$

$a = 3$, the slopes match

$ae^0 + b = 2\cos 0 + 3(0)$ the values match

$3 + b = 2 \implies b = -1.$

So $a=3$ and $b=-1$.

(c) At $x=4$ the left piece has value and slope

$$4^2 - 2(4) = 8, \qquad \frac{d}{dx}\left(x^2-2x\right) = 2x-2 = 6.$$

So g needs $g(4)=8$ and $g'(4)=6$. Check each choice:

$g(x)$	$g(4)$	$g'(x)$	$g'(4)$
$6x$	24	6	6
$x^{3/2}$	8	$\frac{3}{2}x^{1/2}$	3
x^2-8	8	$2x$	8
$\frac{1}{8}x^3$	8	$\frac{3}{8}x^2$	6

Only $g(x) = \frac{1}{8}x^3$ matches both. ($6x$ matches only the slope; $x^{3/2}$ and x^2-8 match only the value.)

Practice

P74

(a) Let $f(x) = \begin{cases} ax^2 - 1 & \text{if } x < 2 \\ k & \text{if } x = 2 \\ bx + 3 & \text{if } x > 2. \end{cases}$ Find the values of a, b and k for which f is differentiable at $x = 2$.

(b) Let $f(x) = \begin{cases} a \sin x + b & \text{if } x \leq \pi \\ x + \cos x & \text{if } x > \pi. \end{cases}$ Find the values of a and b for which f is differentiable at $x = \pi$.

(c) Let $f(x) = \begin{cases} x^2 + 3, & x \leq 1 \\ g(x), & x > 1. \end{cases}$ If f is differentiable, which of the following could define $g(x)$: $4x$, $x^3 + 3$, $4\sqrt{x}$, $2x^2 + 2$?

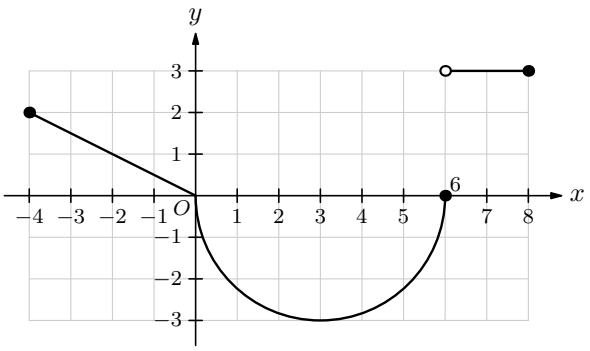
P75 Let g be the function defined by

$$g(x) = \begin{cases} k\sqrt{x} & \text{for } 0 \le x \le 4 \\ mx + 2 & \text{for } 4 < x \le 6, \end{cases}$$

where k and m are constants. If g is differentiable at $x = 4$, what are the values of k and m?

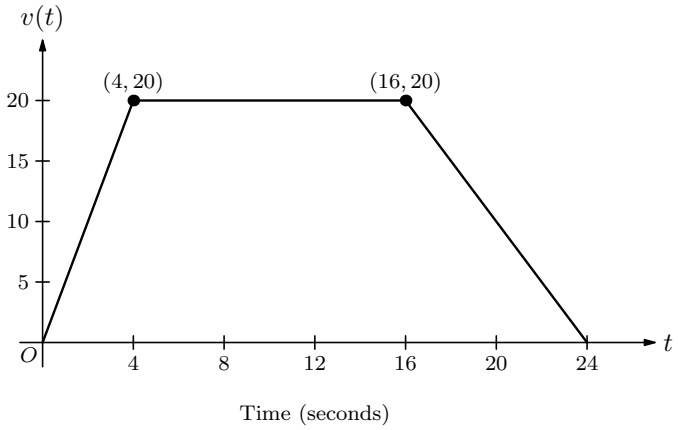
Exercise 3

3.1 [BC 2018 Alt FRQ] The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure. For each of $\lim_{x \to 0^-} g'(x)$ and $\lim_{x \to 0^+} g'(x)$, find the value or state that it does not exist.



Graph of g

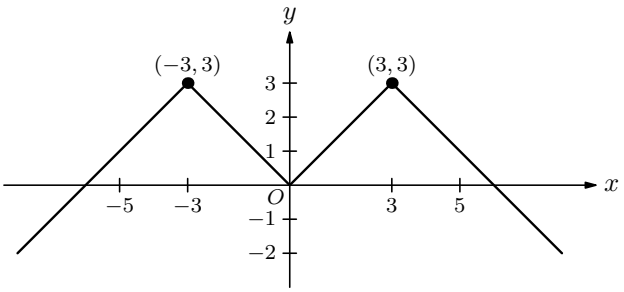
3.2 [AB 2005 FRQ] A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph.



(a) For each of $v'(4)$ and $v'(20)$, find the value or explain why it does not exist. Indicate units of measure.

(b) For $0 < t < 24$, write a piecewise-defined function for $v'(t)$.

3.3 The graph of the even function $y = f(x)$ consists of 4 line segments, as shown. Which of the following statements about f is false?



(A) $\lim\limits_{x\to 0} \dfrac{f(x) - f(0)}{x} = 0$

(B) $\lim\limits_{x\to 0} \dfrac{f(x) - f(-x)}{2x} = 0$

(C) $\lim\limits_{x\to 2} \dfrac{f(x) - f(2)}{x - 2} = 1$

(D) $\lim\limits_{x\to 3} \dfrac{f(x) - f(3)}{x - 3}$ does not exist.

3.4$^{\bigstar}$ Let $f(x) = x\,|x-2|$.

(a) Is f continuous at $x = 2$? Justify your answer.

(b) Use one-sided limits of the difference quotient to show that f is not differentiable at $x = 2$.

(c) Evaluate $\lim\limits_{h\to 0} \dfrac{f(2+h) - f(2-h)}{2h}$. Explain why your answer does not contradict (b).

3.5 For each function, use one-sided limits of the difference quotient to decide whether the function is differentiable at $x = 0$. If it is not, say whether the graph has a corner, a cusp or a vertical tangent at $x = 0$.

(a) $f(x) = \sqrt[3]{x}$

(b) $g(x) = \sqrt[3]{x^2}$

(c) $k(x) = x\,|x|$

3.6 [AB 2012 Intl] Let f be the function defined by $f(x) = \sqrt{|x-2|}$ for all x. Which of the following statements is true?

(A) f is continuous but not differentiable at $x = 2$.

(B) f is differentiable at $x = 2$.

(C) f is not continuous at $x = 2$.

(D) $x = 2$ is a vertical asymptote of the graph of f.

3.7 [AB 2002 FRQ] Let f be a function that is differentiable for all real numbers. The table below gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let g be the function given by $g(x) = \begin{cases} 2x^2 - x - 7 & \text{for } x < 0 \\ 2x^2 + x - 7 & \text{for } x \geq 0. \end{cases}$

The graph of g passes through each of the points $\big(x, f(x)\big)$ given in the table above. Is it possible that f and g are the same function? Give a reason for your answer.

3.8★ The function f is twice differentiable. The table gives values of f and its derivative f' at selected values of x.

x	1	2	4	6
$f(x)$	3	5	5	2
$f'(x)$	4	-1	2	-3

(a) Find $\lim_{x\to 4} \dfrac{f(x)-5}{x-2}$, or state that the limit does not exist.

(b) Find $\lim_{x\to 4} \dfrac{f(x)-5}{x-4}$, or state that the limit does not exist.

(c) Must there be a value c, $1 < c < 2$, for which $f(c) = 4$? Justify your answer.

(d) Explain why there must be at least three values of c, $1 < c < 6$, for which $f'(c) = 0$.

3.9 The function f is defined for all x, and $\lim_{x\to 1} \dfrac{f(x)-5}{x-1} = 3$.

(a) Which of the following statements must be true?

I. $\lim_{x\to 1} f(x) = 5$

II. $f(1) = 5$

III. $f'(1) = 3$

(A) I only (B) I and II only (C) I and III only (D) I, II, and III

(b) Suppose, in addition, that f is continuous at $x = 1$. Find $f(1)$ and $f'(1)$.

3.10 [AB 2013 Intl FRQ] The twice-differentiable function W models the volume of water in a reservoir at time t, where $W(t)$ is measured in gigaliters (GL) and t is measured in days. The table gives values of $W'(t)$ sampled at various times during the time interval $0 \le t \le 30$ days. At time $t = 30$, the reservoir contains 125 gigaliters of water. Explain why there must be at least one time t, other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

3.11 [AB 2022 FRQ] An ice sculpture melts in such a way that it can be modeled as a cone that maintains a conical shape as it decreases in size. The radius of the base of the cone is given by a twice-differentiable function r, where $r(t)$ is measured in centimeters and t is measured in days. The table gives selected values of $r'(t)$, the rate of change of the radius, over the time interval $0 \le t \le 12$. Is there a time t, $0 \le t \le 3$, for which $r'(t) = -6$? Justify your answer.

t (days)	0	3	7	10	12
$r'(t)$ (centimeters per day)	-6.1	-5.0	-4.4	-3.8	-3.5

3.12★ Let

$$f(x) = \begin{cases} ax + b & \text{if } x < 0 \\ \cos x + x & \text{if } 0 \le x \le \pi \\ cx + d & \text{if } x > \pi. \end{cases}$$

Find the values of the constants a, b, c and d for which f is differentiable for all x.

3.13★

(a) Let $f(x) = \begin{cases} x^3 & \text{if } x \le a \\ 12x - 16 & \text{if } x > a. \end{cases}$ Find all values of the constant a for which f is differentiable for all x.

(b) Let $g(x) = \begin{cases} \ln x & \text{if } 0 < x \le a \\ mx & \text{if } x > a, \end{cases}$ where $a > 0$ and m are constants. Find the values of a and m for which g is differentiable for all $x > 0$.

3.14 [BC 2008]

$$f(x) = \begin{cases} cx + d & \text{for } x \le 2 \\ x^2 - cx & \text{for } x > 2 \end{cases}$$

Let f be the function defined above, where c and d are constants. If f is differentiable at $x = 2$, what is the value of $c + d$?

(A) -4 (B) -2 (C) 0 (D) 2

4 Detailed solutions

Practice

Section 1: Derivative Rules

P1

(a)

$$\begin{aligned} y &= 8x^{\frac{1}{3}} + 2x^{-3} - \frac{7}{4}x^{-1} + \sqrt{3} && \text{rewrite every term as } cx^n \\ \frac{dy}{dx} &= 8 \cdot \frac{1}{3}x^{-\frac{2}{3}} + 2(-3)x^{-4} - \frac{7}{4}(-1)x^{-2} + 0 && \text{power rule; } \sqrt{3} \text{ is a constant} \\ &= \frac{8}{3x^{\frac{2}{3}}} - \frac{6}{x^4} + \frac{7}{4x^2}. \end{aligned}$$

(b)

$$\begin{aligned} y &= 4x^4 - 4x^2 + 1 && \text{expand} \\ \frac{dy}{dx} &= 16x^3 - 8x. \end{aligned}$$

(c)

$$\begin{aligned} f(x) &= \frac{x^3}{x^2} - \frac{2x}{x^2} + \frac{5}{x^2} = x - 2x^{-1} + 5x^{-2} && \text{split} \\ f'(x) &= 1 + 2x^{-2} - 10x^{-3} = 1 + \frac{2}{x^2} - \frac{10}{x^3} \\ f'(2) &= 1 + \frac{2}{4} - \frac{10}{8} = \frac{1}{4}. \end{aligned}$$

(The trap answer is $f(2) = \dfrac{8-4+5}{4} = \dfrac{9}{4}$.)

P2

$$\begin{aligned} y &= \frac{1}{2}x^{\frac{4}{5}} - 3x^{-5} && \text{rewrite as powers} \\ \frac{dy}{dx} &= \frac{1}{2} \cdot \frac{4}{5}x^{-\frac{1}{5}} - 3(-5)x^{-6} && \text{power rule: } \frac{4}{5} - 1 = -\frac{1}{5},\ -5 - 1 = -6 \\ &= \frac{2}{5x^{\frac{1}{5}}} + \frac{15}{x^6}. \end{aligned}$$

The answer is (A). Choice (B) lowers the power -5 to -4 instead of -6.

P3

$$f'(x) = 4(-2)x^{-3} + \frac{1}{4}\cdot 2x + 0 = -\frac{8}{x^3} + \frac{x}{2} \qquad \text{power rule}$$
$$f'(2) = -\frac{8}{8} + \frac{2}{2} = -1 + 1 = 0.$$

The answer is (C).

P4

(a) e^2 is a constant.

$$f'(x) = 4(-\sin x) - 5\cos x + 2e^x - 0 = -4\sin x - 5\cos x + 2e^x$$
$$f'(0) = -4(0) - 5(1) + 2(1) = -3. \qquad \sin 0 = 0,\ \cos 0 = 1,\ e^0 = 1$$

(b)

$$y = 3\ln x + 4x^{-1} - \ln 2 \qquad \ln 2 \text{ is a constant}$$
$$\frac{dy}{dx} = \frac{3}{x} - 4x^{-2} = \frac{3}{x} - \frac{4}{x^2}.$$

(c) 2^x is an exponential and e is a constant.

$$\frac{dy}{dx} = 2^x \ln 2 - 2x.$$

(d) Split the ln first.

$$f(x) = \ln 5 + \ln x + 4x^{\frac{1}{2}} \qquad \ln(5x) = \ln 5 + \ln x$$
$$f'(x) = 0 + \frac{1}{x} + 2x^{-\frac{1}{2}} = \frac{1}{x} + \frac{2}{\sqrt{x}} \qquad \ln 5 \text{ is a constant}$$
$$f'(4) = \frac{1}{4} + \frac{2}{2} = \frac{5}{4}.$$

(e)

$$f'(x) = -3\sin x - 2\cos x$$
$$f'\left(\frac{\pi}{6}\right) = -3\cdot\frac{1}{2} - 2\cdot\frac{\sqrt{3}}{2} = -\frac{3}{2} - \sqrt{3}.$$

P5

(a)

$$\frac{dy}{dx} = \frac{d}{dx}\left(x^4\right)\cdot\sin x + \frac{d}{dx}(\sin x)\cdot x^4 \qquad \text{product rule}$$
$$= 4x^3\sin x + x^4\cos x.$$

(b)

$$f'(x) = \frac{d}{dx}\left(e^x\right)\cdot\left(x^2+2x\right) + \frac{d}{dx}\left(x^2+2x\right)\cdot e^x \qquad \text{product rule}$$
$$= e^x\left(x^2+2x\right) + (2x+2)e^x = e^x\left(x^2+4x+2\right) \qquad \text{factor out } e^x$$
$$f'(1) = e(1+4+2) = 7e.$$

(c)

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left(x^{\frac{1}{3}}\right)\cdot \ln x + \frac{d}{dx}(\ln x)\cdot x^{\frac{1}{3}} && \text{product rule}\\ &= \frac{1}{3x^{\frac{2}{3}}}\ln x + \frac{1}{x}\cdot x^{\frac{1}{3}} = \frac{\ln x}{3x^{\frac{2}{3}}} + \frac{1}{x^{\frac{2}{3}}} && \frac{x^{\frac{1}{3}}}{x} = x^{-\frac{2}{3}}\\ &= \frac{\ln x + 3}{3x^{\frac{2}{3}}}.\end{aligned}$$

(d) Two polynomials: expand first.

$$\begin{aligned}y &= 2x^3 - x^2 + 6x - 3 && \text{expand}\\ \frac{dy}{dx} &= 6x^2 - 2x + 6.\end{aligned}$$

(e) Three factors: three terms.

$$\begin{aligned}f'(x) &= \frac{d}{dx}\left(x^3\right)\cdot 2^x \ln x + \frac{d}{dx}\left(2^x\right)\cdot x^3 \ln x + \frac{d}{dx}(\ln x)\cdot x^3\, 2^x\\ &= 3x^2\, 2^x \ln x + 2^x \ln 2\cdot x^3 \ln x + \frac{1}{x}\cdot x^3\, 2^x\\ f'(1) &= 0 + 0 + 1\cdot 2 = 2. && \ln 1 = 0\end{aligned}$$

P6

(a)

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{d}{dx}(x)\cdot e^x - \frac{d}{dx}\left(e^x\right)\cdot x}{\left(e^x\right)^2} && \text{quotient rule}\\ &= \frac{e^x - e^x\, x}{\left(e^x\right)^2} = \frac{e^x(1-x)}{e^x\cdot e^x} = \frac{1-x}{e^x}. && \text{cancel one } e^x\end{aligned}$$

(b)

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{d}{dx}(\cos x)\cdot(1-\sin x) - \frac{d}{dx}(1-\sin x)\cdot\cos x}{(1-\sin x)^2} && \text{quotient rule}\\ &= \frac{(-\sin x)(1-\sin x) - (-\cos x)\cos x}{(1-\sin x)^2}\\ &= \frac{-\sin x + \sin^2 x + \cos^2 x}{(1-\sin x)^2} = \frac{1-\sin x}{(1-\sin x)^2} && \sin^2 x + \cos^2 x = 1\\ &= \frac{1}{1-\sin x}.\end{aligned}$$

(c) One term on the bottom: split.

$$\begin{aligned}y &= \frac{4x^3}{x^2} - \frac{x^{\frac{1}{2}}}{x^2} = 4x - x^{-\frac{3}{2}} && \text{split}\\ \frac{dy}{dx} &= 4 + \frac{3}{2}x^{-\frac{5}{2}}.\end{aligned}$$

(d) 2^x is not a power of x, so use the quotient rule.

$$f'(x) = \frac{\frac{d}{dx}\left(2^x\right) \cdot x - \frac{d}{dx}(x) \cdot 2^x}{x^2} \qquad \text{quotient rule}$$
$$= \frac{2^x \ln 2 \cdot x - 2^x}{x^2} = \frac{2^x(x \ln 2 - 1)}{x^2}$$
$$f'(2) = \frac{4(2 \ln 2 - 1)}{4} = 2 \ln 2 - 1.$$

(e)

$$f'(x) = \frac{\frac{d}{dx}(\ln x) \cdot (2x-1) - \frac{d}{dx}(2x-1) \cdot \ln x}{(2x-1)^2} \qquad \text{quotient rule}$$
$$= \frac{\frac{1}{x}(2x-1) - 2 \ln x}{(2x-1)^2}$$
$$f'(1) = \frac{1 \cdot 1 - 0}{1^2} = 1. \qquad \ln 1 = 0$$

P7

$$f'(x) = \frac{\frac{d}{dx}(5-x) \cdot \left(x^3+2\right) - \frac{d}{dx}\left(x^3+2\right) \cdot (5-x)}{\left(x^3+2\right)^2} \qquad \text{quotient rule}$$
$$= \frac{(-1)\left(x^3+2\right) - 3x^2(5-x)}{\left(x^3+2\right)^2}$$
$$= \frac{-x^3 - 2 - 15x^2 + 3x^3}{\left(x^3+2\right)^2} = \frac{2x^3 - 15x^2 - 2}{\left(x^3+2\right)^2}.$$

The answer is (C).

P8

$$f'(x) = \frac{\frac{d}{dx}(3x-2) \cdot (2x+3) - \frac{d}{dx}(2x+3) \cdot (3x-2)}{(2x+3)^2} \qquad \text{quotient rule}$$
$$= \frac{3(2x+3) - 2(3x-2)}{(2x+3)^2} = \frac{6x + 9 - 6x + 4}{(2x+3)^2} = \frac{13}{(2x+3)^2}.$$

The answer is (C). Choice (A) has the numerator in the wrong order; choice (D) adds instead of subtracting.

P9

$$h'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{\big(g(x)\big)^2} \qquad \text{quotient rule}$$

$$h'(1) = \frac{(6)(6) - (4)(3)}{6^2} = \frac{24}{36} = \frac{2}{3}. \qquad f(1) = 3,\ f'(1) = 2 + 4 = 6,\ g(1) = 6,\ g'(1) = 4$$

The answer is (B).

P10

(a)

$$\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right) = \frac{\frac{d}{dx}(\cos x)\cdot \sin x - \frac{d}{dx}(\sin x)\cdot \cos x}{\sin^2 x} \qquad \text{quotient rule}$$

$$= \frac{(-\sin x)\sin x - \cos x \cos x}{\sin^2 x} = \frac{-\big(\sin^2 x + \cos^2 x\big)}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x} = -\csc^2 x.$$

$$\frac{d}{dx}(\csc x) = \frac{d}{dx}\left(\frac{1}{\sin x}\right) = \frac{\frac{d}{dx}(1)\cdot \sin x - \frac{d}{dx}(\sin x)\cdot 1}{\sin^2 x} \qquad \text{quotient rule}$$

$$= \frac{0\cdot \sin x - \cos x}{\sin^2 x} = -\frac{1}{\sin x}\cdot\frac{\cos x}{\sin x} = -\csc x \cot x.$$

(b) One rule for each term.

$$\frac{dy}{dx} = -4\csc^2 x - 3\csc x \cot x - \sec x \tan x + 2\cos x + \frac{4}{\sqrt{x}} - \frac{3}{x}.$$

(c)

$$\frac{dy}{dx} = \frac{d}{dx}\big(e^x\big)\cdot \sec x + \frac{d}{dx}(\sec x)\cdot e^x \qquad \text{product rule}$$

$$= e^x \sec x + \sec x \tan x\, e^x = e^x \sec x(1 + \tan x).$$

(d) Simplify first.

$$f(x) = \frac{\cos x}{\sin x}\cdot \sin x = \cos x$$

$$f'(x) = -\sin x$$

$$f'\left(\frac{\pi}{6}\right) = -\sin\frac{\pi}{6} = -\frac{1}{2}.$$

(e)

$$\frac{dy}{dx} = \frac{d}{dx}\big(5^x\big)\cdot \tan x + \frac{d}{dx}(\tan x)\cdot 5^x \qquad \text{product rule}$$

$$= 5^x \ln 5\, \tan x + 5^x \sec^2 x = 5^x\big(\ln 5\, \tan x + \sec^2 x\big).$$

(f)

$$f'(x) = -\csc x \cot x + \csc^2 x$$

$$f'\left(\frac{\pi}{6}\right) = -2\sqrt{3} + 2^2 = 4 - 2\sqrt{3}. \qquad \csc\frac{\pi}{6} = 2,\ \cot\frac{\pi}{6} = \sqrt{3}$$

P11

$$\frac{dy}{dx} = \sec^2 x - \big(-\csc^2 x\big) = \sec^2 x + \csc^2 x.$$

The answer is (D). Choice (C) forgets the minus sign in the derivative of $\cot x$.

P12

- (A), (B): the derivative of $\dfrac{1}{\tan x}$ is not $\dfrac{1}{\text{derivative of } \tan x}$.
- (C): the quotient rule with top 1 (derivative 0) and bottom $\tan x$ (derivative $\sec^2 x$), in the correct order. Correct.
- (D): the numerator has a plus sign instead of a minus sign.

(Check: $-\dfrac{\sec^2 x}{\tan^2 x} = -\dfrac{1}{\cos^2 x} \cdot \dfrac{\cos^2 x}{\sin^2 x} = -\dfrac{1}{\sin^2 x} = -\csc^2 x$.) The answer is (C).

P13

Simplify first.

$$\cos x \tan x = \cos x \cdot \frac{\sin x}{\cos x} = \sin x$$
$$\frac{d}{dx}(\cos x \tan x) = \frac{d}{dx}(\sin x) = \cos x.$$

The answer is (B).

P14

(a) Use the row $x = 0$; the middle term is a product of three factors.

$$h'(x) = 4g'(x) - \Big[e^x f(x)\cos x + f'(x)e^x \cos x - \sin x\, e^x f(x)\Big] - 3^x \ln 3$$
$$h'(0) = 4(-4) - \big[(1)(-2)(1) + (3)(1)(1) - 0\big] - \ln 3$$
$$= -16 - 1 - \ln 3 = -17 - \ln 3.$$

(b) Use the row $x = 1$.

- Top $f(x)g(x)$: value $(-1)(3) = -3$; derivative $f'(1)g(1) + g'(1)f(1) = (4)(3) + (-2)(-1) = 14$.
- Bottom $2\sqrt{x} + \ln x$: value $2 + 0 = 2$; derivative $\dfrac{1}{\sqrt{x}} + \dfrac{1}{x} = 1 + 1 = 2$ at $x = 1$.

$$h'(1) = \frac{(14)(2) - (2)(-3)}{2^2} = \frac{28 + 6}{4} = \frac{17}{2}.$$

(c) Use the row $x = 0$.

- Top $f(x) - \sin x$: value $-2 - 0 = -2$; derivative $f'(0) - \cos 0 = 3 - 1 = 2$.
- Bottom $\tan x + g(x)$: value $0 + 1 = 1$; derivative $\sec^2 0 + g'(0) = 1 - 4 = -3$.

$$h'(0) = \frac{(2)(1) - (-3)(-2)}{1^2} = 2 - 6 = -4.$$

Section 2: What the Derivative Means

P15

(a) Replace every x by $x+h$. For $f'(-1)$: $f(-1) = -2 - 1 = -3$.

$$f'(x) = \lim_{h\to 0} \frac{2(x+h)^3 + (x+h) - \big(2x^3 + x\big)}{h},$$
$$f'(-1) = \lim_{h\to 0} \frac{2(-1+h)^3 + (-1+h) + 3}{h} = \lim_{x\to -1} \frac{2x^3 + x + 3}{x+1}.$$

(Check: $f'(x) = 6x^2 + 1$, so $f'(-1) = 7$.)

(b)

$$f'(9) = \lim_{x\to 9} \frac{\sqrt{x} - 3}{x - 9} \qquad f(9) = 3$$
$$= \lim_{x\to 9} \frac{\sqrt{x} - 3}{\big(\sqrt{x} - 3\big)\big(\sqrt{x} + 3\big)} = \lim_{x\to 9} \frac{1}{\sqrt{x} + 3} = \frac{1}{6}.$$

(c)

$$f'(x) = \lim_{h\to 0} \frac{\dfrac{1}{x+h+2} - \dfrac{1}{x+2}}{h}$$
$$= \lim_{h\to 0} \frac{\dfrac{(x+2) - (x+h+2)}{(x+h+2)(x+2)}}{h} \qquad \text{common denominator}$$
$$= \lim_{h\to 0} \frac{-h}{h\,(x+h+2)(x+2)} = \lim_{h\to 0} \frac{-1}{(x+h+2)(x+2)} \qquad \text{cancel } h$$
$$= -\frac{1}{(x+2)^2},$$
$$f'(2) = -\frac{1}{16}.$$

(Check with the quotient rule: $\dfrac{0\cdot(x+2) - 1\cdot 1}{(x+2)^2} = -\dfrac{1}{(x+2)^2}$.)

P16

By the definition, $f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$, and $f(x+h)$ replaces every x by $x+h$:

$$f(x+h) = 3(x+h)^2 + 2(x+h).$$

The answer is (C). Choice (A) adds h outside f (it is $f(x) + h$, not $f(x+h)$); (B) and (D) send $x \to 0$ instead of $h \to 0$.

P17

(a) $f(x) = \cos x$ and $a = \dfrac{\pi}{5}$: the limit is $f'\left(\frac{\pi}{5}\right) = -\sin\dfrac{\pi}{5}$.

(b) The definition of $f'(3)$: $f'(3) = \dfrac{1}{3}$. (The trap answer $\ln 3$ is $f(3)$.)

(c) $2 = \sqrt{4}$, so $f(x) = \sqrt{x}$ and $a = 4$:

$$\lim_{h\to 0} \frac{\sqrt{4+h} - \sqrt{4}}{h} = f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

(d) The step is $-h$, so the bottom must be $-h$ too: $f(x) = \ln x$, $a = 3$.

$$\lim_{h\to 0} \frac{\ln(3-h) - \ln 3}{h} = -\lim_{h\to 0} \frac{\ln\big(3 + (-h)\big) - \ln 3}{-h} = -f'(3) = -\frac{1}{3}.$$

(e) The step on the top is h^2; multiply the top and the bottom by h to make the bottom h^2:

$$\lim_{h\to 0} \frac{\sin\big(\pi + h^2\big) - \sin\pi}{h} = \lim_{h\to 0} h \cdot \frac{\sin\big(\pi + h^2\big) - \sin\pi}{h^2} = 0 \cdot f'(\pi) = 0 \cdot \cos\pi = 0.$$

(f) As $h \to 0$ the quotient tends to $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$, so for h very close to 0 it is approximately $\dfrac{1}{x}$. When $x = 2e$ the limit is $\dfrac{1}{2e}$.

P18

$f(4) = 3$, so $6 - 2f(x) = -2\big(f(x) - f(4)\big)$ and $4 - x = -(x-4)$.

$$\begin{aligned}
\lim_{x\to 4} \frac{6 - 2f(x)}{4 - x} &= \lim_{x\to 4} \frac{-2\big(f(x) - f(4)\big)}{-(x-4)} = 2\lim_{x\to 4} \frac{f(x) - f(4)}{x - 4} \\
&= 2f'(4) && f \text{ is differentiable at } x = 4 \\
&= 2(14) = 28.
\end{aligned}$$

P19

(a)

$$\begin{aligned}
&\lim_{h\to 0} \frac{(x+h)^2 + 4(x+h) - x^2 - 4x}{h} \\
&= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 + 4x + 4h - x^2 - 4x}{h} && \text{expand} \\
&= \lim_{h\to 0} \frac{2xh + h^2 + 4h}{h} = \lim_{h\to 0} (2x + h + 4) && \text{cancel } h \\
&= 2x + 4.
\end{aligned}$$

It is $f'(x)$ for $f(x) = x^2 + 4x$.

(b) This is $f'(2)$ for $f(x) = x^3 + 2x$.

$$\begin{aligned}
&\lim_{h\to 0} \frac{(2+h)^3 + 2(2+h) - 8 - 4}{h} \\
&= \lim_{h\to 0} \frac{8 + 12h + 6h^2 + h^3 + 4 + 2h - 12}{h} && (2+h)^3 = 8 + 12h + 6h^2 + h^3 \\
&= \lim_{h\to 0} \frac{14h + 6h^2 + h^3}{h} = \lim_{h\to 0} \big(14 + 6h + h^2\big) \\
&= 14.
\end{aligned}$$

(Check: $f'(x) = 3x^2 + 2$, $f'(2) = 14$.)

P20

With $h = 0.001$, very close to 0,

$$\frac{\cos(x + 0.001) - \cos x}{0.001} \approx \frac{d}{dx}(\cos x) = -\sin x.$$

So the graph looks like $y = -\sin x$: 0 at $x = 0$, -1 at $x = \dfrac{\pi}{2}$ and 1 at $x = -\dfrac{\pi}{2}$. The answer is (D). Graph (A) is $\cos x$ itself.

P21

(a)

$$\begin{aligned}
g(-1) &= -2 + 1 - 5 = -6 && \text{the point is } (-1, -6) \\
g'(x) &= 6x^2 + 2x \implies g'(-1) = 6 - 2 = 4 && \text{the slope} \\
y + 6 &= 4(x + 1).
\end{aligned}$$

(b) The value is given; the slope comes from the formula for h'.

$$\begin{aligned}
h'(4) &= \frac{4^2 - 2}{4} = \frac{14}{4} = \frac{7}{2} && \text{substitute } x = 4 \\
y - (-3) &= \frac{7}{2}(x - 4), \quad \text{i.e. } y + 3 = \frac{7}{2}(x - 4).
\end{aligned}$$

(c) The line tangent to f: point $(0, -1)$, slope 3, so $y + 1 = 3x$.

For g: substitute $x = 0$, $f(0) = -1$ and $f'(0) = 3$.

$$\begin{aligned}
g'(0) &= \big(3f(0) + f'(0)\big)\cos 0 = \big(3(-1) + 3\big)(1) = 0 && \cos 0 = 1 \\
y - 5 &= 0(x - 0), \quad \text{i.e. } y = 5. && \text{a horizontal tangent}
\end{aligned}$$

(d) The graph of f' passes through $(0, -2)$, so $f'(0) = -2$ (the height of the graph of f', not a slope). The point is $(0, 3)$.

$$y - 3 = -2(x - 0), \quad \text{i.e. } y = 3 - 2x.$$

(e) Requirement 2: value 1 and slope 2 at $x = 2$.

(i) $f'(x) = 2ax$.

$$\begin{aligned}
f(2) = 4a = 1 &\implies a = \frac{1}{4} && \text{the value} \\
f'(2) = 4a = 2 &\implies a = \frac{1}{2} && \text{the slope}
\end{aligned}$$

The two equations need different values of a, so no value of a works.

(ii) $g'(x) = 3cx^2 - \dfrac{x}{2}$.

$$g(2) = 8c - 1 = 1 \implies c = \frac{1}{4} \qquad \text{the value}$$

$$g'(2) = 12c - 1 = 2 \implies c = \frac{1}{4} \qquad \text{the slope}$$

So $c = \dfrac{1}{4}$.

(iii) $h'(x) = \dfrac{nx^{n-1}}{k}$.

$$h(2) = \frac{2^n}{k} = 1 \implies k = 2^n \qquad \text{the value}$$

$$h'(2) = \frac{n \cdot 2^{n-1}}{2^n} = \frac{n}{2} = 2 \implies n = 4 \qquad \text{the slope}$$

So $n = 4$ and $k = 2^4 = 16$, i.e. $h(x) = \dfrac{x^4}{16}$.

Requirement 1: $h(0) = 0$ and $h'(0) = \dfrac{4 \cdot 0^3}{16} = 0$.

(f) Substitute the point $x = -1$, $y = f(-1) = 2$ into $\dfrac{dy}{dx}$.

$$\frac{dy}{dx}\Big|_{(-1,\,2)} = \frac{-(-1)(2)^2}{2} = \frac{4}{2} = 2$$

$$y - 2 = 2(x + 1).$$

(g) (i)

$$f'(-2) = -3 \qquad \text{the slope of } y = -3x + 1$$

$$f(-2) = -3(-2) + 1 = 7. \qquad \text{the value of the line at } x = -2$$

$f(0)$ need not be 1: that is the value of the line at $x = 0$, and the graph of f only touches the line at $x = -2$.

(ii)

$$f'(-1) = \frac{11 - 3}{3 - (-1)} = \frac{8}{4} = 2$$

$$y - 3 = 2(x + 1).$$

P22

(h) Slope: $y = x^{-2}$, so $\dfrac{dy}{dx} = -2x^{-3} = -\dfrac{2}{x^3}$; at $x = w$ the slope is $-\dfrac{2}{w^3}$.

$$\ell\colon \quad y - \frac{1}{w^2} = -\frac{2}{w^3}(x - w)$$

$$\text{at } R\ (y = 0)\colon \quad -\frac{1}{w^2} = -\frac{2}{w^3}(k - w) \implies \frac{w}{2} = k - w \qquad \text{multiply by } -\frac{w^3}{2}$$

$$k = \frac{3w}{2}.$$

When $w = 3$: $k = \dfrac{9}{2}$.

(i) Check the point and the slope at $x = 3$.

- Same point: $f(3) = 4(9) - 27 = 9$, and the line gives $18 - 3(3) = 9$.
- Same slope: $f'(x) = 8x - 3x^2$, so $f'(3) = 24 - 27 = -3$, the slope of $y = 18 - 3x$.

So ℓ passes through $\big(3, f(3)\big)$ with slope $f'(3)$: it is tangent to the graph of f at $x = 3$.

(j) (i) Parallel to $y = 9x - 1$: slope 9.

$$\frac{dy}{dx} = 3x^2 + 6x = 9 \implies x^2 + 2x - 3 = 0 \implies (x+3)(x-1) = 0 \implies x = 1 \text{ or } x = -3.$$

$$y(1) = 1 + 3 - 5 = -1, \qquad y(-3) = -27 + 27 - 5 = -5.$$

The points are $(1, -1)$ and $(-3, -5)$.

(ii) The graph of g has slope -1, so the perpendicular slope is $-\dfrac{1}{-1} = 1$.

$$f'(x) = \frac{1}{2\sqrt{x}} = 1 \implies \sqrt{x} = \frac{1}{2} \implies x = \frac{1}{4}, \qquad f\Big(\frac{1}{4}\Big) = \frac{1}{2}.$$

$P = \Big(\dfrac{1}{4}, \dfrac{1}{2}\Big)$.

(k) (i)

$$\frac{dy}{dx} = 2x - 5 = 3 \implies x = 4 \implies y = 16 - 20 + 1 = -3 \qquad \text{the point of tangency is } (4, -3)$$

$$-3 = 3(4) + k \implies k = -15. \qquad \text{the line passes through } (4, -3)$$

(ii) The line through $(0, 1)$ and $(2, -3)$ has slope $\dfrac{-3 - 1}{2 - 0} = -2$. Parallel: $g'(1) = -2$.

$$g'(x) = 2ax + 3 \implies g'(1) = 2a + 3 = -2 \implies a = -\frac{5}{2}.$$

P23

Both points lie on the tangent line at $x = 1$, so its slope is $f'(1)$.

$$f'(1) = \frac{7 - (-2)}{1 - (-2)} = \frac{9}{3} = 3.$$

The answer is (B). Choice (C) is $f(1)$.

P24

$f'(3)$ is the slope of the tangent line. Read two grid points on the line: $(1, 5)$ and $(5, 3)$.

$$f'(3) = \frac{3 - 5}{5 - 1} = -\frac{1}{2}.$$

The answer is (A). Choice (C) is $f(3) = 4$, choice (D) is the y-intercept of the tangent line, and choice (B) is the negative reciprocal of the slope.

P25

The line crosses the y-axis at -2 and the x-axis at -1.

$$f(0) = -2 \qquad \text{the line meets the graph of } f \text{ at } x = 0$$
$$f'(0) = \frac{0-(-2)}{-1-0} = -2. \qquad \text{the slope of the line}$$

So $f'(0) = f(0)$. The answer is (C).

P26

The slope of the tangent line at $x = -1$ is

$$f'(-1) = \lim_{x\to -1} \frac{f(x)-f(-1)}{x-(-1)} = \lim_{x\to -1} \frac{f(x)-f(-1)}{x+1} = -\frac{3}{5}.$$

The answer is (A). Choice (C) is the limit of $f(x)$, a y-value, not a slope.

P27

The line is $y = -x + k$, slope -1. Find where the curve has slope -1.

$$\frac{dy}{dx} = 2x + 3 = -1 \implies x = -2 \implies y = 4 - 6 + 1 = -1 \qquad \text{the point of tangency is } (-2,-1)$$
$$k = x + y = -2 + (-1) = -3. \qquad \text{the line passes through } (-2,-1)$$

The answer is (A). Choice (B) is x and choice (C) is y at the point of tangency.

P28

The line through $(0,-2)$ and $(3,4)$ has slope $\dfrac{4-(-2)}{3-0} = 2$. Parallel: $g'(-1) = 2$.

$$g'(x) = 2x + b \implies g'(-1) = -2 + b = 2 \implies b = 4.$$

The answer is (D).

P29

(a) (i) $t = 22$ is on the segment from $(20, 15)$ to $(24, 3)$, and $t = 14$ on the segment from $(12, 9)$ to $(16, 15)$. At $t = 2$, f is the formula: differentiate.

$$f'(22) = \frac{3-15}{24-20} = -3 \text{ calories per minute per minute,}$$
$$f'(14) = \frac{15-9}{16-12} = 1.5 \text{ calories per minute per minute,}$$
$$f'(t) = -\frac{3}{4}t^2 + 3t \implies f'(2) = -3 + 6 = 3 \text{ calories per minute per minute.}$$

(ii) From the graph: $f(14) = 12$ (halfway along the segment from $(12, 9)$ to $(16, 15)$) and $f'(14) = 1.5$ from (i).

$$k'(t) = \frac{d}{dt}(t)\cdot f(t) + \frac{d}{dt}\big(f(t)\big)\cdot t = f(t) + f'(t)\,t \qquad \text{product rule}$$
$$k'(14) = 12 + (1.5)(14) = 33.$$

(b) (i) At b the graph rises, so $f'(b) > 0$; c is the peak, so $f'(c) = 0$; at d the graph falls, so $f'(d) < 0$.

$$f'(d) < f'(c) < f'(b).$$

(ii) $f(a) < 0$ (below the x-axis), $f'(c) = 0$, and $f(d) - f(c) < 0$ ($f(c)$ is the highest value). That leaves $f'(a)$ and the secant slope $\dfrac{f(b) - f(a)}{b - a}$, both positive. The graph gets less steep from a to b, so the tangent at a is steeper than the secant from a to b (from the grid, about 3.8 against about 2.8). The greatest is $f'(a)$.

(iii) The secant from $\big(a, f(a)\big) \approx (-2.3, -2.8)$ to $\big(d, f(d)\big) \approx (1.1, 1.8)$ has slope about $\dfrac{4.6}{3.3} \approx 1.4$. The slope of f goes down steadily from left to right (the graph keeps bending down), so it equals 1.4 at exactly one point. At b the slope is about 1.8, more than 1.4; at c it is 0. So there is 1 point, between b and c.

P30

5.5 is between 4 and 7.

$$L'(5.5) \approx \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour.}$$

P31

3.5 is between 2 and 5.

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2} = \frac{52 - 60}{3} = -\frac{8}{3} \approx -2.667 \text{ degrees Celsius per minute.}$$

P32

- At $x = -2$ the graph rises: $f'(-2) > 0$.
- At $x = 0$ the graph is flat (a horizontal tangent): $f'(0) = 0$.
- At $x = 3$ the graph falls steeply: $f'(3) < 0$.

So $f'(3) < f'(0) < f'(-2)$. The answer is (D).

P33

The limit is $f'(c)$, so the question asks for the points where the tangent line is parallel to the secant through $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$. The secant rises gently. Slide a line with this slope across the graph:

- from the valley to the peak on the left of the y-axis, the slope goes up from 0 and back to 0, passing the secant slope twice: 2 points;
- just right of $x = 0$, the graph rises steeply and levels off at a peak: 1 point;
- after the next valley, the graph rises more and more steeply up to b: 1 point.

The graph falls everywhere else, and $x = 0$ does not count ($f'(0)$ does not exist). That makes 4 values of c. The answer is (D).

P34

(a) The derivative of a line is its slope.

- f passes through $(0,1)$ and $(1,2)$: slope 1.
- g passes through $(0,2)$ and $(3,0)$: slope $-\frac{2}{3}$.

$$h'(x) = f'(x) + g'(x) = 1 - \frac{2}{3} = \frac{1}{3}.$$

The answer is (C). Choice (D) is $h(x) = (x+1) + \left(-\frac{2}{3}x + 2\right) = \frac{1}{3}x + 3$.

(b) Read $f'(2) = 1.5$ from the graph of f' (the point $(2, 1.5)$).

$$g'(x) = f'(x) - \frac{1}{x}$$
$$g'(2) = f'(2) - \frac{1}{2} = 1.5 - 0.5 = 1.$$

P35

(a)

$$\frac{f\left(\frac{\pi}{2}\right) - f(0)}{\frac{\pi}{2} - 0} = \frac{e^{\frac{\pi}{2}} \sin\frac{\pi}{2} - e^0 \sin 0}{\frac{\pi}{2}} = \frac{e^{\frac{\pi}{2}}}{\frac{\pi}{2}} = \frac{2e^{\frac{\pi}{2}}}{\pi}. \qquad \sin\frac{\pi}{2} = 1,\ \sin 0 = 0$$

(b) R is itself a rate (gallons per hour), so the answer is in gallons per hour per hour.

$$\frac{R(8) - R(2)}{8 - 2} = \frac{25.553 - 33.929}{6} \qquad \text{calculator, radians}$$
$$= -1.396 \text{ gallons per hour per hour.}$$

(c)

$$\frac{f(b) - f(2)}{b - 2} = 6 \qquad \text{average rate of change over } [2, b]$$
$$\frac{b^2 - 3b + 2}{b - 2} = 6 \qquad f(2) = 4 - 6 = -2$$
$$\frac{(b-1)(b-2)}{b-2} = 6$$
$$b - 1 = 6 \qquad b \neq 2$$
$$b = 7.$$

P36

Only the two endpoint values are needed (no derivative of $\cos(2x)$).

$$\frac{\cos\pi - \cos 0}{\frac{\pi}{2} - 0} = \frac{-1 - 1}{\frac{\pi}{2}} = -\frac{4}{\pi}.$$

The answer is (A).

P37

(a) Use the $f(x)$ row: $\dfrac{f(12)-f(3)}{12-3}=\dfrac{15-(-2)}{9}=\dfrac{17}{9}$.

(b) Now the function is f', so use the $f'(x)$ row: $\dfrac{f'(12)-f'(3)}{12-3}=\dfrac{9-5}{9}=\dfrac{4}{9}$.

(c) Compute each one from the $f(x)$ row.

$$[-2,3]:\ \frac{-2-(-10)}{5}=\frac{8}{5}, \qquad [3,5]:\ \frac{-1-(-2)}{2}=\frac{1}{2},$$
$$[5,12]:\ \frac{15-(-1)}{7}=\frac{16}{7}, \qquad [12,15]:\ \frac{36-15}{3}=7.$$

The greatest is on $[12,15]$.

P38

Read the endpoints: $f(-5)=0$ (the start of the quarter circle) and $f(0)=2$ (the segment from $(-2,0)$ to $(0,2)$).

$$\frac{f(0)-f(-5)}{0-(-5)}=\frac{2-0}{5}=\frac{2}{5}.$$

P39

$x=1$ is in the first piece and $x=5$ in the second: $g(1)=1^2+1=2$ and $g(5)=13-5=8$.

$$\frac{g(5)-g(1)}{5-1}=\frac{8-2}{4}=\frac{3}{2}.$$

The answer is (B). The graph has a corner at $x=3$, but the average rate of change needs only $g(1)$ and $g(5)$, so "Undefined" is a trap.

P40

(a) The best estimate of $P'(4)$ is the average rate of change over the shortest interval containing 4, here $[3.9,4.1]$. The answer is (D). Choice (A) is a number of fish, not a rate.

(b) Average rate of change: $f(5)=125-50+25-16=84$ and $f(0)=-16$, so

$$\frac{f(5)-f(0)}{5-0}=\frac{84-(-16)}{5}=20.$$

Instantaneous rate: $f'(x)=3x^2-4x+5$.

$$3x^2-4x+5=20 \implies 3x^2-4x-15=0 \implies (3x+5)(x-3)=0 \implies x=3 \text{ or } x=-\frac{5}{3}.$$

Reject $-\dfrac{5}{3}$ (not in $[0,5]$). The answer is $x=3$.

P41

The rate at $t = 9$ is $h'(9)$. The best estimate is the average rate of change over the shortest interval containing $t = 9$ whose endpoints are known: $[7, 11]$.

$$h'(9) \approx \frac{30{,}763 - 16{,}859}{11 - 7}.$$

The answer is (D). The other choices use the longer intervals $[0, 15]$, $[5, 15]$ and $[7, 15]$.

P42

$$\frac{f(4) - f(1)}{4 - 1} = \frac{(4 + \ln 4) - (1 + 0)}{3} = 1 + \frac{\ln 4}{3} \approx 1.462 \qquad \text{average rate of change}$$

$$f'(c) = 1 + \frac{1}{c} \qquad \text{instantaneous rate}$$

$$1 + \frac{1}{c} = 1 + \frac{\ln 4}{3} \implies c = \frac{3}{\ln 4} \approx 2.164.$$

The answer is (C). (The calculator is needed only for the decimal; you can also intersect $y = 1 + \dfrac{1}{x}$ with $y = 1.462$.) Choice (B) solves $f(c) = 1.462$, using the value instead of the slope.

P43

(a) The radius is fixed at 3 meters, so

$$V = \pi(3)^2 h = 9\pi h \implies V'(t) = 9\pi h'(t). \qquad \text{constant-multiple rule}$$

$$V'(5) = 9\pi\big(0.5 + 0.2\cos(0.5)\big) = 19.100. \qquad \text{radian mode}$$

The volume is changing at 19.100 cubic meters per hour.

(b) The rate is $R'(25)$.

$$R'(x) = 60 + 1.6x - 0.05x^{1.5} \qquad \text{power rule: } 0.02 \cdot 2.5 = 0.05$$

$$R'(25) = 60 + 40 - 0.05(125) = 93.75. \qquad 25^{1.5} = \left(\sqrt{25}\right)^3 = 125$$

The weekly revenue is changing at 93.75 dollars per kilogram.

P44

$$w'(t) = -2t$$

$$w'(3) = -6.$$

The answer is (D). Choice (B) is $w(3) = 25 - 9 = 16$.

P45

$$v(h) = 3h^{\frac{1}{2}}$$
$$v'(h) = \frac{3}{2}h^{-\frac{1}{2}} = \frac{3}{2\sqrt{h}}$$
$$v'(2) = \frac{3}{2\sqrt{2}}.$$

The answer is (B).

P46

Scratch: `Y1=X^4+(1/2)X^3-5X^2+tan(X/2)`; `nDeriv(Y1,X,`a`)` for each a.

$$f'(-2) = -4.287, \qquad f'(-1) = 8.149, \qquad f'(0) = 0.5, \qquad f'(1) = -3.851.$$

The greatest slope is at $x = -1$. The answer is (B).

P47

Step 1: Where the graph crosses the x-axis.

$$e^{\tan x} - 2 = 0 \implies x = 0.60611\ldots \qquad \texttt{2:zero} \text{ on } [0, 1]$$

Step 2: The slope there.

$$\left.\frac{dy}{dx}\right|_{x=0.60611\ldots} = 2.961. \qquad \texttt{nDeriv(Y1,X,X)}$$

The answer is (C). Choice (A) is the x-value, not the slope.

P48

$t = 4$ is in the second piece, $2 < t \le 5$, so use $r(t) = 115 + 25\sin\left(0.2t^2\right)$.

$$r'(4) = -39.932 \text{ gallons per hour per hour.} \qquad \texttt{nDeriv(115+25sin(0.2X^2),X,4)}$$

(The first piece at $t = 4$ gives the trap answer -18.132.)

P49

The line $y = 1 + 4x$ has slope 4, so solve $f'(x) = 4$ on $[0, 2]$. *Scratch:* `Y1=1-5cos(3X^2-2)`, `Y2=` 4, window $0 \le x \le 2$; count the crossings.

$$1 - 5\cos\left(3x^2 - 2\right) = 4 \implies x = 1.185,\ 1.422,\ 1.871.$$

There are three values. The answer is (C).

P50

Step 1: The y-intercept of ℓ.

The tangent at $(k, \cos k)$ is $y = \cos k - \sin k\,(x - k)$. At $x = 0$:

$$y\text{-intercept} = \cos k + k \sin k.$$

Step 2: Through the origin.

$$\cos k + k \sin k = 0 \implies k = 2.798 \quad \text{in } 0 < k < \pi.$$

The answer is (C).

P51

Step 1: The slope of the tangent to f at $x = 2$.

$$f'(2) = 0.8. \qquad \texttt{nDeriv(ln(X\^2+1),X,2)}$$

Step 2: Equal slopes.

$$\begin{aligned} g'(a) &= f'(2) \\ 5a^4 + 3a^2 &= 0.8 && g'(x) = 5x^4 + 3x^2 \\ a &= 0.447. && a > 0 \end{aligned}$$

The answer is (C). Choice (D) solves $g(a) = 0.8$ instead of $g'(a) = 0.8$.

Section 3: Differentiability and Continuity

P52

(a) (i) f is not differentiable at

- $x = -3$: a vertical tangent;
- $x = -1$: a hole ($f(-1) = 1$, but the graph comes in to the open dot from both sides);
- $x = 1$: a corner (a gentle positive slope on the left, slope -2 on the right);
- $x = 4$: a jump;
- $x = 5$: a cusp.

At $x = 3$ the tangent is horizontal, so $f'(3) = 0$ exists.

(ii) Continuous at $x = -3$, 1 and 5 (vertical tangent, corner, cusp).

(b) (i) At $x = -3$: no, the graph has a corner (slope $\frac{3}{2}$ on the left, $-\frac{1}{2}$ on the right).
At $x = -1$: no, on the right the semicircle has a vertical tangent at $(-1, 2)$.

(ii) The slopes of the segments: $\dfrac{3-0}{-3-(-5)} = \dfrac{3}{2}$ and $\dfrac{2-3}{-1-(-3)} = -\dfrac{1}{2}$.

$$f'(x) = \begin{cases} \dfrac{3}{2} & \text{for } -5 < x < -3 \\ -\dfrac{1}{2} & \text{for } -3 < x < -1 \end{cases}$$

(iii) $\lim_{x\to -1^-} f'(x) = -\frac{1}{2}$, the slope of the segment.
$\lim_{x\to -1^+} f'(x) = \text{DNE}/-\infty$: the lower semicircle starts at $(-1, 2)$ with a vertical tangent, going down.

P53

At $x = 0$: $f(0) = 0$, but the left branch comes in to the open dot at $(0, -1)$. A jump, so f is not continuous and not differentiable at 0.

At $x = 2$: a vertical tangent, so $f'(2)$ does not exist.

At $x = 1$ and $x = 3$: horizontal tangents, so $f'(1) = f'(3) = 0$ exist. The answer is (B).

P54

- a: the two pieces meet at a sharp peak, a corner. Continuous, not differentiable.
- b: a hole with the point plotted above it. Not continuous.
- c: smooth. Differentiable.
- d: a jump. Not continuous.

The answer is (A).

P55

- At r the segment meets the lower semicircle at its left end, where the semicircle has a vertical tangent: $f'(r)$ is undefined. (The slopes are $\frac{5}{3}$ on the left and DNE/$-\infty$ on the right.)
- At s the lower semicircle has its lowest point: a horizontal tangent, $f'(s) = 0$.
- At t both semicircles have vertical tangents: $f'(t)$ is undefined.

The answer is (D).

P56

A horizontal tangent means $f'(4) = 0$: f is differentiable at $x = 4$. Statement (C) is false. The answer is (C).

(A), (B) and (D) are true: a jump, a vertical tangent and a hole all make the derivative fail to exist.

P57

$\lim_{x\to 2} g(x) = g(2)$ means g is continuous at 2. That rules out the jumps in (A) and (D).

Unequal one-sided limits of g' mean the slopes on the two sides of $x = 2$ differ. In (C) the curve is smooth, with the same slope on both sides. In (B) the two segments meet at $(2, 2)$ with slopes 2 and -2. The answer is (B).

P58

- (A): $f(a)$ is the filled dot. True.
- (B): $\lim_{x\to a} f(x)$ is the height of the open dot, not $f(a)$. True.
- (C): both sides approach the open dot. True.
- (D): $f'(x)$ is the positive slope of the left segment for $x < a$ and the negative slope of the right segment for $x > a$, so $\lim_{x\to a^-} f'(x) \neq \lim_{x\to a^+} f'(x)$ and $\lim_{x\to a} f'(x)$ does not exist. Must be false.

The answer is (D).

P59

No. $f(0) = 2$. On $-4 \le x \le 0$ the graph is the segment through $(-3, 0)$ and $(0, 2)$, so $f(x) = \frac{2}{3}(x + 3)$ there.

$$\lim_{h\to 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0^-} \frac{\frac{2}{3}(h+3) - 2}{h} = \lim_{h\to 0^-} \frac{\frac{2}{3}h}{h} = \frac{2}{3}.$$

On $0 < x < 3$ the graph decreases from $(0, 2)$ to $(3, 0)$, so $f(h) < f(0)$ for $0 < h < 3$, and

$$\frac{f(0+h) - f(0)}{h} < 0 \quad \text{for } 0 < h < 3.$$

So $\lim_{h\to 0^+} \frac{f(0+h) - f(0)}{h}$, if it exists, is ≤ 0, and it is not $\frac{2}{3}$. The one-sided limits are not equal, so $f'(0)$ does not exist: f is not differentiable at $x = 0$.

P60

(a) We need to check whether $\lim_{h\to 0} \frac{f(-3+h) - f(-3)}{h}$ exists, where $f(-3) = 0$. Split it into the two one-sided limits.

$$\begin{aligned}
&\lim_{h\to 0^-} \frac{f(-3+h) - f(-3)}{h} \\
&= \lim_{h\to 0^-} \frac{\left|9 - (-3+h)^2\right| - 0}{h} \\
&= \lim_{h\to 0^-} \frac{\left|6h - h^2\right|}{h} \\
&= \lim_{h\to 0^-} \frac{\left|h(6-h)\right|}{h} \\
&= \lim_{h\to 0^-} \frac{-h(6-h)}{h} \\
&= \lim_{h\to 0^-} (-6+h) = -6
\end{aligned}$$

$$\begin{aligned}
&\lim_{h\to 0^+} \frac{f(-3+h) - f(-3)}{h} \\
&= \lim_{h\to 0^+} \frac{\left|9 - (-3+h)^2\right| - 0}{h} \\
&= \lim_{h\to 0^+} \frac{\left|6h - h^2\right|}{h} \\
&= \lim_{h\to 0^+} \frac{\left|h(6-h)\right|}{h} \\
&= \lim_{h\to 0^+} \frac{h(6-h)}{h} \\
&= \lim_{h\to 0^+} (6-h) = 6
\end{aligned}$$

(For h close to 0, $6 - h > 0$, so $h(6 - h)$ has the sign of h.)

$-6 \neq 6$, so $f'(-3)$ does not exist: f is not differentiable at $x = -3$ (a corner).

(b) Continuity: $g(-2) = 0 - 3 = -3$ and $\lim_{x\to -2} g(x) = \sqrt{0} - 3 = -3 = g(-2)$, so g is continuous at $x = -2$.

Differentiability: we need to check whether $\lim_{h\to 0} \frac{g(-2+h) - g(-2)}{h}$ exists. Split it into the two one-sided limits.

$$\lim_{h\to 0^-}\frac{g(-2+h)-g(-2)}{h}$$
$$=\lim_{h\to 0^-}\frac{\sqrt{|h|}-3+3}{h}=\lim_{h\to 0^-}\frac{\sqrt{-h}}{h}$$
$$=\lim_{h\to 0^-}\frac{-1}{\sqrt{-h}}=\text{DNE}/-\infty$$

$$\lim_{h\to 0^+}\frac{g(-2+h)-g(-2)}{h}$$
$$=\lim_{h\to 0^+}\frac{\sqrt{|h|}-3+3}{h}=\lim_{h\to 0^+}\frac{\sqrt{h}}{h}$$
$$=\lim_{h\to 0^+}\frac{1}{\sqrt{h}}=\text{DNE}/\infty$$

The one-sided limits are not finite, so g is continuous but not differentiable at $x=-2$ (a cusp).

(c) We need to check whether $\lim_{h\to 0}\dfrac{f(3+h)-f(3)}{h}$ exists, where $f(3)=9-3=6$. For $h<0$ use x^2-x; for $h>0$ use $5x-9$.

$$\lim_{h\to 0^-}\frac{f(3+h)-f(3)}{h}$$
$$=\lim_{h\to 0^-}\frac{(3+h)^2-(3+h)-6}{h}$$
$$=\lim_{h\to 0^-}\frac{5h+h^2}{h}$$
$$=\lim_{h\to 0^-}(5+h)=5$$

$$\lim_{h\to 0^+}\frac{f(3+h)-f(3)}{h}$$
$$=\lim_{h\to 0^+}\frac{5(3+h)-9-6}{h}$$
$$=\lim_{h\to 0^+}\frac{5h}{h}$$
$$=\lim_{h\to 0^+}5=5$$

Both are 5, so f is differentiable at $x=3$, and $f'(3)=5$.

(d) $f(2)=4-2+1=3$, from x^2-x+1.

$$\lim_{x\to 2^-}\frac{f(x)-f(2)}{x-2}=\lim_{x\to 2^-}\frac{(3x+1)-3}{x-2}$$
$$=\lim_{x\to 2^-}\frac{3x-2}{x-2}=\text{DNE}/-\infty,\qquad \text{top}\to 4,\ \text{bottom}\to 0^-$$
$$\lim_{x\to 2^+}\frac{f(x)-f(2)}{x-2}=\lim_{x\to 2^+}\frac{\left(x^2-x+1\right)-3}{x-2}=\lim_{x\to 2^+}\frac{(x-2)(x+1)}{x-2}$$
$$=\lim_{x\to 2^+}(x+1)=3.$$

The left limit does not exist, so f is not differentiable at $x=2$. The pieces do not meet ($3(2)+1=7\neq 3$), even though both pieces have slope 3 there.

P61

$f(3)=2$ and $f(3+h)-f(3)=|h|$.

$$\lim_{h\to 0^-}\frac{|h|}{h}=\lim_{h\to 0^-}\frac{-h}{h}=-1,\qquad \lim_{h\to 0^+}\frac{|h|}{h}=\lim_{h\to 0^+}\frac{h}{h}=1.$$

$-1\neq 1$, so $f'(3)$ does not exist (a corner). The answer is (D).

P62

I and II: $\lim_{x\to 3^-}(x+2)=5$ and $\lim_{x\to 3^+}(4x-7)=5$, so the limit exists and equals $f(3)=5$. I and II are true.

III: $f(3)=5$.

$$\lim_{x\to 3^-}\frac{f(x)-f(3)}{x-3}=\lim_{x\to 3^-}\frac{(x+2)-5}{x-3}=1,$$
$$\lim_{x\to 3^+}\frac{f(x)-f(3)}{x-3}=\lim_{x\to 3^+}\frac{(4x-7)-5}{x-3}=\lim_{x\to 3^+}\frac{4(x-3)}{x-3}=4.$$

$1\neq 4$: a corner, so III is false. The answer is (C), I and II only.

P63

For $x\neq 2$, $f(x)=\dfrac{(x-2)(x+2)}{x-2}=x+2$.

I: $\lim_{x\to 2}f(x)=\lim_{x\to 2}(x+2)=4$. True.

II: $f(2)=1\neq 4$. False.

III: $f(2)=1$, so

$$\lim_{x\to 2}\frac{f(x)-f(2)}{x-2}=\lim_{x\to 2}\frac{(x+2)-1}{x-2}=\lim_{x\to 2}\frac{x+1}{x-2}:$$

the top tends to 3 and the bottom to 0, so the quotient blows up on both sides. False.

The answer is (A), I only.

P64

At $x=-1$: $f(-1)=1-3=-2$

$$\lim_{x\to -1^-}\frac{f(x)-f(-1)}{x-(-1)}=\lim_{x\to -1^-}\frac{\frac{2}{x}+2}{x+1}=\lim_{x\to -1^-}\frac{\frac{2(x+1)}{x}}{x+1}=\lim_{x\to -1^-}\frac{2}{x}=-2,$$
$$\lim_{x\to -1^+}\frac{f(x)-f(-1)}{x-(-1)}=\lim_{x\to -1^+}\frac{\left(x^2-3\right)+2}{x+1}=\lim_{x\to -1^+}(x-1)=-2.$$

Equal: differentiable at $x=-1$.

At $x=2$: $f(2)=4-3=1$

$$\lim_{x\to 2^+}\frac{f(x)-f(2)}{x-2}=\lim_{x\to 2^+}\frac{(4x-3)-1}{x-2}=\lim_{x\to 2^+}\frac{4x-4}{x-2}=\text{DNE}/\infty,$$

since the top tends to 4 and the bottom to 0 through positive values. So f is not differentiable at $x=2$ (the pieces do not meet: $4(2)-3=5\neq 1$), although both pieces have slope 4 there.

The answer is (B), $x=2$ only.

P65

(a) I and II only.

- I: f is differentiable at $-1 \implies f$ is continuous at $-1 \implies \lim_{x\to -1} f(x) = f(-1) = 2$, and then the one-sided limit is 2 as well. True.
- II: $2 = f(-1)$ and $x + 1 = x - (-1)$, so this is the definition of $f'(-1)$. True.
- III: this says f' is continuous at -1, which is not given. Could be false.

(b) I and III only.

- I: continuous at -2 means $\lim_{x\to -2} g(x) = g(-2)$, so the limit exists. True.
- II: continuous does not give differentiable: $g(x) = |x+2|$ is continuous at -2, but $g'(-2)$ does not exist. Could be false.
- III: this limit is the definition of $g'(-2)$, so if $g'(-2)$ does not exist, the limit does not exist. True.

(c) $f(2) = -1$, so $\big(f(x)\big)^2 - 1 = \big(f(x)+1\big)\big(f(x)-1\big)$, and $f(x) + 1 = f(x) - f(2)$.

$$\begin{aligned}\lim_{x\to 2} \frac{\big(f(x)\big)^2 - 1}{x-2} &= \lim_{x\to 2} \frac{f(x)-f(2)}{x-2} \cdot \lim_{x\to 2} \big(f(x)-1\big) \\ &= f'(2) \cdot \big(f(2)-1\big) && \text{definition of } f'(2);\ f \text{ is continuous at } 2 \\ &= (2)(-1-1) = -4.\end{aligned}$$

(d) (i) P is twice differentiable $\implies P'$ is differentiable $\implies P'$ is continuous on $[5, 8]$.

$$P'(8) = 55 < 65 < 90 = P'(5).$$

By the Intermediate Value Theorem there is a time t in $(5, 8)$ with $P'(t) = 65$; this t is not 2. (The interval $[0, 2]$ gives only $t = 2$ itself.)

(ii) Yes. P' is continuous on $[0, 2]$ (as in (i)), and

$$P'(0) = 40 < 50 < 65 = P'(2).$$

By the Intermediate Value Theorem there is a time t in $(0, 2)$ with $P'(t) = 50$.

P66

- I: false. At a removable discontinuity $f(a)$ is defined, but $\lim_{x\to a} f(x) \neq f(a)$.
- II: true. This is the definition of continuity.
- III: true. Differentiable at $a \implies$ continuous at a.

The answer is (C).

P67

- (A): f is differentiable at 3 $\implies$ continuous at 3, so both one-sided limits equal $f(3) = 8$. Must be true.
- (B) and (C): $8 = f(3)$, so both are the definition of $f'(3) = 5$. Must be true.
- (D): this says f' is continuous at 3, which is not given. Could be false.

The answer is (D).

P68

- (A) and (B): $f(x) = |x - 5|$ is not differentiable at 5, but it is continuous there and $\lim_{x \to 5} f(x) = 0$. Not necessarily true.
- (C): "not differentiable at 5" means exactly that this limit, the definition of $f'(5)$, does not exist. Must be true.
- (D): the graph could instead have a cusp, a vertical tangent, a jump or a hole at $x = 5$. Not necessarily true.

The answer is (C).

P69

Continuous at 3 means $\lim_{x \to 3} f(x) = f(3)$, that is, both one-sided limits equal $f(3)$. The answer is (B).

- (A) says the opposite of (B).
- (C): continuous does not give differentiable; $f(x) = |x - 3|$ has no derivative at 3.
- (D): $f(x) = x$ is continuous at 3 with $f'(x) = 1 > 0$ everywhere.

P70

- (A): a limit can exist at a hole, where $f(1)$ is not defined; for example $f(x) = \dfrac{x^2 - 1}{x - 1}$ has $\lim_{x \to 1} f(x) = 2$, but $f(1)$ is not defined.
- (B): continuity at 1 needs $f(1)$ (it says $\lim_{x \to 1} f(x) = f(1)$).
- (C): $f'(1) = \lim_{x \to 1} \dfrac{f(x) - f(1)}{x - 1}$ uses $f(1)$. (Also, differentiable $\implies$ continuous.)
- (D): the tangent line passes through the point $\big(1, f(1)\big)$ on the graph.

The answer is (A).

P71

At $x = a$ the graph has a hole (open dot) on the curve and the value $f(a)$ (filled dot) below it.

- From both sides $f(x)$ approaches the height of the hole, so $\lim_{x\to a} f(x)$ exists ((B) and (D) are false), but it is not $f(a)$. So (C) is true.
- (A): f is not continuous at $a \implies f$ is not differentiable at a, so the limit in (A), the definition of $f'(a)$, does not exist.

The answer is (C).

P72

f is (twice) differentiable, so f is continuous at 2: $\lim_{x\to 2} f(x) = f(2) = 3$. The bottom tends to $2 \neq 0$, so substitute.

$$\lim_{x\to 2} \frac{f(x)}{x} = \frac{f(2)}{2} = \frac{3}{2}.$$

The value $f'(2) = 4$ is not needed: the limit is not $0/0$.

P73

Check continuity first:

$$\lim_{x\to 5^-} f(x) = \lim_{x\to 5^-} \left(x^2 - 20\right) = 5,$$
$$f(5) = \lim_{x\to 5^+} f(x) = \lim_{x\to 5^+} \left(-x^2 + 20\right) = -5.$$

$5 \neq -5$: a jump. f is not continuous at 5, so it is not differentiable there. The answer is (A).

Choice (B) is a trap: the slopes $2x = 10$ and $-2x = -10$ differ, but the pieces do not meet, so the graph has no corner.

P74

(a) Match the value and the slope at $x = 2$:

$$4a - 1 = k = 2b + 3 \qquad \text{the value}$$
$$4a = b \qquad \text{the slopes: } g'(x) = 2ax,\ h'(x) = b$$

So $4a - 1 = 2(4a) + 3$, which gives $a = -1$, then $b = -4$ and $k = -5$. (Method 2 gives the same equations: the top of each one-sided quotient must tend to 0, then $\lim_{x\to 2^-} \dfrac{a(x^2-4)}{x-2} = 4a$ and $\lim_{x\to 2^+} \dfrac{b(x-2)}{x-2} = b$.)

(b) Slopes first.

$$\lim_{x\to \pi^-} f'(x) = \lim_{x\to \pi^-} a\cos x = -a, \qquad \lim_{x\to \pi^+} f'(x) = \lim_{x\to \pi^+} (1 - \sin x) = 1,$$
$$-a = 1 \implies a = -1, \qquad \text{the slopes match}$$
$$a\sin\pi + b = \pi + \cos\pi \qquad \text{the values match}$$
$$b = \pi - 1.$$

So $a = -1$ and $b = \pi - 1$.

(c) At $x = 1$ the left piece has value $1 + 3 = 4$ and slope $2x = 2$. So g needs $g(1) = 4$ and $g'(1) = 2$.

$g(x)$	$g(1)$	$g'(x)$	$g'(1)$
$4x$	4	4	4
x^3+3	4	$3x^2$	3
$4\sqrt{x}$	4	$\dfrac{2}{\sqrt{x}}$	2
$2x^2+2$	4	$4x$	4

Only $g(x)=4\sqrt{x}$ matches both.

P75

Here both equations contain both unknowns. $\dfrac{d}{dx}\big(k\sqrt{x}\big)=\dfrac{k}{2\sqrt{x}}$ and $\dfrac{d}{dx}(mx+2)=m$.

$$k\sqrt{4}=4m+2 \implies 2k=4m+2 \qquad \text{the values match}$$

$$\frac{k}{2\sqrt{4}}=m \implies m=\frac{k}{4} \qquad \text{the slopes match}$$

$$2k=4\cdot\frac{k}{4}+2=k+2 \implies k=2, \qquad m=\frac{2}{4}=\frac{1}{2}.$$

So $k=2$ and $m=\dfrac{1}{2}$. (Check: both pieces give 4 at $x=4$, and both slopes are $\dfrac{1}{2}$.)

Exercises

Section 1: Derivative Rules

1.1

$$f'(x)=3x^2-2x+1$$
$$f'(2)=12-4+1=9.$$

The answer is (C). Choice (A) is $f(2)=8-4+2-1$.

1.2

$f(2)=8+4=12$ is a number, and the derivative of a number is 0:

$$\frac{d}{dx}\big(f(2)\big)=\frac{d}{dx}(12)=0.$$

The answer is (A). Choice (B) is $f(2)$; choice (C) is $f'(2)=3(2)^2+2=14$.

1.3

$$\begin{aligned} f'(x) &= \frac{d}{dx}\big(e^x\big)\cdot\sin x + \frac{d}{dx}(\sin x)\cdot e^x && \text{product rule} \\ &= e^x\sin x + e^x\cos x = e^x(\sin x+\cos x). \end{aligned}$$

The answer is (C). Choice (A) is the product of the derivatives.

1.4

$$\begin{aligned} f'(x) &= \frac{\frac{d}{dx}(\cos x)\cdot\ln x - \frac{d}{dx}(\ln x)\cdot\cos x}{(\ln x)^2} && \text{quotient rule} \\ &= \frac{(-\sin x)\ln x - \dfrac{1}{x}\cos x}{(\ln x)^2} \\ &= \frac{-x\sin x\ln x - \cos x}{x(\ln x)^2}. && \text{multiply the top and the bottom by } x \end{aligned}$$

The answer is (B).

1.5

$$\frac{d}{dx}(\cot x) = -\csc^2 x.$$

The answer is (B).

1.6

(a) e^2 and $\ln 3$ are constants.

$$\begin{aligned} y &= 4x^{\frac{1}{2}} - 3x^{-2} + 2\sin x - \tan x + 5e^x - \ln x + 2^x + e^2 + \ln 3 \\ \frac{dy}{dx} &= \frac{2}{\sqrt{x}} + \frac{6}{x^3} + 2\cos x - \sec^2 x + 5e^x - \frac{1}{x} + 2^x\ln 2. \end{aligned}$$

(b) Three factors: three terms.

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\big(x^3\big)\cdot e^x\cos x + \frac{d}{dx}\big(e^x\big)\cdot x^3\cos x + \frac{d}{dx}(\cos x)\cdot x^3e^x \\ &= 3x^2e^x\cos x + x^3e^x\cos x - x^3e^x\sin x. \end{aligned}$$

(c)

$$\begin{aligned} f'(x) &= \frac{\frac{d}{dx}\big(e^x+\sin x\big)\cdot\big(x^2+\cos x\big) - \frac{d}{dx}\big(x^2+\cos x\big)\cdot\big(e^x+\sin x\big)}{\big(x^2+\cos x\big)^2} && \text{quotient rule} \\ &= \frac{\big(e^x+\cos x\big)\big(x^2+\cos x\big) - \big(2x-\sin x\big)\big(e^x+\sin x\big)}{\big(x^2+\cos x\big)^2} \\ f'(0) &= \frac{(1+1)(0+1)-(0-0)(1+0)}{1^2} = 2. \end{aligned}$$

(d)

$$f'(x) = \frac{d}{dx}(x)\cdot \ln x + \frac{d}{dx}(\ln x)\cdot x - 3^x \ln 3 = \ln x + 1 - 3^x \ln 3 \qquad \text{product rule on } x\ln x$$
$$f'(1) = 0 + 1 - 3\ln 3 = 1 - 3\ln 3.$$

$f(1) = 0 - 3 = -3$ is a number, so $\frac{d}{dx}\big(f(1)\big) = 0$.

1.7

$$g'(x) = \frac{d}{dx}(2x-1)\cdot f(x) + \frac{d}{dx}\big(f(x)\big)\cdot(2x-1) \qquad \text{product rule}$$
$$= 2f(x) + f'(x)(2x-1)$$
$$g'(3) = 2f(3) + f'(3)\cdot 5 = 2(7) + (-1)(5) = 9.$$

1.8

The top is $f(x) + g(x)$, with derivative $f'(x) + g'(x)$.

$$M'(x) = \frac{\big(f'(x)+g'(x)\big)h(x) - h'(x)\big(f(x)+g(x)\big)}{\big(h(x)\big)^2} \qquad \text{quotient rule}$$
$$M'(0) = \frac{(-2+5)(2) - (7)(-1+3)}{2^2} = \frac{6-14}{4} = -2.$$

The answer is (A).

1.9

(a) $g(1) = 4$ is a constant, so its derivative is 0. Use the row $x = 0$.

- Top $f(x)g(x)$: value $(2)(3) = 6$; derivative $f'(0)g(0) + g'(0)f(0) = (-1)(3) + (1)(2) = -1$.
- Bottom $e^x + x$: value $1 + 0 = 1$; derivative $e^x + 1 = 2$ at $x = 0$.

$$h'(0) = \frac{(-1)(1) - (2)(6)}{1^2} + 0 = -13.$$

(b) Use the row $x = 1$.

$$k'(x) = \Big[3^x \ln 3\cdot f(x) + f'(x)\cdot 3^x\Big] - \Big[\frac{1}{2\sqrt{x}}\cdot g(x) + g'(x)\cdot\sqrt{x}\Big] \qquad \text{product rule, twice}$$
$$k'(1) = \big[3\ln 3\,(-1) + (2)(3)\big] - \Big[\frac{1}{2}(4) + (-3)(1)\Big]$$
$$= -3\ln 3 + 6 - (-1) = 7 - 3\ln 3.$$

(c) $f(0)g(0) = (2)(3) = 6$ is a number, so $\frac{d}{dx}\big(f(0)g(0)\big) = 0$.

1.10

The bottom is $x^{\frac{1}{3}}$, with derivative $\dfrac{1}{3}x^{-\frac{2}{3}}$.

$$g'(x) = \frac{f'(x)\,x^{\frac{1}{3}} - \frac{1}{3}x^{-\frac{2}{3}} \cdot f(x)}{x^{\frac{2}{3}}} \qquad \text{quotient rule; } \left(x^{\frac{1}{3}}\right)^2 = x^{\frac{2}{3}}$$

$$g'(8) = \frac{5 \cdot 2 - \frac{1}{3} \cdot \frac{1}{4} \cdot 2}{4} = \frac{10 - \frac{1}{6}}{4} = \frac{59}{24}. \qquad 8^{\frac{1}{3}} = 2,\ 8^{\frac{2}{3}} = 4$$

The answer is (A).

1.11

(a) Two polynomials: expand.

$$y = x^4 - 1 \qquad (a+b)(a-b) = a^2 - b^2$$
$$\frac{dy}{dx} = 4x^3.$$

(b) One term on the bottom: split.

$$y = x^2 - 4x^{-\frac{1}{2}} \qquad \frac{x^{\frac{1}{2}}}{x} = x^{-\frac{1}{2}}$$
$$\frac{dy}{dx} = 2x + 2x^{-\frac{3}{2}} = 2x + \frac{2}{x^{\frac{3}{2}}}.$$

(c) Two terms on the bottom: quotient rule.

$$\frac{dy}{dx} = \frac{\frac{d}{dx}(x) \cdot \left(x^2+1\right) - \frac{d}{dx}\left(x^2+1\right) \cdot x}{\left(x^2+1\right)^2}$$
$$= \frac{1 \cdot \left(x^2+1\right) - 2x \cdot x}{\left(x^2+1\right)^2} = \frac{1-x^2}{\left(x^2+1\right)^2}.$$

(d) $\sin^2 x = \sin x \cdot \sin x$: product rule.

$$\frac{dy}{dx} = \frac{d}{dx}(\sin x) \cdot \sin x + \frac{d}{dx}(\sin x) \cdot \sin x$$
$$= \cos x \sin x + \cos x \sin x = 2 \sin x \cos x.$$

(e) A constant over a power: power rule, not the quotient rule.

$$y = \frac{5}{2}x^{-3}$$
$$\frac{dy}{dx} = \frac{5}{2}(-3)x^{-4} = -\frac{15}{2x^4}.$$

Section 2: What the Derivative Means

2.1

Step 1: Turn the question into an equation.

$$\frac{f(6)-f(a)}{6-a}=0 \iff f(a)=f(6)=1.$$

So count the values of a with $-4 \le a < 6$ and $f(a)=1$.

Step 2: Look at each piece of the graph.

- On $[-4,0]$ the graph is the segment through $(-3,0)$ and $(0,2)$, so $f(x)=\frac{2}{3}(x+3)$. Then $\frac{2}{3}(a+3)=1$ gives $a=-\frac{3}{2}$.
- On $(0,3)$, f decreases from 2 to 0 and is continuous, so it takes the value 1 exactly once (Intermediate Value Theorem).
- On $[3,6)$, f increases from 0 towards 1 and stays below 1; it reaches 1 only at $x=6$, which is excluded.

Answer.

Two values of a.

2.2

$f'(x)=4x^3-5$.

(a) $\dfrac{f(3)-f(0)}{3-0}=\dfrac{(81-15)-0}{3}=\dfrac{66}{3}=22.$

(b) $f'(1)=4-5=-1.$

(c) The 6 is a hidden $f(2)$: $f(2)=16-10=6$. So the limit is $f'(2)=4(8)-5=27$.

(d) $f(2)=6$ is a number, so $\dfrac{d}{dx}\big(f(2)\big)=\dfrac{d}{dx}(6)=0$. (It is not $f'(2)=27$.)

(e) The definition of $g'\big(\frac{\pi}{4}\big)$: $\sec^2\dfrac{\pi}{4}=\big(\sqrt{2}\big)^2=2.$

2.3★

- I: $3=f(2)$, so this is $\lim\limits_{h\to0}\dfrac{f(2+h)-f(2)}{h}=f'(2)=5$. True.
- II: multiply the top and the bottom by -1: $\dfrac{3-f(x)}{2-x}=\dfrac{f(x)-f(2)}{x-2}$, so the limit is $f'(2)=5$. True.
- III: add and subtract $f(2)$ on the top.

$$\begin{aligned}\lim_{h\to0}\frac{f(2+h)-f(2-h)}{h}&=\lim_{h\to0}\frac{f(2+h)-f(2)}{h}+\lim_{h\to0}\frac{f(2)-f(2-h)}{h}\\&=f'(2)+f'(2)=10.\end{aligned}$$

the second limit: $k=-h$

Not 5.

The answer is (B).

2.4

(a) Substitute the point $x = 1$, $y = f(1) = 4$.

$$f'(1) = \frac{3(1)^2 + 1}{2(4)} = \frac{4}{8} = \frac{1}{2}.$$

(b) Find $g(1)$ and $g'(1)$, using $f(1) = 4$ and $f'(1) = \frac{1}{2}$.

$$\begin{aligned} g(1) &= 1^2 \cdot f(1) = 4 \\ g'(x) &= \frac{d}{dx}\big(x^2\big) \cdot f(x) + \frac{d}{dx}\big(f(x)\big) \cdot x^2 = 2xf(x) + f'(x)\, x^2 \qquad \text{product rule} \\ g'(1) &= 2(1)(4) + \frac{1}{2}(1) = \frac{17}{2} \\ y - 4 &= \frac{17}{2}(x - 1). \end{aligned}$$

(c) $g(1) = 4$ is a number, and the derivative of a number is 0:

$$\frac{d}{dx}\big(g(1)\big) = \frac{d}{dx}(4) = 0.$$

It is not $g'(1) = \frac{17}{2}$.

2.5★

Step 1: the tangent line.

$\frac{dy}{dx} = nx^{n-1}$, so the slope at $x = 1$ is $n \cdot 1^{n-1} = n$.

$$\ell: \quad y - 1 = n(x - 1).$$

Step 2: where ℓ meets the x-axis.

$$0 - 1 = n(x - 1) \implies x - 1 = -\frac{1}{n} \implies x = 1 - \frac{1}{n}.$$

Step 3: the area.

T has vertices $\left(1 - \frac{1}{n}, 0\right)$, $(1, 0)$ and $(1, 1)$: a right triangle with base $1 - \left(1 - \frac{1}{n}\right) = \frac{1}{n}$ and height 1.

$$\text{Area of } T = \frac{1}{2} \cdot \frac{1}{n} \cdot 1 = \frac{1}{2n}.$$

2.6

- At $x = -1$ the graph is at the bottom of a valley: $f'(-1) = 0$.
- At $x = 0$ the graph rises: $f'(0) > 0$.
- At $x = 1$ the graph falls: $f'(1) < 0$.

So $f'(1) < f'(-1) < f'(0)$. The answer is (C).

2.7

(a) $S(t)$ counts the shoppers who have entered so far, so the rate at which they enter is $S'(t)$.

$$S'(t) = 2t^3 - 48t^2 + 288t$$
$$S'(3) = 54 - 432 + 864 = 486.$$

Shoppers are entering the store at 486 shoppers per hour.

(b) $S(3)$ is a number ($S(3) = 40.5 - 432 + 1296 = 904.5$), and the derivative of a number is 0:

$$\frac{d}{dt}\big(S(3)\big) = 0.$$

It is not $S'(3)$.

(c) Find the average rate, then solve $S'(t) =$ that value.

Step 1: The average rate of change.

$$\frac{S(12) - S(0)}{12 - 0} = \frac{3456 - 0}{12} = 288. \qquad S(12) = 10368 - 27648 + 20736$$

Step 2: Solve $S'(t) = 288$.

$$2t^3 - 48t^2 + 288t = 288 \implies t = 1.245,\ 7.666 \quad \text{in } 0 < t < 12.$$

Scratch: `Y1=2X^3-48X^2+288X`, `Y2=` 288, window $0 \le x \le 12$, `5:intersect` twice.

(The third solution, $t = 15.089$, is outside the interval.)

2.8

(a)

$$W'(v) = -22.1(0.16)v^{-0.84} = -3.536v^{-0.84} \qquad \text{power rule}$$
$$W'(20) = -0.286.$$

$W'(20) = -0.286$ degrees Fahrenheit per mph.

(b) Find the average rate, then solve $W'(v) =$ that value.

Step 1: The average rate of change.

$$\frac{W(60) - W(5)}{60 - 5} = -0.254 \text{ degrees Fahrenheit per mph.}$$

Step 2: Solve $W'(v) = -0.254$ (stored value).

$$-3.536v^{-0.84} = -0.25380\ldots \implies v = 23.011.$$

Scratch: store the average rate in `A`; `Y1=-3.536X^(-0.84)`, `Y2=A`, `5:intersect`.

2.9

(a) Horizontal tangent: $g'(x) = 0$.

$$\sin\Big(x + \frac{1}{x}\Big) = 0 \implies x = 0.163,\ 0.359 \quad \text{in } 0.12 \le x \le 1.$$

Scratch: `Y1=sin(X+1/X)`, window $0.12 \le x \le 1$, `2:zero` twice.

(Check: $x + \dfrac{1}{x} = 2\pi$ and π there; $x = 1$ gives $\sin 2 \neq 0$.)

(b) Parallel tangent lines have equal slopes: $f'(x) = h'(x)$.

$$2x - 2 = 1 + \pi\cos(\pi x) \qquad f'(x) = 2x - 2$$
$$x = 0.676. \qquad \texttt{5:intersect} \text{ on } 0 < x < 1$$

Section 3: Differentiability and Continuity

3.1

For $-4 < x < 0$ the graph is the segment from $(-4, 2)$ to $(0, 0)$:

$$\lim_{x\to 0^-} g'(x) = \frac{0-2}{0-(-4)} = -\frac{1}{2}.$$

For $0 < x < 6$ the graph is the lower semicircle with centre $(3, 0)$ and radius 3. At $(0, 0)$ its tangent is vertical, and it goes down, so

$$\lim_{x\to 0^+} g'(x) = \text{DNE}/-\infty.$$

3.2

(a) $v'(4)$ does not exist. The graph has a corner at $(4, 20)$:

$$\lim_{h\to 0^-} \frac{v(4+h)-v(4)}{h} = \frac{20-0}{4-0} = 5, \qquad \lim_{h\to 0^+} \frac{v(4+h)-v(4)}{h} = 0,$$

and $5 \neq 0$.

$v'(20)$ is the slope of the segment from $(16, 20)$ to $(24, 0)$:

$$v'(20) = \frac{0-20}{24-16} = -\frac{5}{2} \text{ meters per second per second.}$$

(b) The slopes of the three segments:

$$v'(t) = \begin{cases} 5 & \text{for } 0 < t < 4 \\ 0 & \text{for } 4 < t < 16 \\ -\dfrac{5}{2} & \text{for } 16 < t < 24 \end{cases}$$

$v'(t)$ is undefined at $t = 4$ and $t = 16$ (corners).

3.3

f is even, so $f(-x) = f(x)$ and $\dfrac{f(x) - f(-x)}{2x} = 0$ for every $x \neq 0$: (B) is true.

At $x = 0$ the graph has a corner (slope -1 on the left, 1 on the right), so

$$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x} = -1 \neq 1 = \lim_{x \to 0^+} \frac{f(x) - f(0)}{x},$$

and the limit in (A) does not exist. (A) is false.

(C): near $x = 2$ the slope is $\frac{3-0}{3-0} = 1$. True. (D): $x = 3$ is a corner (slopes 1 and -1). True.

The answer is (A). The symmetric quotient in (B) has a limit even though $f'(0)$ does not exist.

3.4★

(a) Yes. $f(2) = 2 \cdot 0 = 0$ and $\lim_{x \to 2} x\,|x - 2| = 2 \cdot 0 = 0 = f(2)$.

(b) $f(2 + h) - f(2) = (2 + h)\,|h|$.

$$\lim_{h \to 0^-} \frac{(2+h)|h|}{h} = \lim_{h \to 0^-} \frac{(2+h)(-h)}{h} = \lim_{h \to 0^-} \big(-(2+h)\big) = -2$$

$$\lim_{h \to 0^+} \frac{(2+h)|h|}{h} = \lim_{h \to 0^+} \frac{(2+h)h}{h} = \lim_{h \to 0^+} (2+h) = 2$$

$-2 \neq 2$, so $f'(2)$ does not exist (a corner).

(c) $f(2 + h) = (2 + h)|h|$ and $f(2 - h) = (2 - h)|-h| = (2 - h)|h|$.

$$\lim_{h \to 0} \frac{f(2+h) - f(2-h)}{2h} = \lim_{h \to 0} \frac{(2+h)|h| - (2-h)|h|}{2h} = \lim_{h \to 0} \frac{2h\,|h|}{2h} = \lim_{h \to 0} |h| = 0.$$

No contradiction: this symmetric quotient is not the definition of $f'(2)$. It averages the slopes on the two sides, $\frac{1}{2}(-2 + 2) = 0$, and can have a limit at a corner.

3.5

(a) $\dfrac{f(h) - f(0)}{h} = \dfrac{h^{1/3}}{h} = \dfrac{1}{h^{2/3}}$, and $h^{2/3} = \left(\sqrt[3]{h}\right)^2 > 0$ on both sides:

$$\lim_{h \to 0^-} \frac{1}{h^{2/3}} = \text{DNE}/\infty, \qquad \lim_{h \to 0^+} \frac{1}{h^{2/3}} = \text{DNE}/\infty.$$

Not differentiable; both sides go to ∞: a vertical tangent.

(b) $\dfrac{g(h) - g(0)}{h} = \dfrac{h^{2/3}}{h} = \dfrac{1}{h^{1/3}}$, which has the sign of h:

$$\lim_{h \to 0^-} \frac{1}{h^{1/3}} = \text{DNE}/-\infty, \qquad \lim_{h \to 0^+} \frac{1}{h^{1/3}} = \text{DNE}/\infty.$$

Not differentiable; the sides go to $-\infty$ and ∞: a cusp.

(c) $\dfrac{k(h) - k(0)}{h} = \dfrac{h\,|h|}{h} = |h|$, and $\lim_{h \to 0^-} |h| = \lim_{h \to 0^+} |h| = 0$. Differentiable, with $k'(0) = 0$ (the graph is smooth at the origin, although $|x|$ has a corner there).

3.6

$f(2) = 0$ and $\lim_{x\to 2} \sqrt{|x-2|} = 0$, so f is continuous at 2 (this rules out (C) and (D)). With $f(2+h) - f(2) = \sqrt{|h|}$,

$$\lim_{h\to 0^+} \frac{\sqrt{|h|}}{h} = \lim_{h\to 0^+} \frac{1}{\sqrt{h}} = \text{DNE}/\infty,$$

so $f'(2)$ does not exist (a cusp). The answer is (A).

3.7

Is g differentiable at $x = 0$?

$g(0) = -7$, from $2x^2 + x - 7$.

$$\lim_{x\to 0^-} \frac{g(x) - g(0)}{x - 0} = \lim_{x\to 0^-} \frac{\left(2x^2 - x - 7\right) + 7}{x} = \lim_{x\to 0^-} (2x - 1) = -1,$$
$$\lim_{x\to 0^+} \frac{g(x) - g(0)}{x - 0} = \lim_{x\to 0^+} \frac{\left(2x^2 + x - 7\right) + 7}{x} = \lim_{x\to 0^+} (2x + 1) = 1.$$

$-1 \neq 1$, so g is not differentiable at $x = 0$.

Answer

No. f is differentiable for all real numbers (and $f'(0) = 0$), but g is not differentiable at $x = 0$. So f and g cannot be the same function.

3.8★

f is twice differentiable $\implies$ f and f' are both differentiable $\implies$ f and f' are both continuous.

(a) f is continuous at 4: $\lim_{x\to 4} f(x) = f(4) = 5$. The bottom tends to $4 - 2 = 2 \neq 0$, so substitute.

$$\lim_{x\to 4} \frac{f(x) - 5}{x - 2} = \frac{f(4) - 5}{4 - 2} = \frac{0}{2} = 0.$$

(b) Now the bottom tends to 0. The 5 is a hidden $f(4)$, so this is the definition of $f'(4)$:

$$\lim_{x\to 4} \frac{f(x) - 5}{x - 4} = \lim_{x\to 4} \frac{f(x) - f(4)}{x - 4} = f'(4) = 2.$$

(c) Yes. f is continuous on $[1, 2]$ and $f(1) = 3 < 4 < 5 = f(2)$. By the Intermediate Value Theorem there is a c in $(1, 2)$ with $f(c) = 4$.

(d) f' is continuous on $[1, 6]$. Use the Intermediate Value Theorem with the value 0 on three intervals that do not overlap:

$$f'(2) = -1 < 0 < 4 = f'(1) \qquad \text{a } c \text{ in } (1, 2)$$
$$f'(2) = -1 < 0 < 2 = f'(4) \qquad \text{a } c \text{ in } (2, 4)$$
$$f'(6) = -3 < 0 < 2 = f'(4) \qquad \text{a } c \text{ in } (4, 6)$$

The three intervals do not overlap, so these are three different values of c, $1 < c < 6$, with $f'(c) = 0$.

3.9

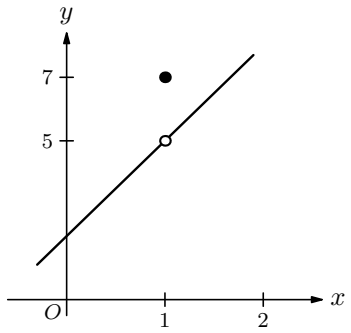
(a) I only.

- I: write $f(x) - 5$ as the given quotient times $x - 1$ (the proof in the notes):

$$\lim_{x\to 1}\big(f(x) - 5\big) = \lim_{x\to 1} \frac{f(x) - 5}{x - 1} \cdot (x - 1) = 3 \cdot 0 = 0,$$

  so $\lim\limits_{x\to 1} f(x) = 5$. True.

- II and III: the limit does not use the value $f(1)$. Take $f(x) = 3x + 2$ for $x \neq 1$ and $f(1) = 7$. Then $\dfrac{f(x) - 5}{x - 1} = \dfrac{3(x - 1)}{x - 1} = 3$ for $x \neq 1$, so the limit is 3; but $f(1) = 7 \neq 5$, and f is not continuous at 1, so $f'(1)$ does not exist. Both could be false.



$f(x) = 3x + 2$ for $x \neq 1$, $f(1) = 7$: the limit is 3, but $f(1) \neq 5$.

The answer is (A).

(b) f is continuous at 1, so $f(1) = \lim\limits_{x\to 1} f(x) = 5$ by (a). Then the 5 is $f(1)$, and the given limit is the definition of $f'(1)$:

$$f'(1) = \lim_{x\to 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x\to 1} \frac{f(x) - 5}{x - 1} = 3.$$

So $f(1) = 5$ and $f'(1) = 3$.

3.10

W is twice differentiable $\implies$ W' is differentiable $\implies$ W' is continuous on $[22, 30]$.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22).$$

By the Intermediate Value Theorem there is a time t in $(22, 30)$ with $W'(t) = 0.7$ GL/day; this t is not 10. (The interval $[0, 10]$ gives only $t = 10$ itself. The interval $[10, 22]$ does not work either: 0.7 is an endpoint value there.)

3.11

Yes. r is twice differentiable $\implies$ r' is differentiable $\implies$ r' is continuous on $[0, 3]$.

$$r'(0) = -6.1 < -6 < -5.0 = r'(3).$$

By the Intermediate Value Theorem there is a time t in $(0, 3)$ with $r'(t) = -6$.

3.12★

At each join the values and the slopes must match. The middle piece has $\dfrac{d}{dx}(\cos x + x) = -\sin x + 1$.

At $x = 0$

$$a = -\sin 0 + 1 = 1 \qquad \text{the slopes match}$$
$$a(0) + b = \cos 0 + 0 \implies b = 1 \qquad \text{the values match}$$

At $x = \pi$

$$c = -\sin \pi + 1 = 1 \qquad \text{the slopes match}$$
$$c\pi + d = \cos \pi + \pi \implies \pi + d = \pi - 1 \implies d = -1 \qquad \text{the values match}$$

Answer

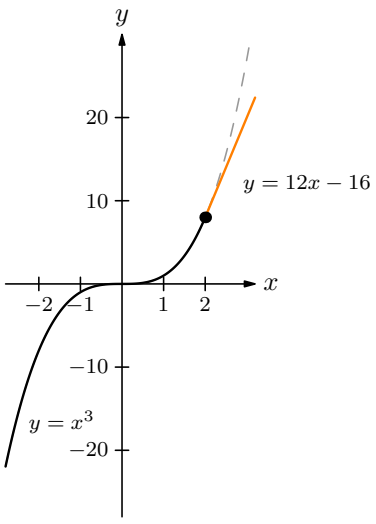
$a = 1$, $b = 1$, $c = 1$, $d = -1$. (The outer pieces $y = x+1$ and $y = x-1$ are the tangent lines to $y = \cos x + x$ at $x = 0$ and $x = \pi$.)

3.13★

(a) Each piece is differentiable, so only the join $x = a$ matters. Slopes first.

$$3a^2 = 12 \implies a = 2 \text{ or } a = -2 \qquad \text{the slopes match}$$
$$a = 2: \quad 2^3 = 8, \quad 12(2) - 16 = 8 \qquad \text{the values match}$$
$$a = -2: \quad (-2)^3 = -8, \quad 12(-2) - 16 = -40 \qquad \text{the values differ: a jump}$$

So $a = 2$ only. The line $y = 12x - 16$ is the tangent line to $y = x^3$ at $x = 2$; at $x = -2$ the slopes match but the graph jumps.



(b) $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$ and $\dfrac{d}{dx}(mx) = m$.

$$\frac{1}{a} = m \qquad \text{the slopes match}$$
$$\ln a = ma = \frac{1}{a} \cdot a = 1 \qquad \text{the values match}$$
$$a = e, \qquad m = \frac{1}{e}.$$

So $a = e$ and $m = \dfrac{1}{e}$: the line $y = \dfrac{x}{e}$ is the tangent line to $y = \ln x$ at $(e, 1)$, and it passes through the origin.

3.14

$$
\begin{aligned}
c &= 2(2) - c && \text{slopes: } \tfrac{d}{dx}(cx+d) = c,\ \tfrac{d}{dx}\left(x^2 - cx\right) = 2x - c \\
c &= 2, \\
2c + d &= 2^2 - 2c && \text{the values match} \\
4 + d = 0 &\implies d = -4.
\end{aligned}
$$

So $c + d = 2 + (-4) = -2$. The answer is (B). Choice (A) is d and (D) is c.